Further Potts Zero-Free Regions from Coupling Independence

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Abstract

The main article [SS26] establishes a transfer principle turning Hamming coupling independence of the antiferromagnetic Potts model into graph-uniform zero-free neighborhoods of activity intervals, and proves coupling independence above the Vigoda threshold $q > 11\Delta/6$. The present companion article proves coupling independence in seven further regimes — near-Vigoda, Carlson–Vigoda, large-girth, high-temperature, BBR-interval, edge-Potts, and girth-five — and records, through the same transfer principle, the resulting zero-free neighborhoods for all normalized pinned partition functions.

AI Disclosure. The coupling-independence proofs in this article were produced by AI models — GPT-5.6 Sol, except the girth-five results, which were proved by GPT-6 Astra. A Lean formalization, developed with assistance from GPT-6 Astra and Claude Opus 5.5, proves every numbered theorem, lemma, proposition and corollary of this article, some in an equivalent form [Shi26]. The Lean statements take only the displayed mathematical hypotheses. It also proves, in the form the proofs use, the results cited in the large-girth, BBR-interval, and girth-five regimes, [CLMM23, Lemmas 5.13 and 8.7 and Equation (10)] and [BBR25, Proposition 2.6(i) and Theorem 2.5], with Proposition 2.6(i) also for $\Delta \geq q + 2$. At the twenty critical pairs $(\Delta, q) = (6j, 11j)$, $1 \leq j \leq 20$, in place of [CFG+25, Theorem 20] it extends the Carlson–Vigoda contraction of Theorem 5.1 to $q \geq 11\Delta/6$ for every $\Delta \geq 6$. The authors assume responsibility for all claims.

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1 Introduction

The main article [SS26] proves that on any induced-subgraph-closed class of graphs of maximum degree at most Δ with $q \geq \Delta + 1$, Hamming coupling independence at $x = 0$ together with a uniform coupling bound on each $[\delta, 1]$, $\delta > 0$, yields a graph-uniform zero-free neighborhood of $[0,1]$ for all normalized pinned Potts partition functions (Theorem 2.4 below). This article supplies the coupling input in seven further regimes; Table 1 lists the resulting zero-free regions, and Section 4 states them precisely.

Regime	Parameters	Zero-free near
near-Vigoda	$\Delta \geq 2$, $q \geq (11/6 - 1/84000)\Delta$	$[0, 1]$
Carlson–Vigoda	$\Delta \geq 125$, $q \geq 1.809\Delta$	$[0, 1]$
large-girth	$\Delta \geq 3$, $q \geq \Delta + 3$; $\mathrm{girth}(G^\tau) \geq g_*(q, \Delta)$	$[0, 1]$
high-temperature	$\Delta \geq 2$, $q > \frac{11}{6}(1 - x_*)\Delta$, $x_* \in (0, 1]$	$[x_*, 1]$
BBR interval	$q \geq 3$, $\Delta/q \geq (e - 1/2)/(e - 1)$; $\mathrm{girth}(G^\tau) \geq g_{\mathrm{BBR}}(q, \Delta)$	$[x_0, 1]$
edge-Potts	$\Delta \geq 2$, $q \geq 3\Delta$	$[0, 1]$
girth-five	$q \geq (1 + \delta)\Delta$, $0 < \delta \leq 1$; $\Delta \geq \Delta_5(\delta)$, $\mathrm{girth}(G^\tau) \geq 5$	$[0, 1]$

In every row the conclusion is $\widehat{Z}^\tau_G(z) \neq 0$ for all z in an ε-neighborhood of the stated interval, with $\varepsilon > 0$ depending only on the listed parameters, uniformly over graphs and pinnings; the unnormalized pinned polynomial has no zeros in the same neighborhood except possibly at $z = 0$, of order $m_G(\tau)$. In the edge-Potts row the polynomial is $\widehat{Z}^\tau_{L(G)}$, with τ an edge-color pinning, and $m_G(\tau)$ is replaced by $m_{L(G)}(\tau)$. In the BBR row, $x_0 := 1 - \frac{q}{\Delta}\left(1 - \frac{k}{\Delta}\right)^2 \frac{\Delta - k}{\Delta - k/2}$ with $k := e\frac{\Delta - q/2}{\Delta - q}$ (and $x_0 := 3/4$ if $(q, \Delta) = (3, 4)$). The girth-five row assumes no bounds on local short-cycle counts.

Table 1: Zero-free regions proved in this article.

The coupling input of each row originates as follows: near-Vigoda, the strict flip coupling of [SS26] continued across the critical line, with [CFG+25, Theorem 20] at the critical pairs $(\Delta, q) = (6j, 11j)$; Carlson–Vigoda, the contraction of [CV25] extended to $x \in [0, 1]$; large-girth, the mechanism of [CLMM23] extended to $x \in [0, 1]$; high-temperature, Proposition 3.2; BBR, the certificate of [BBR25] at large girth; edge-Potts, a slot-lift coupling after [CWZZ25]; girth-five, a covariance recursion built on an all-pinning spectral gap proved in Section 9.2, converted to a coupling by the sphere-to-coupling theorem of [CLMM23, Lemma 5.13]; the girth-five setting parallels the spectral local-to-global framework of [CL26], from which no input is taken.

Organization. Section 2 fixes the pinning framework and recalls the transfer principle from the main article. Section 3 recalls the strict Vigoda-threshold coupling results and the soft-flip interface used throughout. The remaining sections prove the seven regimes in turn.

2 Framework

This section fixes the notation of [SS26] and recalls the transfer principle used in every subsequent section.

2.1 Pinnings and partition functions

Throughout, q and Δ are positive integers, $[q] = \{1, \ldots, q\}$, $x \in [0, 1]$ is a real base point, and $z \in \mathbb{C}$ is the complex edge-activity variable. We write $\mathbf{1}_{[A]}$ for the indicator of an event A, and use the convention $0^0 = 1$. For a finite graph G, write

$$\Delta(G) := \max_{v \in V(G)} \deg_G(v)$$

for its maximum degree; the symbol Δ denotes a fixed degree bound. Let

$$\mathcal{G}_\Delta := \{G : G \text{ is finite, simple, and } \Delta(G) \leq \Delta\},$$

and, for $g \geq 3$, let

$$\mathcal{G}_{\Delta, g} := \{G \in \mathcal{G}_\Delta : \mathrm{girth}(G) \geq g\}.$$

Both classes are closed under induced subgraphs. We use the convention that an acyclic graph, including the empty graph, has girth $+\infty$.

Fix $G = (V, E) \in \mathcal{G}_\Delta$. For a pinning $\tau\colon \Lambda \to [q]$, write

$$V^\tau := V \setminus \Lambda, \qquad G^\tau := G[V^\tau],$$

and call G^τ the *free graph*. For $u \in V^\tau$ and $c \in [q]$, define the boundary count

$$b_u^\tau(c) := \left|\{p \in \Lambda : up \in E,\ \tau_p = c\}\right|. \tag{1}$$

Boundary counts record pinned neighbors separately, even when several have the same color. The normalized pinned partition function has the factorized form

$$\widehat{Z}_G^\tau(z) = \sum_{\varphi\colon V^\tau \to [q]} \prod_{u \in V^\tau} z^{b_u^\tau(\varphi_u)} \prod_{uv \in E(G^\tau)} \left(z\mathbf{1}_{[\varphi_u = \varphi_v]} + \mathbf{1}_{[\varphi_u \neq \varphi_v]}\right). \tag{2}$$

Writing $m_G(\tau)$ for the number of monochromatic edges of $G[\Lambda]$, the full and normalized pinned polynomials are related by

$$Z_G^\tau(q; z) = z^{m_G(\tau)} \widehat{Z}_G^\tau(q; z); \tag{3}$$

no division by z is involved. For the empty pinning, $\widehat{Z}_G^\tau = Z_G$. We suppress q when fixed and write $\mathcal{U}_\varepsilon(I) = \{z \in \mathbb{C} : \operatorname{dist}(z, I) < \varepsilon\}$.

Lemma 2.1 (Pinned-leaf realization)**.** *For every $G \in \mathcal{G}_\Delta$ and pinning τ, there is a graph $\widetilde{G} \in \mathcal{G}_\Delta$ with a pinning $\widetilde{\tau}$ whose domain consists of leaves, such that*

$$\widehat{Z}_G^\tau(z) = \widehat{Z}_{\widetilde{G}}^{\widetilde{\tau}}(z), \qquad \operatorname{girth}(\widetilde{G}) = \operatorname{girth}(G^\tau).$$

Proof. Start from G^τ. For every edge $up \in E$ with $u \in V^\tau$ and $p \in \Lambda$, attach a distinct leaf ℓ_{up} to u and pin it to τ_p. Each leaf reproduces the corresponding boundary factor in (2), so the normalized polynomials agree. The degree of each free vertex is unchanged from its degree in G, and adding leaves creates no cycle. □

Whenever $\widehat{Z}_G^\tau(x) > 0$, let $\mu_{G,x}^\tau$ be the Gibbs law on $[q]^{V^\tau}$:

$$\mu_{G,x}^\tau(\varphi) := \frac{1}{\widehat{Z}_G^\tau(x)} \prod_{u \in V^\tau} x^{b_u^\tau(\varphi_u)} \prod_{uv \in E(G^\tau)} \left(x\mathbf{1}_{[\varphi_u = \varphi_v]} + \mathbf{1}_{[\varphi_u \neq \varphi_v]}\right).$$

For $x > 0$, this is the usual conditional Gibbs law after pinned-only edge contributions have been removed. At $x = 0$, it is the normalized hard pinned law, which remains meaningful even when τ is improper.

For $r \in V^\tau$ and $a \in [q]$, write

$$\tau^{r=a} := \tau \cup \{r \mapsto a\}.$$

Then $V^{\tau^{r=a}} = V^\tau \setminus \{r\}$, and

$$\widehat{Z}_G^\tau(z) = \sum_{a \in [q]} z^{b_r^\tau(a)} \widehat{Z}_G^{\tau^{r=a}}(z). \tag{4}$$

The multiplier $z^{b_r^\tau(a)}$ is determined when r is pinned. The normalized function $\widehat{Z}_G^{\tau^{r=a}}$ remains well defined even when this multiplier vanishes at $z = 0$.

2.2 Hard-coloring feasibility

A color c is blocked at a free vertex u when $b_u^\tau(c) > 0$. Its hard effective list is

$$L^\tau(u) := \{c \in [q] : b_u^\tau(c) = 0\}.$$

A coloring $\varphi\colon V^\tau \to [q]$ is *hard-feasible* if it is proper on G^τ and $\varphi_u \in L^\tau(u)$ for every $u \in V^\tau$.

Lemma 2.2 (Hard feasibility)**.** *Assume $q \geq \Delta + 1$. For every $G \in \mathcal{G}_\Delta$, every pinning τ, and every $u \in V^\tau$,*

$$|L^\tau(u)| \geq \deg_{G^\tau}(u) + 1.$$

Consequently $\widehat{Z}_G^\tau(0) > 0$.

Proof. At most $|N_G(u) \cap \Lambda|$ colors are blocked at u. Since

$$\deg_{G^\tau}(u) + |N_G(u) \cap \Lambda| = \deg_G(u) \leq \Delta,$$

we have

$$|L^\tau(u)| \geq q - |N_G(u) \cap \Lambda| \geq \deg_{G^\tau}(u) + 1.$$

A greedy list-coloring of G^τ, in any vertex ordering, leaves an available color at every step, so a hard-feasible coloring exists and $\widehat{Z}_G^\tau(0) > 0$. □

2.3 Coupling independence and Wasserstein distance

Let $(\mathcal{X}, d)$ be a finite metric space. For probability measures ν_0, ν_1 on $\mathcal{X}$,

$$W_{1,d}(\nu_0, \nu_1) = \inf_{\pi \in \Pi(\nu_0, \nu_1)} \mathbb{E}_\pi[d(X, Y)], \tag{5}$$

where $\Pi(\nu_0, \nu_1)$ denotes the set of couplings of ν_0 and ν_1, and X, Y are the two coordinate projections under π. On a finite space the infimum is attained. For configurations on a common site set,

$$\mathrm{Ham}(\sigma, \sigma') := \left|\{u : \sigma_u \neq \sigma'_u\}\right|, \qquad W_{1,\mathrm{Ham}} := W_{1,d} \text{ with } d = \mathrm{Ham}.$$

All Hamming distances below are taken on the common free vertex set.

Definition 2.3 (Potts coupling independence)**.** Let $C \in [0, \infty)$, $\mathcal{G} \subseteq \mathcal{G}_\Delta$, and $J \subseteq [0, 1]$. We say that $\mathcal{G}$ satisfies *C-coupling independence* on J if, for every $G \in \mathcal{G}$, every pinning τ, every $r \in V^\tau$, all $a, b \in [q]$, and every $x \in J$, the two pinned laws are well defined and

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}\right) \leq C. \tag{6}$$

They live on the common free set $V^\tau \setminus \{r\}$. At $x = 0$ they are the normalized hard pinned laws. When $\mathcal{G} = \mathcal{G}_\Delta$, we simply say that the bounded-degree Potts model satisfies C-coupling independence.

2.4 The transfer principle

The following theorem, proved in the main article [SS26], converts the real coupling bounds into complex zero-freeness.

Theorem 2.4 (Potts transfer)**.** *Fix integers q, Δ with $\Delta \geq 2$ and $q \geq \Delta + 1$, and let $\mathcal{G} \subseteq \mathcal{G}_\Delta$ be closed under taking induced subgraphs. Assume:*

(i) *$\mathcal{G}$ satisfies coupling independence at $x = 0$ with constant C_0;*

(ii) *for every $\delta \in (0, 1]$, it satisfies coupling independence on $[\delta, 1]$ with a finite constant C_δ.*

Then there is $\varepsilon > 0$ such that, for every $G \in \mathcal{G}$ and every pinning τ,

$$\widehat{Z}_G^\tau(z) \neq 0 \qquad \text{whenever } \operatorname{dist}(z, [0, 1]) < \varepsilon.$$

Consequently, $Z_G^\tau(z) = z^{m_G(\tau)} \widehat{Z}_G^\tau(z)$ has only the additional forced zero at $z = 0$ when $m_G(\tau) \geq 1$.

Its positive-temperature induction, also proved in [SS26], is used directly in some sections below.

Lemma 2.5 (Positive-base Potts induction)**.** *Fix integers q, Δ with $\Delta \geq 2$, let $\delta \in (0, 1]$ and $C \in [0, \infty)$, and let $\mathcal{G} \subseteq \mathcal{G}_\Delta$ be closed under taking induced subgraphs and satisfy C-coupling independence on $[\delta, 1]$. There are $\alpha, \varepsilon > 0$, depending only on (q, Δ, δ, C), such that for every $G \in \mathcal{G}$, every pinning τ, every $x \in [\delta, 1]$, and every $|z - x| < \varepsilon$,*

$$\widehat{Z}_G^\tau(z) \neq 0,$$

and, for every $v \in V^\tau$ and $a, b \in [q]$,

$$\left| \operatorname{Log} \frac{\widehat{Z}_G^{\tau^{v=a}}(z) / \widehat{Z}_G^{\tau^{v=a}}(x)}{\widehat{Z}_G^{\tau^{v=b}}(z) / \widehat{Z}_G^{\tau^{v=b}}(x)} \right| \leq \alpha. \tag{7}$$

The logarithm is continued from $z = x$.

3 Recalled coupling tools

This section recalls, without proof, the strict Vigoda-threshold results of the main article [SS26] and the soft-flip interface on which several later sections are built.

Theorem 3.1 (Hamming coupling independence on $[0, 1]$)**.** *Let q, Δ be positive integers with $\Delta \geq 2$ and $q > 11\Delta/6$. Put*

$$\gamma_0 := \frac{q}{\Delta} - \frac{11}{6} > 0.$$

Then, for every $G \in \mathcal{G}_\Delta$, every pinning τ, every $x \in [0, 1]$, every $r \in V^\tau$, and all $a, b \in [q]$,

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}\right) \leq \frac{2(1-x)\Delta}{q - (11/6)(1-x)\Delta} \leq \frac{2}{\gamma_0}.$$

The two laws are on $V^\tau \setminus \{r\}$. At $x = 0$ they are well defined by Lemma 2.2; indeed, the hypotheses imply $q \geq \Delta + 2$.

Proposition 3.2 (Vigoda-profile coupling for newly pinned laws)**.** *Let q, Δ be positive integers with $\Delta \geq 2$. For every $G \in \mathcal{G}_\Delta$, every pinning τ, every $r \in V^\tau$, all $a, b \in [q]$, and every $x \in (0, 1]$ satisfying $q > (11/6)(1 - x)\Delta$,*

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}\right) \leq \frac{2(1-x)\Delta}{q - (11/6)(1-x)\Delta}. \tag{8}$$

If $q > 11\Delta/6$, the same formula holds at $x = 0$.

Corollary 3.3 (Critical-line coupling input)**.** *Let q, Δ be integers with $\Delta \geq 2$ and $q = 11\Delta/6$. Then $\mathcal{G}_\Delta$ satisfies coupling independence at $x = 0$ with a finite constant, and for every $\delta \in (0, 1]$ it satisfies coupling independence on $[\delta, 1]$ with constant $12/(11\delta)$.*

3.1 Soft-flip tools

The soft-flip kernel $K_{x,\rho}^{G,\tau}$ of a profile $\rho = (\rho_s)_{s\geq 1}$ samples independent Bernoulli$(1-x)$ coins on the labeled inequality constraints of the pinned system, runs the flip chain with profile ρ on the resulting hard list-coloring instance, and forgets the sampled constraints. The following four lemmas, proved in [SS26], are the interface used below.

Lemma 3.4 (Soft flip kernels)**.** *For every profile ρ and $x \in (0,1]$, $K_{x,\rho}^{G,\tau}$ is reversible with stationary law $\mu_{G,x}^{\tau}$. If $\rho_1 > 0$, it is irreducible on $[q]^{V^\tau}$.*

Lemma 3.5 (The two hard instances differ only near v)**.** *For adjacent colorings X, Y differing at v, with $X_v = a \neq b = Y_v$, draw one independent Bernoulli$(1-x)$ coin per labeled constraint and activate that constraint on either side exactly when the coin is one and the inequality is satisfied there. This common-coin construction has the correct activation marginals. The two instances have the same induced graph and lists off v. Their incident edge sets can differ only at neighbors of color a or b, and their root lists can differ only in those two colors.*

Lemma 3.6 (Effect of adding one labeled free–pinned edge)**.** *Let two pinned systems have the same free vertex set W, of size $m \geq 1$, and the same labeled constraints except for one additional free–pinned edge in the plus system. For a profile with $R_\rho := \sup_{s\geq 1} s\rho_s < \infty$, their soft kernels satisfy*

$$W_{1,\mathrm{Ham}}(K_-(\sigma,\cdot), K_+(\sigma,\cdot)) \leq \frac{(1-x)R_\rho}{mq} \qquad (\sigma \in [q]^W). \tag{9}$$

For hard list kernels, deleting one color from one list gives the bound $R_\rho/(mq)$ on colorings feasible for the smaller list. Adding k labeled free–pinned edges (for hard list kernels, deleting k colors from lists) multiplies the respective bound by k.

Lemma 3.7 (Finite-state continuity of normalized laws)**.** *On a finite space Ω, let two families of nonnegative weights be continuous at $x = 0$, with positive total weights there. Their normalized laws $\mu_{0,x}, \mu_{1,x}$ converge in total variation as $x \downarrow 0$, and for every metric d on Ω,*

$$W_{1,d}(\mu_{0,x}, \mu_{1,x}) \longrightarrow W_{1,d}(\mu_{0,0}, \mu_{1,0}).$$

Thus a uniform positive-activity coupling bound extends to $x = 0$.

4 Zero-free consequences of the transfer principle

This section collects zero-free consequences of the Potts transfer theorem. The first subsection treats the full interval $[0,1]$; the second treats high-temperature intervals; the third gives the girth-five consequence. For fixed q, Δ, the full-interval results provide the zero-free input for deterministic approximation of normalized pinned partition functions via Barvinok interpolation and the coefficient-computation framework of [Bar16, PR17]. The coupling inputs are recalled in Section 3.1 and proved in the subsequent sections, except for the independent hard-coloring input on the critical line $q = 11\Delta/6$.

Unless explicitly redefined, this article uses the pinning and normalized-child conventions of Section 2. For probability laws ν, ν' on a finite set Ω, we use

$$\|\nu - \nu'\|_{\mathrm{TV}} := \frac{1}{2}\sum_{\omega\in\Omega} |\nu(\omega) - \nu'(\omega)|.$$

For finite-dimensional vectors, $\|\cdot\|_2$ is the Euclidean norm; $\|\cdot\|_{2\to 2}$ is its induced matrix norm. The local L^2 conventions used for the spectral-gap proof are specified in Section 9.2.

4.1 Zero-free neighborhoods of $[0,1]$

Theorem 4.1 (Additional Potts zero-free regimes)**.** *Let q, Δ be integers with $\Delta \geq 2$, and assume one of the following:*

(i) $q \geq (11/6 - 1/84000)\Delta$*;*

(ii) $\Delta \geq 125$ *and* $q \geq 1.809\Delta$*;*

(iii) $\Delta \geq 3$ *and* $q \geq \Delta + 3$*.*

There exists $\varepsilon = \varepsilon(q,\Delta) > 0$, and in regime (iii) *there also exists $g_* = g_*(q,\Delta) \geq 3$, such that the following holds. For every finite simple graph G of maximum degree at most Δ and every pinning τ, with the additional condition* $\operatorname{girth}(G^\tau) \geq g_*$ *in regime* (iii)*,*

$$\widehat{Z}_G^\tau(z) \neq 0 \qquad \text{for every } z \in \mathcal{U}_\varepsilon([0,1]).$$

Consequently, $Z_G^\tau(z)$ has no zeros in the same neighborhood except possibly at $z = 0$; its order of vanishing there is $m_G(\tau)$.

Remark 4.2 (Literature and algorithmic use of the three $[0,1]$ regimes). Regime (i) continues the strict $11\Delta/6$ coupling argument proved in the main article [SS26]. On the critical line, the endpoint input is the hard-coloring coupling of Chen, Feng, Guo, Zhang, and Zou, with the list-contraction estimates originating in the bounded-flip work of Chen, Delcourt, Moitra, Perarnau, and Postle; see [CFG+25, Theorem 20 and Proposition 22] and [CDM+19, Section 6 and Appendix E]. Regime (ii) is the Carlson–Vigoda 1.809Δ improvement, extended here from the hard endpoint to the full activity interval; compare [CV25]. Regime (iii) uses the large-girth coupling mechanism of [CLMM23], extended here to the soft activity interval.

In contrast, the BBR result below concerns only $[x_0, 1]$, not the full activity interval; its conclusion is a high-temperature zero-free neighborhood.

The following elementary observation isolates the finitely many integer pairs on the critical line that occur in regime (i).

Lemma 4.3 (Integer reduction)**.** *Let Δ and q be integers with $3 \leq \Delta \leq 124$ and*

$$q \geq \left(\frac{11}{6} - \frac{1}{84000}\right)\Delta.$$

Then either $q > 11\Delta/6$, or $(\Delta, q) = (6j, 11j)$ for some integer $1 \leq j \leq 20$.

Proof. The interval $[11\Delta/6 - \Delta/84000, 11\Delta/6]$ has length less than $1/6$. The fractional part of $11\Delta/6$ belongs to $\{0, 1/6, \ldots, 5/6\}$, so this interval contains an integer only when $11\Delta/6$ is itself an integer. Since $\gcd(11,6) = 1$, this means $\Delta = 6j$ and $q = 11j$; the bound $\Delta \leq 124$ gives $j \leq 20$. □

Remark 4.4 (The hard endpoint on the critical line). Every integer pair with $q = 11\Delta/6$ and $\Delta \geq 2$ has $(\Delta, q) = (6j, 11j)$ for some $j \geq 1$. Thus $\Delta \geq 6$, and the hard-coloring input of [CFG+25, Theorem 20 and Proposition 22] applies along the entire critical line. Its list-contraction input comes from [CDM+19, Section 6 and Appendix E]. To match the pinning conventions, recall that the normalized law at $x = 0$ is the uniform proper list-coloring law on G^τ, with

$$L^\tau(u) = \{c \in [q] : b_u^\tau(c) = 0\}, \qquad |L^\tau(u)| - \deg_{G^\tau}(u) \geq q - \Delta = \frac{5}{6}\Delta.$$

The conditional weights in [CFG+25, Section 2] likewise omit edges within the pinned set, so this correspondence allows improper τ. Apply its Theorem 20 to $\tau^{r=a}$ and $\tau^{r=b}$, and restrict the coupling to $V^\tau \setminus \{r\}$, to obtain Definition 2.3 at $x = 0$. This supplies the independent endpoint input in Corollary 3.3; the positive-temperature estimate there does not provide a constant uniform as $x \downarrow 0$.

Theorem 4.5 (Coupling inputs in the three additional Potts regimes)**.** *Let q, Δ be integers with $\Delta \geq 2$, and assume one of the three regimes in Theorem 4.1. In regimes* (i) *and* (ii)*, put $\mathcal{G} = \mathcal{G}_\Delta$. In regime* (iii)*, let $g_* = g_*(q, \Delta)$ be supplied by Theorem 6.1 and put $\mathcal{G} = \mathcal{G}_{\Delta, g_*}$. There exists $C_0 = C_0(q, \Delta) < \infty$ such that $\mathcal{G}$ satisfies C_0-coupling independence at $x = 0$, and for every $\delta \in (0, 1]$ there exists $C_\delta = C_\delta(q, \Delta) < \infty$ such that it satisfies C_δ-coupling independence on $[\delta, 1]$.*

In regimes (ii) *and* (iii)*, one finite constant may be used for every $x \in [0, 1]$. The same is true in the noncritical cases of regime* (i)*. At the exceptional pairs*

$$(\Delta, q) = (6j, 11j), \qquad 1 \leq j \leq 20, \quad j \in \mathbb{Z},$$

the endpoint bound is supplied independently by the hard-coloring coupling theorem, while one may take $C_\delta = 12/(11\delta)$. No single uniform-in-x constant is asserted at these critical pairs.

Proof. Regime (ii) is Theorem 5.1, and regime (iii) is Theorem 6.1; both theorems give one constant on $[0, 1]$.

For regime (i), first suppose $\Delta \geq 125$. Since

$$\frac{11}{6} - \frac{1}{84000} - 1.809 = \frac{681}{28000} > 0,$$

the assertion again follows from Theorem 5.1. If $\Delta = 2$, the integer hypothesis forces $q \geq 4 > 11\Delta/6$, so Theorem 3.1 applies. Finally, let $3 \leq \Delta \leq 124$. By Lemma 4.3, either $q > 11\Delta/6$, when Theorem 3.1 applies, or $(\Delta, q) = (6j, 11j)$ for $1 \leq j \leq 20$. In the latter case Corollary 3.3 supplies both the endpoint bound and the stated C_δ. □

Proof of Theorem 4.1. Every listed regime has $q \geq \Delta + 1$. In regimes (i) and (ii), apply Theorem 2.4 to $\mathcal{G}_\Delta$ using Theorem 4.5.

In regime (iii), the same two theorems first give a uniform radius for every graph in $\mathcal{G}_{\Delta, g_*}$, where g_* is the threshold supplied by Theorem 6.1. For a pair (G, τ) with $\mathrm{girth}(G^\tau) \geq g_*$, apply Lemma 2.1. The resulting graph $\widetilde{G}$ belongs to $\mathcal{G}_{\Delta, g_*}$ and has the same normalized pinned polynomial, so the established radius applies. The statement for unnormalized pinnings follows from (3). □

4.2 High-temperature zero-free neighborhoods

The tree contraction of Bencs, Berrekkal, and Regts [BBR25] gives a further region when the number of colors may be smaller than the maximum degree. Retaining the unrounded contraction parameter gives the following interval.

Theorem 4.6 (A BBR-certified interval at large girth)**.** *Let q, Δ be integers such that*

$$q \geq 3, \qquad \frac{\Delta}{q} \geq \frac{e - 1/2}{e - 1}.$$

If $(q, \Delta) = (3, 4)$, put $x_0 := 3/4$. Otherwise put

$$k := e\frac{\Delta - q/2}{\Delta - q}, \qquad x_0 := 1 - \frac{q}{\Delta}\left(1 - \frac{k}{\Delta}\right)^2 \frac{\Delta - k}{\Delta - k/2}. \tag{10}$$

Then $0 < x_0 < 1$, and there are $g_{\mathrm{BBR}} = g_{\mathrm{BBR}}(q,\Delta) \geq 3$ and $\varepsilon = \varepsilon(q,\Delta) > 0$ such that, for every finite simple graph G of maximum degree at most Δ and every pinning τ with $\operatorname{girth}(G^\tau) \geq g_{\mathrm{BBR}}$,

$$\widehat{Z}_G^\tau(z) \neq 0 \qquad \text{for every } z \in \mathcal{U}_\varepsilon([x_0,1]).$$

After decreasing ε, if necessary, so that $\varepsilon < x_0$, the same zero-free statement holds for the unnormalized pinned polynomial.

Theorem 4.6 addresses the zero-free-neighborhood question in [BBR25, Section 5]. Outside $(q,\Delta) = (3,4)$, their endpoint is obtained from (10) by replacing k with e^2. Since $k \leq e^2$ and $s \mapsto (1-s/\Delta)^3/(1-s/(2\Delta))$ is decreasing on $(0,\Delta)$, the interval above contains theirs whenever the latter is nonempty. The exceptional pair has an empty rounded interval and is covered separately here. Indeed, writing $t = \Delta - q$, the ratio hypothesis gives $t/q \geq 1/[2(e-1)]$, and hence

$$k = e\left(1 + \frac{q}{2t}\right) \leq e^2.$$

Proof of Theorem 4.6. For $(q,\Delta) = (3,4)$, the assertion $0 < x_0 < 1$ is immediate. Otherwise the ratio hypothesis and integrality imply $\Delta \geq q+2$: if $\Delta = q+1$, then the ratio hypothesis forces $q = 3$ and $\Delta = 4$. Writing $t = \Delta - q \geq 2$, we have

$$\Delta(\Delta - q) - e(\Delta - q/2) = t(t-e) + q(t - e/2) > 0,$$

where for $t = 2$ the right side is positive already at $q = 3$, and for $t \geq 3$ both summands are positive. Hence $0 < k < \Delta$. Both factors following q/Δ in (10) lie in $(0,1)$, so again $0 < x_0 < 1$.

Let g_{BBR} and C_{BBR} be supplied by Theorem 7.6. Apply Lemma 2.5 to $\mathcal{G}_{\Delta,g_{\mathrm{BBR}}}$, with $\delta = x_0$ and $C = C_{\mathrm{BBR}}$. This gives one radius throughout $[x_0,1]$ for every graph in this class and every pinning.

For the free-graph formulation, apply Lemma 2.1 to (G,τ). The resulting graph lies in $\mathcal{G}_{\Delta,g_{\mathrm{BBR}}}$ and has the same normalized pinned partition function, so the radius just obtained applies. Finally shrink it below $x_0/2$, so the possible monomial zero of an unnormalized pinning is outside the neighborhood. □

Corollary 4.7 (General-graph high-temperature intervals)**.** *Let q,Δ be integers with $\Delta \geq 2$, and fix $x_* \in (0,1]$. If*

$$q > \frac{11}{6}(1 - x_*)\Delta,$$

then there is $\varepsilon = \varepsilon(q,\Delta,x_*) > 0$ such that, for every finite simple graph G of maximum degree at most Δ and every pinning τ,

$$\widehat{Z}_G^\tau(z) \neq 0 \qquad \text{for every } z \in \mathcal{U}_\varepsilon([x_*,1]).$$

After decreasing ε, if necessary, so that $\varepsilon < x_*$, the same statement holds for the unnormalized pinned polynomial.

Proof. For $x \in [x_*,1]$, Proposition 3.2 gives coupling independence with the following uniform constant, since its bound is decreasing in x:

$$C_* := \frac{2(1-x_*)\Delta}{q - (11/6)(1-x_*)\Delta}.$$

Apply Lemma 2.5 to $\mathcal{G}_\Delta$, with $\delta = x_*$ and $C = C_*$. This gives the normalized statement. Shrinking the radius below x_* and using (3) gives the unnormalized statement. □

4.3 Girth-five zero-free neighborhoods at the $(1+\delta)\Delta$ threshold

Throughout this subsection, $\delta \in (0,1]$ is the slack parameter in $q \geq (1+\delta)\Delta$. The unrestricted girth-five coupling theorem Theorem 9.4 applies to every residual instance satisfying the degree budget and makes no local short-cycle sparsity assumption.

Theorem 4.8 (Unrestricted girth-five zero-freeness)**.** *Fix $0 < \delta \leq 1$, let $\Delta \geq \Delta_5(\delta)$ be as in Theorem 9.4, and let $q \geq (1+\delta)\Delta$. There exists $\varepsilon_5 = \varepsilon_5(q, \Delta, \delta) > 0$ such that, for every $G \in \mathcal{G}_\Delta$ and every pinning τ with $\operatorname{girth}(G^\tau) \geq 5$,*

$$\widehat{Z}_G^\tau(z) \neq 0 \qquad \text{for every } z \in \mathcal{U}_{\varepsilon_5}([0,1]).$$

The radius is independent of G and τ; no bounds on local five-cycle or six-cycle counts are assumed. The unnormalized polynomial $Z_G^\tau(z)$ has no zeros in this neighborhood except possibly at $z = 0$, where its order of vanishing is $m_G(\tau)$. In particular, $Z_G(z) \neq 0$ throughout the same neighborhood when G itself has girth at least five.

Proof. Since $q \geq \Delta + 1$, Theorems 2.4 and 9.4 give one zero-free radius on the hereditary class $\mathcal{G}_{\Delta,5}$. For the stated free-graph condition, apply Lemma 2.1 to (G, τ): the resulting graph belongs to $\mathcal{G}_{\Delta,5}$ and has the same normalized pinned polynomial. The same radius therefore applies. The statement for the unnormalized polynomial follows from (3), since the normalized polynomial is nonzero at $z = 0$. □

Remark 4.9 (Scope and relation to the spectral-gap input). The spectral local-to-global framework of Chen and Liu [CL26] concerns spectral gaps and rapid mixing on girth-five graphs. Here the all-pinning gap is supplied by Theorem 9.7; the additional coupling-independence argument and zero-free transfer are given in Theorem 9.4 and above. Thus the zero-free statement is a consequence proved here, not a coupling-independence claim attributed to that source.

5 Coupling independence for $q \geq 1.809\,\Delta$, $\Delta \geq 125$

Set

$$m_0 := \frac{1724}{1809}, \qquad \delta_{\mathrm{CV}} := \frac{59}{226125}. \tag{11}$$

Theorem 5.1 (Hamming coupling independence at $1.809\,\Delta$)**.** *Let q, Δ be positive integers with $\Delta \geq 125$ and $q \geq 1.809\,\Delta$. For every $G \in \mathcal{G}_\Delta$, every pinning τ, every $x \in [0,1]$, every $r \in V^\tau$, and all $a, b \in [q]$,*

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}\right) \leq \frac{2}{m_0 \delta_{\mathrm{CV}}} = \frac{409060125}{50858} < 8043.19. \tag{12}$$

The laws live on $V^\tau \setminus \{r\}$. At $x = 0$, they are the normalized hard pinned laws.

At $x = 0$, the underlying list-coloring contraction is due to Carlson and Vigoda [CV25, Theorem 4.12 and the proof of Corollary 1.2]. Combined with the stationary-law comparison of [BCC+22, Lemma 4.3], it already yields the corresponding bounded conditional Hamming couplings; the explicit coupling-independence formulation appears in [CFG+25, Theorem 20 and Proposition 21]. This section extends that input to positive activities.

The proof has three stages. Random active constraints first turn the soft Potts law into a mixture of hard list-coloring laws, on which we use the Carlson–Vigoda profile. A coupling matrix for adjacent configurations then combines the Carlson–Vigoda component matches with a product completion, and a geometric path metric gives uniform contraction. Finally, stationary comparison through the root-deleted instance proves Theorem 5.1; the low-multiplicity estimate is certified exactly in Lemma 5.20.

5.1 Soft Carlson–Vigoda kernel

The aggregate flip parameters of [CV25, Eq. (1)] are

$$\begin{aligned} &P_0 = 0, \quad P_1 = 1, \quad P_2 = \frac{81}{250}, \quad P_3 = \frac{77}{500}, \\ &P_4 = \frac{11}{125}, \quad P_5 = \frac{11}{250}, \quad P_6 = \frac{11}{1000}, \quad P_j = 0 \quad (j \geq 7), \end{aligned} \tag{13}$$

and we write $\boldsymbol{P} := (P_j)_{j \geq 0}$. A feasible alternating component of size j is accepted with probability P_j/j from each of its j proposal vertices, so the aggregate move has transition probability $P_j/(nq)$ on an n-vertex instance. We shall use

$$\max_j jP_j = 1, \qquad \max_j (j-2)P_j = \frac{22}{125}.$$

Throughout the contraction proof, fix $G \in \mathcal{G}_\Delta$, a pinning τ, $x \in (0,1]$, and $n := |V^\tau| \geq 1$; assume $\Delta \geq 125$, $q \geq (1809/1000)\Delta$, and set $\theta := 1 - x$. We use the free graph G^τ and boundary counts $b_u^\tau(c)$ from Section 2, which satisfy

$$\deg_{G^\tau}(u) + \sum_{c \in [q]} b_u^\tau(c) = \deg_G(u) \leq \Delta.$$

The free–pinned constraints retain their individual edge labels, as in Section 3.1, so repeated boundary colors are counted with multiplicity.

Definition 5.2 (Soft-CV kernel)**.** For $G \in \mathcal{G}_\Delta$, a pinning τ, and $x \in (0,1]$, define

$$K_{x,\mathrm{CV}}^{G,\tau} := K_{x,\boldsymbol{P}}^{G,\tau}.$$

This is the soft profile kernel of Section 3.1 specialized to (13).

The endpoint $x = 0$ is added in Theorem 5.1 by convergence of the finite-state child laws.

Because component flips are involutions and P_j/j compensates the j proposal vertices, the profile-parametrized argument of Lemma 3.4 shows that $K_{x,\mathrm{CV}}^{G,\tau}$ is irreducible and reversible with stationary law $\mu_{G,x}^\tau$. For adjacent configurations we use the coupled activations of Lemma 3.5; in each marginal the subsequent move is the ordinary hard list-coloring flip transition.

5.2 Geometric path metric

Let $X, Y \in [q]^{V^\tau}$ be adjacent with disagreement root v, $X_v = a$, $Y_v = b$, $T = \{a, b\}$. A neighbor $u \in N_{G^\tau}(v)$ is *regular* if $X_u = Y_u \notin T$. For regular u define

$$r_u := \left|\{y \in N_{G^\tau}(u) \setminus \{v\} : X_y \in T\}\right|, \qquad e_u := b_u^\tau(a) + b_u^\tau(b), \qquad s_u := \theta\, x^{\,r_u + e_u}. \tag{14}$$

The survival probability s_u is exactly the probability that, in the active instance, the root edge vu is active, every free T-blocker at u is inactive, and neither target color is deleted at u. Thus s_u is a deterministic function of the input pair; the probability is over the subsequently sampled active set. Expectations below average both the active-set draw and the flip transition. Write $\mathcal{U}_u$ for this activation event, so that $\Pr(\mathcal{U}_u) = s_u$.

Definition 5.3 (Geometric soft-CV metric)**.** For adjacent X, Y as above set

$$\ell_x^{G,\tau}(X,Y) := 1 - \frac{17}{200\,q} \sum_{u \text{ regular}} s_u, \tag{15}$$

and let $d_x^{G,\tau}$ be the shortest-path metric generated by these one-coordinate edge lengths on the graph with vertex set $[q]^{V^\tau}$, in which two configurations are adjacent exactly when they differ at one vertex. The coefficient is $(P_2 - P_3)/2$ per unit of q, matching the hard metric of [CV25, Eq. (2)]. Its limit as $x \downarrow 0$ is the hard metric: a neighbor has limiting survival probability one exactly when it has no blocker and no target deletion, and otherwise has limiting survival probability zero.

Lemma 5.4 (Metric comparison; adjacent geodesics)**.** *If $q \geq 1.809\,\Delta$, then $m_0 \operatorname{Ham}(\sigma,\sigma') \leq d_x^{G,\tau}(\sigma,\sigma') \leq \operatorname{Ham}(\sigma,\sigma')$ for all $\sigma,\sigma' \in [q]^{V^\tau}$, where m_0 is given in (11); for adjacent pairs the direct edge is a geodesic.*

Proof. There are at most Δ regular neighbors and each survival probability is at most 1, so every edge length lies in $[1 - 17\Delta/(200q), 1] \subseteq [1724/1809, 1]$. Any path makes at least $\operatorname{Ham}(\sigma,\sigma')$ coordinate changes, and changing disagreements one at a time costs at most 1 each. A path avoiding the direct adjacent edge uses at least two edges, of cost at least $2m_0 > 1$. □

For $\sigma,\sigma' \in [q]^{V^\tau}$ and a disagreement y, put $T_y = \left\{\sigma_y, \sigma'_y\right\}$. Call w an *endpoint-common neighbor* of y when

$$w \in N_{G^\tau}(y), \qquad \sigma_w = \sigma'_w \notin T_y.$$

For such y, w, define the union blocker count

$$\begin{aligned} b_{\sigma,\sigma'}(y,w) := {} & \left|\{s \in N_{G^\tau}(w) \setminus \{y\} : \sigma_s \in T_y \text{ or } \sigma'_s \in T_y\}\right| \\ & + \sum_{c \in T_y} b_w^\tau(c). \end{aligned}$$

Thus each free-edge factor is counted once even if it is a blocker in both endpoint configurations, and the edge yw is excluded. Define

$$\mathcal{S}_{\text{out}}(\sigma,\sigma') := \sum_{y:\sigma_y \neq \sigma'_y} \theta \sum_{\substack{w \in N_{G^\tau}(y) \\ \sigma_w = \sigma'_w \notin T_y}} x^{b_{\sigma,\sigma'}(y,w)}. \tag{16}$$

Lemma 5.5 (A path upper bound for the geometric metric)**.** *For all $\sigma,\sigma' \in [q]^{V^\tau}$,*

$$d_x^{G,\tau}(\sigma,\sigma') \leq \operatorname{Ham}(\sigma,\sigma') - \frac{17}{200q}\mathcal{S}_{\text{out}}(\sigma,\sigma'). \tag{17}$$

Proof. Change the disagreements in a fixed order. At the step changing y, every blocker present in the intermediate configuration carries a color taken from σ or from σ', so its blocker count at w is at most $b_{\sigma,\sigma'}(y,w)$. Since $m \mapsto x^m$ is nonincreasing for $x \in (0,1]$, each step's edge length is at most $1 - \frac{17}{200q}\theta \sum_w x^{b_{\sigma,\sigma'}(y,w)}$. Sum along the path. □

5.3 Conditional coupling under fixed activations

5.3.1 Common off-root components

Fix adjacent $X, Y \in [q]^{V^\tau}$ and an outcome of the coupled activations from Lemma 3.5. Write the two active hard instances as

$$F_X = (V^\tau, E_X, L_X), \qquad F_Y = (V^\tau, E_Y, L_Y).$$

Here E_X, E_Y are the active free-edge sets, and $L_X = (L_{X,u})_{u \in V^\tau}$, $L_Y = (L_{Y,u})_{u \in V^\tau}$ are the corresponding effective-list families. The two instances agree away from v. Call every $c \notin T$ a *regular color*, and put

$$\Gamma_c := \{u \in N_{G^\tau}(v) : X_u = Y_u = c,\ vu \in E_X \cap E_Y\}, \qquad k_c := |\Gamma_c|.$$

Lemma 3.5 implies that an edge vu with common color $c \notin T$ is active on one side if and only if it is active on the other. For such a regular color, say that c is *root-allowed* when

$$c \in L_{X,v} \cap L_{Y,v}.$$

The two membership tests are equivalent because the root lists can differ only in the colors a, b. Write

$$\mathsf{A}_c := \mathbf{1}_{[c \text{ is root-allowed}]}.$$

For a root target $c_0 \in T$, root-allowed means $a \in L_{Y,v}$ when $c_0 = a$, and $b \in L_{X,v}$ when $c_0 = b$. Equivalently it again means $c_0 \in L_{X,v} \cap L_{Y,v}$, since the other starting configuration already uses c_0.

Lemma 5.6 (Components near the disagreement vertex)**.** *For each regular color $c \notin T$:*

1. *Deleting v from the root-containing $\{a, c\}$-component in F_X gives exactly the off-root $\{a, c\}$-components in F_Y that meet Γ_c, with the same post-swap assignments. The analogous statement holds with a, X, Y replaced by b, Y, X. Consequently a root-containing move is feasible exactly when c is root-allowed and all the corresponding off-root moves are feasible.*

2. *For a fixed off-root proposal vertex and a fixed unordered color pair, the components grown on the two sides have the same vertex set and the same post-swap feasibility whenever both avoid v. If they differ, at least one contains v, and its flip recolors v.*

Proof. By Lemma 3.5, every off-root edge and list is the same in F_X and F_Y. A one-sided root edge has neighbor color a or b, and a one-sided root deletion deletes a or b. Thus all edges and lists relevant to a color $c \notin T$ are common away from the root.

Hence the $\{a, c\}$-component through v on the X-side is v together with exactly the off-root $\{a, c\}$-components attached through Γ_c. These are the same components on the Y-side, because changing the color at v from a to b does not change the induced $\{a, c\}$ graph away from v. The post-swap assignments agree at every off-root vertex. At v the root-containing swap assigns c; the possible extra deletion of the old color a is therefore harmless. Thus the only additional root list check is that c is root-allowed, proving part 1. The same off-root identity proves part 2: the component grown from an off-root proposal vertex can first differ only when one growth reaches v, and any nontrivial two-color flip containing v changes its color. □

Lemma 5.7 (Partition of the aggregate moves)**.** *After common off-root aggregate flips are coupled synchronously, every remaining nonidentity aggregate move belongs uniquely to one of the regular-color families $c \notin T$ or to one of the two root-color families a, b.*

More explicitly, delete v and consider the common off-root $\{a,b\}$-components. Let $\mathcal{B}_a$ be those containing an active a-colored neighbor joined to v on the Y-side, and let $\mathcal{B}_b$ be those containing an active b-colored neighbor joined to v on the X-side. Then

block class	*assigned moves*
$\mathcal{B}_a \setminus \mathcal{B}_b$	*a-family: Y-root and X-off-root*
$\mathcal{B}_b \setminus \mathcal{B}_a$	*b-family: X-root and Y-off-root*
$\mathcal{B}_a \cap \mathcal{B}_b$	*contained in the two root moves, with no off-root copy*
neither	*synchronously coupled.*

Proof. When the corresponding components on both sides avoid v, they have the same flipped set, feasibility, and post-swap assignments on that set by Lemma 5.6; these moves are coupled synchronously. Every remaining move either contains v or is an off-root counterpart of a component that contains v. The root-containing component has color pair $\{a,c\}$ on the X-side or $\{b,c\}$ on the Y-side. If $c \notin T$, this determines the regular c-family uniquely.

It remains to consider the pair $\{a,b\}$. Away from v, its components are common to F_X and F_Y. A block reaches v on the Y-side exactly when it belongs to $\mathcal{B}_a$, and reaches v on the X-side exactly when it belongs to $\mathcal{B}_b$. The four cases in the table follow. In particular, a block meeting both types of root neighbor is part of both root-containing components and is not also counted as an off-root move. This proves both exhaustion and uniqueness. □

5.3.2 Component matching and coupling completion

Definition 5.8 (Conditional coupling matrix)**.** Let $\mathcal{M}_X$ and $\mathcal{M}_Y$ be the side-labeled feasible component moves from X and Y, each augmented by a holding move $\varnothing$. For $M \in \mathcal{M}_X$ and $N \in \mathcal{M}_Y$, write X^M, Y^N for the outputs and $|M|, |N|$ for the numbers of flipped vertices, with $X^\varnothing = X$, $Y^\varnothing = Y$, and $|\varnothing| = 0$. Denote the two move probabilities by $\kappa_X(M)$ and $\kappa_Y(N)$. In particular, a feasible nonholding move of size j has probability $P_j/(nq)$.

A partial coupling matrix is a nonnegative array $\boldsymbol{\pi} = (\pi_{M,N})$ satisfying

$$\sum_N \pi_{M,N} \le \kappa_X(M), \qquad \sum_M \pi_{M,N} \le \kappa_Y(N).$$

Initialize it at zero. Match every common off-root move with its counterpart, with the full probability on that entry: the flipped sets and their post-swap assignments agree, but the full configurations still differ at v. Leave the holding row and column unassigned. The remaining entries are the aggregate-parameter greedy matches of [CV25, §5.1], specified as follows.

Fix $c \notin T$. If $k_c = 0$ and c is root-allowed, put the full probability $1/(nq)$ of the two singleton root moves in their common entry; their outputs agree at v. If c is not root-allowed, neither move is feasible. For $k_c \ge 1$, fix an order of Γ_c and assign every distinct off-root component to the first vertex of Γ_c that it contains.

If the X-side root-containing $\{a,c\}$-move is feasible, put its full probability in the entry joining it to a largest opposite off-root $\{a,c\}$-move. Do the same for the Y-side root-containing $\{b,c\}$-move. For each family, let its maximum-index set consist of the indices to which a largest component is assigned. If the two sets intersect, choose both entries at one common index; otherwise choose any maximizer in each family. Root feasibility implies feasibility of every constituent off-root move by Lemma 5.6, and the profile is nonincreasing on positive component sizes, so neither entry overfills its opposite row or column.

Subtract these entries, scan Γ_c in the fixed order, and at each vertex put the largest possible entry between the two current residual component probabilities assigned there. Each subtraction is

made from the component's global residual, so a component meeting several vertices is never reused. The two flipped sets in such an entry share the indexing vertex and therefore save at least one unit relative to their separate size bounds.

For root color a, the root-containing move is the Y-side $\{a,b\}$-move through v, and its possible opposite moves are the X-side off-root $\{a,b\}$-components meeting an active a-colored root neighbor but not v; interchange X, Y for root color b. If there is exactly one such neighbor and both moves are feasible, put the full root-move probability in the root/off-root entry. Otherwise put a feasible root move in the entry with holding on the opposite side. That entry removes the original root disagreement.

Lemma 5.9 (Component matching and product completion)**.** *The entries in Definition 5.8 extend to a coupling matrix with row sums κ_X and column sums κ_Y.*

Proof. The regular-color capacity checks are part of the construction. In particular, a component is recorded at its first, or canonical, vertex of Γ_c, and every later match subtracts from its global residual. This covers the case in which one component meets several vertices while the components on the other side are distinct. When the construction uses a root/off-root entry for a root color with one active neighbor, its eligible partner C satisfies that the root component is $\{v\} \cup C$. Root feasibility implies feasibility of the off-root move and $P_{|C|+1} \le P_{|C|}$, so the root/off-root entry fits. In every other root-color case, at most one root move on each side is joined to holding. The proposals whose proposed color is the current color give each side holding probability at least $1/q$, while a root move has probability at most $1/(nq)$. Hence the holding entries also fit.

Let the residual row and column probabilities be

$$\alpha_M := \kappa_X(M) - \sum_N \pi_{M,N}, \qquad \beta_N := \kappa_Y(N) - \sum_M \pi_{M,N}. \tag{18}$$

They are nonnegative by the preceding checks. Both move distributions have total mass one, and each specified entry removes the same mass from a row and a column, so

$$\sum_M \alpha_M = 1 - \sum_{M,N} \pi_{M,N} = \sum_N \beta_N =: R.$$

If $R > 0$, add

$$\frac{\alpha_M \beta_N}{R}$$

to every entry (M, N); if $R = 0$, nothing remains. This product completion has residual row sums α_M and column sums β_N, and therefore gives the required coupling matrix. □

Lemma 5.10 (Cost of the completed coupling)**.** *For the residual masses in (18), the product-completion entries have total Hamming increment at most*

$$\sum_M \alpha_M |M| + \sum_N \beta_N |N|.$$

An entry matching a common off-root move has zero increment. An explicit root-color move paired with holding saves two units relative to the flipped-set-size bound. Every nonidentity move occurs in exactly one component family.

Proof. For every pair of moves,

$$\mathrm{Ham}(X^M, Y^N) - 1 \le |M| + |N|,$$

because away from the two flipped sets only the original disagreement can remain. Consequently the product-completion cost is at most

$$\sum_M \alpha_M |M| + \sum_N \beta_N |N| .$$

This proves the displayed residual bound. A common off-root match leaves only the original disagreement and hence has zero increment. An explicit root-color move paired with holding changes the root to its color on the opposite side. For such an X-side move M, $\mathrm{Ham}(X^M, Y) - 1 \le |M| - 2$, and symmetrically on the other side. This is the stated two-unit saving relative to the size bound. Finally, Lemma 5.7 ensures that every nonidentity move occurs in exactly one family. Thus, even when a product-completion entry joins moves from different families, its two size terms are charged separately to the unique families of its row move and column move; no residual mass is omitted or counted twice. □

For the rest of this section, let (X_1, Y_1) denote the outputs drawn from the completed coupling matrix.

For every $c \notin T$, define its physical root multiplicity by

$$m_c := |\{u \in N_{G^\tau}(v) : X_u = Y_u = c\}| ,$$

and recall that $k_c = |\Gamma_c|$ is its active root multiplicity. For the current active outcome, put

$$\mathcal{C}_{\mathrm{low}} := \{c \notin T : c \text{ is root-allowed},\ m_c \in \{1, 2\}\} ,$$

and let E_v be the event that at least one selected component move recolors v.

Lemma 5.11 (Conditional component budgets)**.** *Fix a coupled active outcome for adjacent X, Y, with disagreement root v, $X_v = a$, $Y_v = b$, and put $T = \{a, b\}$. The following facts hold for the component coupling above.*

1. Root-event budget. *Conditionally on the active outcome,*

$$nq \Pr(E_v \mid F_X, F_Y) \le q - P_2 \sum_{c \in \mathcal{C}_{\mathrm{low}}} k_c. \tag{19}$$

2. Output-discount bound. *Fix a regular neighbor u and $w \in N_{G^\tau}(u) \setminus \{v\}$ which is not initially T-colored. For each $t \in T$, collect the active free edges from w to a t-colored vertex and the active pinned-neighbor constraints at w deleting t. After root-recoloring moves have been charged to E_v, the total transition probability that can recolor w into a color of T is at most $(1 + P_2)/(nq)$ whenever at least one collected constraint is active. If none is active and the root edge vu is active while all off-root T-blockers and target deletions at u are inactive, the same-neighbor matrix entry is at least $(1 - P_2)/(nq)$ and flips only u on either side. Credits obtained this way for different ordered pairs (u, w) refer to different summands of (17).*

Proof. For part 1, assign the X-side $\{a, c\}$ and Y-side $\{b, c\}$ root-containing flips to the regular color $c \notin T$; the two remaining $\{a, b\}$ root flips form the a- and b-families. These assignments are disjoint. A color that is not root-allowed contributes no root-changing move, except possibly a or b; for those exceptional colors the extra proposal is the current color on its witness side and hence is a stay proposal, while it is forbidden on the other side. Thus it contributes zero. For $c \notin T$ with $k_c = 0$, the two root-containing components are both the singleton $\{v\}$ and are coupled together

with probability $1/(nq)$. If $k_c \geq 1$, each root-containing component has size at least $k_c + 1$, so their combined probability is at most $2P_{k_c+1}/(nq)$. When $k_c = 1$,

$$2P_2 \leq 1 - P_2 \qquad \text{because} \qquad 3P_2 = \frac{243}{250} \leq 1,$$

and when $k_c = 2$,

$$2P_3 \leq 1 - 2P_2 \qquad \text{because} \qquad 2P_3 + 2P_2 = \frac{239}{250} \leq 1.$$

Hence every $c \in \mathcal{C}_{\text{low}}$ saves at least $P_2 k_c$ from its unit baseline. All other regular colors cost at most their unit baseline. Assign the X-side root-containing $\{a, b\}$-move to the b-baseline and the Y-side root-containing move to the a-baseline. Each has probability at most $P_1/(nq) = 1/(nq)$, so summing the q baselines and multiplying by nq proves (19).

For part 2, fix a target $t \in T$ and compare the two $\{X_w, t\}$-components from w. If both avoid v, by Lemma 5.6 they are identical and hence synchronously coupled, so they contribute only one copy of their component probability. If a component contains v, its move is already charged to E_v; after removing all such moves, again at most one off-root component remains. Its probability is at most $\max_j P_j/(nq) = 1/(nq)$. If an active free edge joins w to an off-root t-colored vertex, that component has size at least two and its probability is at most $P_2/(nq)$; if an active pinned-neighbor constraint deletes t at w, the component is infeasible. In either case one target contributes at most $P_2/(nq)$ and the other at most $1/(nq)$. The remaining possibility is that the active constraint is the edge wv. Then w is itself a regular root neighbor. After the two root-containing moves are assigned to E_v, the possible off-root moves are the two opposite-target components through w. If both are singletons, let r_1, r_2 be their remaining probabilities multiplied by nq, after the root/off-root entries have been assigned, taking zero for an infeasible move. Since each original move has probability at most $1/(nq)$, we have $0 \leq r_1, r_2 \leq 1$. The same-neighbor rule couples residual mass $\min\{r_1, r_2\}/(nq)$. Hence, outside E_v, the probability that at least one of these moves occurs is at most

$$\frac{r_1 + r_2 - \min\{r_1, r_2\}}{nq} = \frac{\max\{r_1, r_2\}}{nq} \leq \frac{1}{nq}.$$

If either is not a singleton, its probability is at most $P_2/(nq)$, while the other is at most $1/(nq)$. Thus this case also has total probability at most $(1 + P_2)/(nq)$, proving the assertion for every active collected constraint.

Under the safe event in the last assertion, the two target components at u are feasible singletons. Each has probability $P_1/(nq) = 1/(nq)$. A root-containing component meeting the active edge vu has size at least two and consumes at most $P_2/(nq)$ of the corresponding singleton in the root/off-root entry. The same-neighbor entry therefore has probability at least $(1 - P_2)/(nq)$ on both sides. It flips only u, to opposite colors in T, and leaves every other vertex unchanged. Thus its output correction in (17) is indexed by the ordered pair (u, w). Different ordered pairs are different terms of the double sum there (even if they share one endpoint), so summing these credits cannot reuse one metric summand. The same coupled move may realize several such savings simultaneously, in which case its path-metric correction is exactly the sum of the corresponding (u, w)-summands. □

5.4 Geometric discount estimates

Recoloring a unique blocker to a fresh color creates discount; recoloring the root, the regular neighbor, or an off-root neighbor can destroy it. The following three lemmas bound these effects.

Put

$$\alpha_{\text{fr}} := q - \Delta - 2. \tag{20}$$

For a regular neighbor u, let SB_u be the activation event that vu is active, exactly one of the r_u other free T-blocker edges at u is active, and every remaining T-blocker edge and T-colored pinned-neighbor constraint at u is inactive. Write

$$b_u^{\mathrm{SB}} := \Pr(\mathsf{SB}_u),$$

where the probability is over the active-constraint draw.

Lemma 5.12 (Fresh-color gain)**.** *Fix a regular neighbor u with $R = r_u$ free target blockers and $e = e_u$ target pinned-neighbor constraints. The expected increase of the output u-discount relative to its deterministic input value s_u, caused by singleton recolorings of its blockers, is at least*

$$\frac{\alpha_{\mathrm{fr}}}{nq}\, b_u^{\mathrm{SB}},$$

and

$$b_u^{\mathrm{SB}} = \begin{cases} R\,\theta^2 x^{R+e-1}, & R \geq 1, \\ 0, & R = 0. \end{cases}$$

Proof. The displayed probability is over the active-set draw: the root edge is active, exactly one blocker edge is active, and all other blockers and all e target-deleting pinned-neighbor constraints are inactive — independent activation bits of distinct constraints. Equivalently, the factor $R\theta^2 x^{R+e-1}$ is the sum, over the R possible unique blocker incidences, of the deterministic discount increment available on the corresponding activation event; it is not an additional probability factor for the singleton flip below. Fix a blocker incidence (u, s). At s, exclude the two target colors, every color of its full free neighborhood, and every color forbidden by one of its pinned-neighbor constraints: at least

$$q - 2 - \left(\deg_{G^\tau}(s) + \sum_{c \in [q]} b_s^\tau(c) \right) \geq \alpha_{\mathrm{fr}}$$

colors remain, and for each the alternating component of s is a list-feasible singleton in *every* active outcome, accepted with probability $P_1 = 1$ and transition mass $1/(nq)$. The flip removes one blocker incidence and raises the corresponding output u-discount relative to s_u by $\theta x^{R+e-1} - \theta x^{R+e} = \theta^2 x^{R+e-1}$. Summing over the α_{fr} colors and R incidences gives the claim; if one blocker vertex serves several regular neighbors, the single flip increases the total discount by the sum of the individual increments. Incidence summation therefore counts each discount contribution, while the flip probability is counted once. □

For a regular root-neighbor color c, recall that m_c is its physical multiplicity, and recall that $b_v^\tau(c)$ counts the root pinned-neighbor constraints forbidding c. Define

$$L_x := \sum_{\substack{c \notin T: \\ m_c \in \{1,2\}}} m_c\, x^{b_v^\tau(c)}, \qquad \phi_x := \frac{L_x}{\Delta}. \tag{21}$$

Since L_x is a weighted subcount of the physical root neighbors, $0 \leq \phi_x \leq 1$. The factor $x^{b_v^\tau(c)}$ is the probability that c survives in the active root list, and root-edge activation is independent of root boundary activation, so the expected value of $\sum_{c \in \mathcal{C}_{\mathrm{low}}} k_c$ is exactly θL_x. Averaging (19) from Lemma 5.11 therefore gives

$$nq\, \Pr(E_v) \;\leq\; q - P_2\, \theta L_x, \tag{22}$$

where E_v is the root-recoloring event. This is the soft analogue of [CV25, Eq. (18)], proved directly for the two active hard instances.

Lemma 5.13 (Ordered-incidence discount bound)**.** *Fix a regular neighbor u and $w \in N_{G^\tau}(u) \setminus \{v\}$ not initially a target blocker, so $X_w = Y_w \notin T$. Let $\mathcal{O}_w$ be the following disjoint set of physical constraint factors:*

$$\mathcal{O}_w := \{wy \in E(G^\tau) : y \neq u,\ X_y \in T \text{ or } Y_y \in T\} \sqcup \{(w,t,j) : t \in T,\ 1 \le j \le b^\tau_w(t)\},$$

and put $t_w = |\mathcal{O}_w|$. The contribution of the ordered incidence (u,w) to input discount loss minus certified output discount is at most $(1+P_2)\, s_u/(nq)$.

Proof. Let $\mathcal{E}_w$ be the event that all of $\mathcal{O}_w$ is inactive. Given the active outcome, for each target $c \in T$ the operations recoloring w into c flip the unique active $\{X_w, c\}$-component of w in one marginal, with transition probability at most $\max_j P_j/(nq) = 1/(nq)$, so their total probability is at most $2/(nq)$ always. When $\mathcal{E}_w$ fails, some constraint of $\mathcal{O}_w$ is active at w: an active edge to a c-colored vertex forces the $\{X_w, c\}$-component to have size at least two, capping its transition probability at $P_2/(nq)$, and an active pinned-neighbor constraint forbidding c makes the swap list-infeasible. Part 2 of Lemma 5.11 shows directly that root-containing moves are already charged to E_v and common off-root components are coupled synchronously, so the remaining probability is at most $(1+P_2)/(nq)$. This proves in the two-instance setting the case-(i) estimate corresponding to [CV25, §4.3]. The averaged probability is thus at most $\big((1+P_2)+(1-P_2)x^{t_w}\big)/(nq)$, and each new blocker incidence costs at most $s_u - s_u x = \theta s_u$ of discount.

The activation bits of $\mathcal{U}_u$ are disjoint from those of $\mathcal{E}_w$, so $\Pr(\mathcal{U}_u \cap \mathcal{E}_w) = s_u x^{t_w}$. On this event both target components at u — the $\{X_u, a\}$-move on one side and the $\{X_u, b\}$-move on the other — are feasible singletons; root-containing components meeting the active root edge have transition probability at most $P_2/(nq)$, so after the root/off-root entry each side retains probability at least $(1-P_2)/(nq)$, and the same-neighbor entry pairs the two moves: the coupled flip recolors u to a on one side and to b on the other, creating a new disagreement at u with pair $\{a,b\}$. Its blocker incidences at w are exactly the t_w constraints of $\mathcal{O}_w$ (no other vertex changes), so the (u,w) output discount in (17) is θx^{t_w}. The certified credit is at least $(1-P_2)\, s_u \theta x^{2t_w}/(nq)$. The net cost is at most

$$\frac{s_u\,\theta}{nq}\Big[(1+P_2)+(1-P_2)\big(x^{t_w}-x^{2t_w}\big)\Big] \le \frac{(1+P_2)\, s_u}{nq},$$

since for $t_w \ge 1$, $x^{t_w} - x^{2t_w} \le x$ and $(1-x)[(1+P_2)+(1-P_2)x] \le 1+P_2$, with slack $x[2P_2+(1-P_2)x] \ge 0$; for $t_w = 0$ the extra term vanishes. Distinct ordered incidences certify distinct summands of (17), so no credit is reused. □

Lemma 5.14 (Expected loss of the geometric discount)**.** *Fix the adjacent input pair and a deterministic order of the off-root neighbors of each regular u. For each coupled outcome, process in order the root recoloring, a recoloring of u into T, and then the off-root neighbors in the fixed order. Let Loss_u be the telescoping decrease of the input u-discount caused by these operations, charging each decrease at its first cause. Let Credit_u be the sum of the distinct (u,w)-indexed output-score summands certified by the same-neighbor matches in Lemmas 5.11 and 5.13. Thus every input loss is charged once and every output summand is credited at most once. Expectations are over both the coupled activation draw and the coupled hard update. For every regular neighbor u,*

$$\mathbb{E}[\mathsf{Loss}_u] - \mathbb{E}[\mathsf{Credit}_u] \ \le\ \frac{\beta_x^+}{nq}\, s_u,$$
$$\beta_x^+ := q - P_2\theta L_x + (1+P_2)\Delta + 2.$$

Proof. The discount at u can disappear in three ways: the root is recolored — probability bounded by (22); the root survives but u is recolored into a target color — by Lemma 5.6, common off-root

components are coupled synchronously, whereas every non-common component meeting the root is already charged to E_v; after those moves are removed, each target contributes probability at most $1/(nq)$, hence at most $2/(nq)$ in total; or a neighbor w creates a new blocker — at most Δ ordered incidences, each bounded by Lemma 5.13. The three types can overlap on a single operation, but each charge separately upper-bounds the part of the discount loss it names and the total loss telescopes over the triggered types, so summing the three expected charges is conservative. □

Lemma 5.15 (Disjointness of the output credits)**.** *For every coupled outcome, the sum of all output-score summands retained as credits in Lemmas 5.12 to 5.14 is at most* $\mathcal{S}_{\mathrm{out}}(X_1, Y_1)$.

Proof. The ordered-incidence credit associated with (u, w) is the summand of (16) indexed by the output disagreement at u and its endpoint-common neighbor w. The deterministic ordering in Lemma 5.14 retains each such ordered pair at most once. The fresh-color credit at a regular root neighbor u is instead the summand indexed by the surviving root disagreement (v, u). These summands have first coordinate v, whereas every ordered-incidence summand has first coordinate $u \neq v$; the two collections are therefore disjoint. If a single coupled flip realizes several credits, they are distinct summands in the defining double sum for $\mathcal{S}_{\mathrm{out}}$. Their total is consequently bounded by the full output score, outcome by outcome. □

5.5 Per-color charge estimates

Set $\varrho = q/\Delta$, $\eta(\varrho) = \frac{P_2 - P_3}{2\varrho} = \frac{17}{200\varrho}$, and

$$\gamma_{\mathrm{gain}} := \eta(\varrho)\Big(\varrho - 1 - \frac{2}{\Delta}\Big), \qquad \gamma_{\mathrm{loss}} := \eta(\varrho)\Big(\varrho - P_2\theta\phi_x + 1 + P_2 + \frac{2}{\Delta}\Big), \tag{23}$$

the per-Δ metric corrections supplied by Lemmas 5.12 and 5.14. On $\varrho \geq 1809/1000$, $\Delta \geq 125$, these coefficients lie in the half-open range

$$(\gamma_{\mathrm{gain}}, \gamma_{\mathrm{loss}}) \in \left[\frac{13481}{361800}, \frac{17}{200}\right) \times \left[0, \frac{799}{5400}\right], \qquad \frac{799}{5400} < P_2.$$

Indeed,

$$\gamma_{\mathrm{gain}} = \frac{17}{200}\left(1 - \frac{1 + 2/\Delta}{\varrho}\right), \qquad \gamma_{\mathrm{loss}} = \frac{17}{200}\left(1 + \frac{1 + P_2 + 2/\Delta - P_2\theta\phi_x}{\varrho}\right).$$

Over the displayed continuous parameter domain, the first expression has its minimum at $(\varrho, \Delta) = (1809/1000, 125)$ and approaches $17/200$ without reaching it; the second has its maximum at $(\varrho, \Delta, \theta\phi_x) = (1809/1000, 125, 0)$. For the finite maximization, close the open endpoint and use the closed bounding rectangle

$$\mathcal{R}_\gamma := \left[\frac{13481}{361800}, \frac{17}{200}\right] \times \left[0, \frac{799}{5400}\right]. \tag{24}$$

For any coefficient pair in this rectangle, set

$$\begin{aligned} &\Lambda_1 := 2 - P_2 + \gamma_{\mathrm{loss}}, && \Lambda_2 := 2 - P_3 - \gamma_{\mathrm{gain}}, \\ &\Lambda_3 := 2 - P_2 + 2P_3 - 2P_4, && \Lambda_{\mathrm{low}} := \max\{\Lambda_1, \Lambda_2, \Lambda_3\}. \end{aligned} \tag{25}$$

5.5.1 Finite component encoding

Definition 5.16 (Finite component records)**.** Fix a regular color c with $k_c \geq 1$, order $\Gamma_c = \{u_1, \ldots, u_{k_c}\}$. At u_i, record the size of the Y-side off-root $\{a, c\}$-component and the size of the X-side off-root $\{b, c\}$-component. A component meeting several u_i is recorded at its first listed incidence; its later appearances receive the marker 0, which represents no move and has $P_0 = 0$.

Replace every positive recorded size at least 7 by 7, and denote the resulting record sizes by a_i, b_i. These are truncated sizes, not necessarily the physical sizes. The truncation changes no transition mass: such an off-root component has mass zero, and a root-containing component containing it has size at least 8 and also has mass zero. Put

$$R_X = 1 + \sum_i a_i, \qquad R_Y = 1 + \sum_i b_i.$$

Let χ_i, υ_i be the available flags of the a- and b-records, respectively. Let

$$\xi_i := \mathbf{1}_{[\text{the } Y\text{-side } \{a,c\}\text{-flip recorded by } a_i \text{ is feasible}]}, \qquad \zeta_i := \mathbf{1}_{[\text{the } X\text{-side } \{b,c\}\text{-flip recorded by } b_i \text{ is feasible}]}.$$

For a positive record, the availability flags are explicitly

$$\chi_i = \mathbf{1}_{[a \in L_{X,u_i}]} = \mathbf{1}_{[a \in L_{Y,u_i}]}, \qquad \upsilon_i = \mathbf{1}_{[b \in L_{X,u_i}]} = \mathbf{1}_{[b \in L_{Y,u_i}]},$$

where the equalities use the common off-root lists. If $a_i = 0$, set $\chi_i = \xi_i = 0$; if $b_i = 0$, set $\upsilon_i = \zeta_i = 0$. Moreover,

$$\xi_i \leq \chi_i, \qquad \zeta_i \leq \upsilon_i, \qquad a_i = 1 \Longrightarrow \xi_i = \chi_i, \qquad b_i = 1 \Longrightarrow \zeta_i = \upsilon_i.$$

Indeed, a feasible swap must place its target color legally at the recording neighbor, while for a singleton this is its only post-swap list check. Thus every realized record satisfies these restrictions.

Let $w_X = \mathsf{A}_c \prod_{i:a_i>0} \xi_i$ and $w_Y = \mathsf{A}_c \prod_{i:b_i>0} \zeta_i$ be the root-component feasibility weights. Following Definition 5.8, choose

$$i_X \in \arg\max_{i:a_i>0} a_i, \qquad i_Y \in \arg\max_{i:b_i>0} b_i,$$

using the same index whenever a common maximizer exists. Among the permitted pairs, choose one that maximizes the right-hand side of (26) below. All move masses in this record are multiplied by nq; thus a feasible size-j move has scaled mass P_j. Define the *record upper charge* for this arbitrary multiplicity:

$$\mathcal{H}_c = w_X(R_X - a_{i_X} - 1)P_{R_X} + w_Y(R_Y - b_{i_Y} - 1)P_{R_Y} + \sum_i (a_i\alpha_i + b_i\beta_i - \min\{\alpha_i, \beta_i\}), \tag{26}$$

where

$$\alpha_i = \xi_i P_{a_i} - w_X \mathbf{1}_{[i=i_X]} P_{R_X}, \qquad \beta_i = \zeta_i P_{b_i} - w_Y \mathbf{1}_{[i=i_Y]} P_{R_Y}$$

are the remaining scaled off-root masses. The first two terms are the root/off-root matches; after their masses are subtracted, the sum is the residual match through the same u_i. Thus (26) is the conservative record used to bound the coupling above. It is defined for every $k_c \geq 1$ and both values of A_c. The cases $\mathsf{A}_c = 1$, $k_c \in \{1, 2\}$ are handled by Lemma 5.20. Finally, define

$$N_{11}(c) := |\{i : a_i = b_i = 1,\ \chi_i = \upsilon_i = 1\}|, \qquad N_{12}(c) := |\{i : \{a_i, b_i\} = \{1, 2\},\ \chi_i = \upsilon_i = 1\}|.$$

The first index set uses the ordered pair $(1, 1)$, whereas the second uses $(1, 2)$ or $(2, 1)$. They are therefore disjoint, and hence

$$N_{11}(c) + N_{12}(c) \leq k_c. \tag{27}$$

A repeated appearance is never counted by $N_{11}(c)$. Omitting a possible $N_{12}(c)$ credit at such an appearance is conservative because that term enters with a minus sign.

For a regular color c with $k_c = 0$, set

$$\mathcal{H}_c := -\mathsf{A}_c, \qquad N_{11}(c) := N_{12}(c) := 0.$$

Lemma 5.17 (Graph-to-record domination)**.** *For every regular color c, its contribution to the conditional expected Hamming increment, after multiplication by nq, is at most $\mathcal{H}_c$. When $k_c = 0$, this means that the contribution is at most $-\mathsf{A}_c$, which is the convention used below.*

Proof. Suppose first that $k_c \geq 1$. A root-containing component consists of the root and its disjoint off-root components. Matching it to the opposite component indexed by i_X, or by i_Y, therefore leaves at most $R_X - a_{i_X} - 1$, or $R_Y - b_{i_Y} - 1$, new disagreements per unit of matched mass. These give the first two terms of (26).

After those masses are subtracted, the two residual moves recorded at u_i have masses α_i, β_i and flipped-set sizes a_i, b_i. Their maximal match has mass $\min\{\alpha_i, \beta_i\}$. Both flipped sets contain u_i, so its Hamming increment is at most

$$a_i\alpha_i + b_i\beta_i - \min\{\alpha_i, \beta_i\}.$$

If the components share additional vertices, the actual increment is smaller; this is why $\mathcal{H}_c$ is an upper charge rather than an exact cost. Every remaining one-sided probability is passed to the product completion and bounded by its flipped-set size. Summing over the canonical indices gives (26); a repeated appearance has the zero marker and contributes no second copy.

If $k_c = 0$ and c is root-allowed, the two singleton root moves are coupled and remove the sole input disagreement, giving increment -1. If c is unavailable, there is no move in this family and its contribution is 0. These are exactly the two values of $-\mathsf{A}_c$. □

The geometric coefficients range over the rectangle (24), but the finite verification need not inspect a subdivision of that rectangle. The following elementary observation reduces it to one coefficient pair.

Lemma 5.18 (Balancing-point domination)**.** *For real numbers $g_{\mathrm{gain}}, g_{\mathrm{loss}}$, put*

$$\mathcal{L}(g_{\mathrm{gain}}, g_{\mathrm{loss}}) := \max\left\{2 - P_2 + g_{\mathrm{loss}},\ 2 - P_3 - g_{\mathrm{gain}},\ \Lambda_3\right\}$$

and, for a fixed component record of a regular color c, define its excess by

$$\mathcal{E}_c(g_{\mathrm{gain}}, g_{\mathrm{loss}}) := \mathcal{H}_c + g_{\mathrm{loss}}N_{11}(c) - g_{\mathrm{gain}}N_{12}(c) + 1 - k_c\mathcal{L}(g_{\mathrm{gain}}, g_{\mathrm{loss}}).$$

Then

$$\mathcal{E}_c(g_{\mathrm{gain}}, g_{\mathrm{loss}}) \leq \mathcal{E}_c\left(\frac{19}{500}, \frac{33}{250}\right).$$

At this balancing pair, all three terms in the maximum defining $\mathcal{L}$ equal $226/125$; the pair belongs to $\mathcal{R}_\gamma$ in (24).

Proof. Set

$$u_{\mathrm{loss}} := g_{\mathrm{loss}} - \frac{33}{250}, \qquad u_{\mathrm{gain}} := \frac{19}{500} - g_{\mathrm{gain}}, \qquad M := \max\left\{0, u_{\mathrm{loss}}, u_{\mathrm{gain}}\right\}.$$

The identities

$$2 - P_2 + \frac{33}{250} = 2 - P_3 - \frac{19}{500} = \Lambda_3 = \frac{226}{125}$$

give

$$\mathcal{L}(g_{\text{gain}}, g_{\text{loss}}) = \frac{226}{125} + M.$$

Consequently,

$$\begin{aligned}\mathcal{E}_c(g_{\text{gain}}, g_{\text{loss}}) - \mathcal{E}_c\left(\frac{19}{500}, \frac{33}{250}\right) &= u_{\text{loss}} N_{11}(c) + u_{\text{gain}} N_{12}(c) - k_c M \\ &\le M\big(N_{11}(c) + N_{12}(c) - k_c\big) \le 0,\end{aligned}$$

where the last inequality is (27). □

Two observations connect the finite expression to the geometric estimates.

Loss charge. The geometric charge γ_{loss} is levied only at the $N_{11}(c)$ locations. This covers the discount budget of Lemma 5.14: a location with a record of size at least two carries an *active* T-blocker, hence lies outside $\mathcal{U}_u$, the safe event defining s_u, and no discount is at risk there; and across the availability split, $\theta\big[x^{b_v^\tau(c)} x^{r_u+e_u} + (1 - x^{b_v^\tau(c)})\big] \ge \theta x^{r_u+e_u} = s_u$. The unavailable contribution appears separately as $(1 - \mathsf{A}_c)k_c$ below.

Gain credit. The $N_{12}(c)$ credit γ_{gain} is claimed only at locations with exactly one active blocker edge and both targets surviving — an event contained in the fresh-gain event of Lemma 5.12; distinct colors claim disjoint incidences, so the credited mass never exceeds the gain supplied.

For $c_0 \in T = \{a, b\}$, let $C_{c_0}(F_X, F_Y)$ denote nq times the charged Hamming increment, including the coalescence saving, to the $\{a, b\}$-component family assigned to target c_0 by Definition 5.8 and Lemma 5.10. Put

$$\mathcal{H}_T(F_X, F_Y) := C_a(F_X, F_Y) + C_b(F_X, F_Y).$$

Lemma 5.19 (Drift decomposition by color)**.** *For coupled active instances F_X, F_Y and each $c \notin T$, retain $\mathsf{A}_c = \mathbf{1}_{[c \text{ is root-allowed}]}$. Let $\mathcal{H}_c(F_X, F_Y)$ be the record upper charge (26) when $k_c \ge 1$, with root-containing masses set to zero when $\mathsf{A}_c = 0$; for $k_c = 0$ put $\mathcal{H}_c = -\mathsf{A}_c$. With $\mathcal{H}_T(F_X, F_Y)$ as defined above, put*

$$\mathcal{D} := \mathbb{E}[d_x^{G,\tau}(X_1, Y_1)] - d_x^{G,\tau}(X, Y).$$

All expectations in this statement are over the coupled activation draw and the coupled hard transition from the fixed adjacent input pair. Then the one-step metric drift satisfies

$$nq\,\mathcal{D} \le \mathbb{E}\left[\mathcal{H}_T(F_X, F_Y) + \sum_{c \notin T}\Big\{\mathcal{H}_c + \gamma_{\text{loss}}(\mathsf{A}_c N_{11}(c) + (1 - \mathsf{A}_c)k_c) - \gamma_{\text{gain}}\mathsf{A}_c N_{12}(c)\Big\}\right]. \tag{28}$$

Here $N_{11}(c) = N_{12}(c) = 0$ when $k_c = 0$.

Proof. Put $\omega = 17/(200q)$. Lemmas 5.7, 5.9 and 5.10 partition, complete, and bound the cost of all aggregate moves, while Lemma 5.17 bounds the actual regular-color Hamming contribution by its record upper charge. Write

$$S_{\text{in}} := \sum_{u \text{ regular}} s_u, \qquad S_{\text{out}} := \mathcal{S}_{\text{out}}(X_1, Y_1).$$

Thus Lemma 5.5 first gives

$$nq\,\mathcal{D} \le \mathbb{E}\left[\mathcal{H}_T(F_X, F_Y) + \sum_{c \notin T} \mathcal{H}_c\right] + nq\,\omega\big(S_{\text{in}} - \mathbb{E}[S_{\text{out}}]\big). \tag{29}$$

Fix the output-discount allocation as follows. The root, target-recoloring, and ordered-incidence terms used in Lemma 5.14 are allocated first; they do not use the fresh increments of Lemma 5.12. Allocate those fresh increments next, and discard any remaining output discount. This is legitimate outcome by outcome by Lemma 5.15. With $b_u^{\mathrm{SB}} = \Pr(\mathsf{SB}_u)$ as defined above, the two lemmas then give

$$S_{\mathrm{in}} - \mathbb{E}[S_{\mathrm{out}}] \le \frac{\beta_x^+}{nq} \sum_{u \text{ regular}} s_u - \frac{\alpha_{\mathrm{fr}}}{nq} \sum_{u \text{ regular}} b_u^{\mathrm{SB}}.$$

By (23),

$$\omega\beta_x^+ = \gamma_{\mathrm{loss}}, \qquad \omega\alpha_{\mathrm{fr}} = \gamma_{\mathrm{gain}}.$$

It remains to turn these two averaged discount bounds into the pointwise color terms in (28). For a physical regular neighbor u of color c, let ι_u indicate that vu is active, and let $N_{c,u}^j$ be one exactly when u is the first recorded incidence of its component and is counted by $N_{1j}(c)$, and zero otherwise, for $j = 1, 2$. Thus $N_{c,u}^1$ contributes to $N_{11}(c)$ and $N_{c,u}^2$ contributes to $N_{12}(c)$. Root-list survival is independent of all root-edge and blocker bits, and

$$\mathbb{E}\Big[\mathsf{A}_c N_{c,u}^1 + (1 - \mathsf{A}_c)\iota_u\Big] = x^{b_v^\tau(c)} s_u + (1 - x^{b_v^\tau(c)})\theta \ge s_u.$$

Moreover, $\mathsf{A}_c N_{c,u}^2 = 1$ entails exactly one active target blocker at u and survival of both targets, so $\mathbb{E}[\mathsf{A}_c N_{c,u}^2] \le b_u^{\mathrm{SB}}$. Summing over neighbors, using $k_c = \sum_{u:X_u=Y_u=c} \iota_u$, and applying the two coefficient identities proves (28).

There is no duplicated charge or credit: Lemma 5.7 partitions the hard moves; A_c and $1 - \mathsf{A}_c$ are disjoint; $N_{11}(c)$ and $N_{12}(c)$ are disjoint; and each $N_{12}(c)$ credit is attached to its own blocker incidence. If one flip improves several such incidences, Lemma 5.12 counts the corresponding sum of distinct metric-discount increments. □

5.5.2 Small multiplicities: exact finite verification

Lemma 5.20 (Finite low-multiplicity estimate)**.** *For every regular color c with $\mathsf{A}_c = 1$ and $k_c \in \{1, 2\}$, every component state encoded above and every $(\gamma_{\mathrm{gain}}, \gamma_{\mathrm{loss}})$ in the rectangle (24) satisfy*

$$\mathcal{H}_c + \gamma_{\mathrm{loss}} N_{11}(c) - \gamma_{\mathrm{gain}} N_{12}(c) \le -1 + k_c \Lambda_{\mathrm{low}}. \tag{30}$$

The maximum excess of the left-hand side over the right-hand side of (30), over the stated records and coefficient pairs, is 0, attained at the balancing pair.

Computer-assisted proof. By Lemma 5.18, it is enough to check the single coefficient pair

$$(\gamma_{\mathrm{gain}}, \gamma_{\mathrm{loss}}) = \left(\frac{19}{500}, \frac{33}{250}\right), \qquad \Lambda_{\mathrm{low}} = \frac{226}{125}.$$

Exact rational enumeration gives maximum excess zero over all 400 structural states and 400 permitted maximizer choices for $k_c = 1$, and all 193600 structural states and 196964 permitted maximizer choices for $k_c = 2$. These include every admissible availability and feasibility pattern, repeated-component marker, truncated size, and choice among tied maximum components. For example, equality at the balancing pair is attained by the one-index record $a_1 = 2$, $b_1 = 1$ with all four availability and feasibility flags equal to 1: then $\mathcal{H}_c = 423/500$, $N_{11}(c) = 0$, and $N_{12}(c) = 1$.

Every component configuration maps to one of these records: order its listed neighbor incidences as in Definition 5.16, write 0 after the first listed incidence of a repeated component, truncate every size at 7, and retain its actual availability and feasibility flags. The permitted records use one state

at size 0, two at size 1, and three at each size $2, \dots, 7$, exactly as justified in that definition. The enumeration also allows some records that no graph realizes; this only enlarges the maximum and is therefore conservative.

The companion Lean development proves this finite estimate and the encoding of the actual component configurations in `CI2ZF/Coupling/CV/Certificate.lean` and its dependent modules; the finite arithmetic is checked by Lean's kernel without importing Python output [Shi26]. As an independent check, the Python verifier `anc/python/soft_cv_certificate_verifier.py` checks the larger eleven-pair test specified in `anc/CERTIFICATE_SPEC.md`. This includes the balancing pair above, every permitted maximum-component tie, and an equality witness. The Python source and frozen certificate are covered by `anc/SHA256SUMS`. For reproducibility, the exhaustive checks are run from the repository root with

```
python3 anc/python/soft_cv_certificate_verifier.py --check
```

and the reported state counts are part of the certificate specification. □

Lemma 5.21 (High multiplicity and missing colors)**.** *Set*

$$\Lambda_{\mathrm{bulk}} := \tfrac{4}{3} + P_2 + \tfrac{4}{3}P_4 = \tfrac{1331}{750}.$$

For a root-allowed regular color c with $k_c \geq 3$ active neighbors,

$$\mathcal{H}_c + \gamma_{\mathrm{loss}} N_{11}(c) \leq -1 + \Lambda_{\mathrm{bulk}} k_c.$$

For a regular color c absent from the active root list,

$$\mathcal{H}_c + \gamma_{\mathrm{loss}} k_c \leq \frac{39743}{27000} k_c \leq \Lambda_{\mathrm{bulk}} k_c.$$

The second inequality is strict when $k_c > 0$.

Proof. The root-containing components have size at least $k_c + 1 \geq 4$, so in (26) each root/off-root term is at most $\max_{r \geq 4}(r-2)P_r = 2P_4$, giving $4P_4$ for the two sides. For the contribution at one regular neighbor, $\max_{1 \leq r,s \leq 7}(rP_r + sP_s - \min\{P_r, P_s\}) = 1 + P_2$. Indeed, $\max_j jP_j = 1$ and $\max_j (j-1)P_j = P_2$; subtracting the smaller of P_r, P_s leaves one term of each type. Equality occurs at $(r,s) = (2,1)$. A location receives the geometric charge $\gamma_{\mathrm{loss}} \leq P_2$ only when both records are singletons, in which case its component term is at most 1. Either way the location contributes at most $1 + P_2$. Hence the total is at most $4P_4 + k_c(1+P_2)$, and

$$4P_4 + k_c(1+P_2) \leq -1 + \Lambda_{\mathrm{bulk}} k_c,$$

with slack $(k_c - 3)(1 + 4P_4)/3 \geq 0$; the same bound in the aggregate-parameter hard setting is [CV25, Lemma 4.6]. If the color is missing from the active root list there are no root-containing moves, and the exact per-neighbor maximum is $1 + P_2$ by the same calculation, matching [CV25, Lemma 4.7]; adding the largest geometric correction $\gamma_{\mathrm{loss}} \leq 799/5400$ gives $39743/27000 < \Lambda_{\mathrm{bulk}}$. □

Lemma 5.22 (The two root colors)**.** *Fix $c_0 \in \{a, b\}$. For the two root colors, define*

$$k_a := \big|\{u \in N_{G^\tau}(v) : X_u = Y_u = a,\ vu \in E_Y\}\big|,$$
$$k_b := \big|\{u \in N_{G^\tau}(v) : X_u = Y_u = b,\ vu \in E_X\}\big|.$$

Thus each multiplicity is counted on the side where the corresponding root-containing move starts. The cost $C_{c_0}(F_X, F_Y)$ defined above for the corresponding $\{a,b\}$-component moves satisfies

$$C_{c_0}(F_X, F_Y) \leq -\mathbf{1}_{[c_0 \text{ is root-allowed}]} + \Lambda_{\mathrm{bulk}} k_{c_0}.$$

Proof. If c_0 is unavailable there is no root-containing move and the off-root components meet disjoint sets of the k_{c_0} neighbors. After the common denominator nq is cleared, each contributes at most one, so $C_{c_0}(F_X, F_Y) \le k_{c_0} \le \Lambda_{\text{bulk}} k_{c_0}$. Let c_0 be available. If $k_{c_0} = 0$, the singleton root move paired with holding probability contributes -1. If $k_{c_0} \ge 2$, an infeasible root move leaves cost at most $k_{c_0} \le \Lambda_{\text{bulk}} k_{c_0} - 1$; a feasible one paired with holding probability has total cost at most

$$k_{c_0} + \max_j (j-2)P_j = k_{c_0} + \tfrac{22}{125},$$

including the off-root component moves, and $2\Lambda_{\text{bulk}} - 1 - (2 + \frac{22}{125}) = \frac{28}{75} > 0$ settles the weakest case $k_{c_0} = 2$. If $k_{c_0} = 1$, let the unique component left after deleting v from the root-containing component have size s. By the block classification in Lemma 5.7, on the opposite side it either avoids v and is the unique off-root partner in this family, or it also reaches v and belongs to the other root-color move, leaving no off-root copy here. In the first case, because the target color is root-allowed, the root-containing flip is feasible exactly when its sole off-root component is feasible. Thus an infeasible root move contributes 0. A feasible root move with no feasible off-root partner is paired with holding probability at cost $\max_s (s-1)P_{s+1} = \frac{22}{125}$; and a feasible pair costs $sP_s - (s+1)P_{s+1} \le \frac{44}{125}$. All are below $\Lambda_{\text{bulk}} - 1$, with slack at least $\frac{317}{750}$. □

5.6 Averaging and degree bounds

5.6.1 Averaging over active multiplicities

A physical regular root color c of multiplicity $m_c \ge 3$ has random active multiplicity $K_c \sim \text{Bin}(m_c, \theta)$. Thus K_c denotes the random multiplicity before the activations are exposed, whereas $k_c = |\Gamma_c|$ denotes its value for a fixed active outcome. In the root-allowed case, the corrected per-color contribution for $1 \le K_c \le 2$ is bounded by (30), whose positive coefficient is at most

$$\overline{\Lambda}_{\text{low}} := 2 - P_2 + \tfrac{799}{5400} = \tfrac{49247}{27000};$$

for $K_c \ge 3$ use Λ_{bulk} from Lemma 5.21 and discard the negative credit. In the root-missing case the rate is $39743/27000 < \Lambda_{\text{bulk}}$ for every K_c. After the allowed case's -1 baseline is separated, the expected positive charge per *physical* neighbor is therefore at most

$$\Lambda_{\text{bulk}}\theta + (\overline{\Lambda}_{\text{low}} - \Lambda_{\text{bulk}})\theta x^{m_c-2}(1 + (m_c-2)\theta) \le \Lambda_{\text{bulk}},$$

since $\theta x^{m_c-3}(1+(m_c-2)\theta) \le 2$ and $3\Lambda_{\text{bulk}} - 2\overline{\Lambda}_{\text{low}} = 22627/13500 > 0$. Thus both availability cases cost at most Λ_{bulk} per physical neighbor after the baseline is separated. The baseline is accounted for in Lemma 5.23.

5.6.2 Degree budget

Let

$$L := \sum_{\substack{c \notin T \\ m_c \in \{1,2\}}} m_c, \qquad H := \deg_{G^\tau}(v) - L, \qquad B_v^\partial := \sum_{c \in [q]} b_v^\tau(c).$$

Then $L + H + B_v^\partial \le \Delta$ and, because $L_x \le L$,

$$(L - L_x) + H + B_v^\partial \le \Delta - L_x. \tag{31}$$

Every distinct color missing from the active root list is witnessed by an active pinned-neighbor constraint forbidding that color. Choosing one witness for each missing color is injective, so their number is at most the number of active root pinned-neighbor constraints, and hence at most B_v^∂.

Lemma 5.23 (The drift envelope)**.** *Let $\mathcal{D}$ be the expected one-step drift of $d_x^{G,\tau}$ under the coupling above. Then*

$$nq\,\mathcal{D} \;\le\; -q + \theta\Lambda_{\text{low}} L_x + \Lambda_{\text{bulk}}(\Delta - L_x). \tag{32}$$

Proof. Apply the drift decomposition (28). Its regular-color summand is exactly the expression controlled by Lemmas 5.20 and 5.21; the root-color summand has no geometric correction because the input metric discounts only regular neighbors.

Condition on the root pinned-neighbor constraint coins; they determine the active root list and are independent of all root-edge and blocker bits. Every color c contributes a baseline: an available color contributes -1 — through Lemma 5.10 when it has no active neighbor, through (30) when $1 \le k_c \le 2$, through Lemma 5.21 when $k_c \ge 3$, and through Lemma 5.22 for the two target colors — while a missing color contributes 0. The witness injection above charges each missing baseline to a distinct root pinned-neighbor constraint.

For each allowed low-multiplicity color c, root availability is independent of the root-edge bits, so $\mathbb{E}[k_c \mathbf{1}_{[\text{allowed}]}] = \theta m_c x^{b_v^\tau(c)}$, and each active neighbor carries at most Λ_{low} by (30). Their total is at most $\theta\Lambda_{\text{low}} L_x$. Neighbors of *unavailable* low colors have expected count

$$\theta \sum_{\substack{c \notin T \\ m_c \in \{1,2\}}} m_c(1 - x^{b_v^\tau(c)}) \le L - L_x$$

and rate at most $39743/27000 < \Lambda_{\text{bulk}}$ each. Physical high-multiplicity regular neighbors are charged at most Λ_{bulk} each by the binomial averaging above, in both availability cases; root-color neighbors at most Λ_{bulk} each by Lemma 5.22. Finally, missing baselines cost at most one per pinned-neighbor constraint, and $1 < \Lambda_{\text{bulk}}$. Hence

$$nq\,\mathcal{D} \le -q + \theta\Lambda_{\text{low}} L_x + \Lambda_{\text{bulk}}\big[(L - L_x) + H + B_v^\partial\big],$$

and (31) proves (32). □

5.7 Numerical bound and contraction

Lemma 5.24 (Final numerical inequality)**.** *For $\varrho \ge 1809/1000$, $\Delta \ge 125$, and $\theta, \phi \in [0,1]$, interpret $\phi_x = \phi$ in (23). Then*

$$-\varrho + \Lambda_{\text{bulk}}(1-\phi) + \theta\phi \max\{\Lambda_1, \Lambda_2, \Lambda_3\} \;\le\; -\frac{59}{226125},$$

where the rates in (25) are evaluated at the actual coefficients (23); in particular $\Lambda_3 = 226/125$.

Proof. Write $\eta_0 = 17/200$, $y = \theta\phi$, and $C_\Delta = 1 + P_2 + 2/\Delta$. For $j = 1, 2, 3$, set

$$F_j := -\varrho + \Lambda_{\text{bulk}}(1-\phi) + y\Lambda_j.$$

For the first case this expression is

$$F_1 = -\varrho + \Lambda_{\text{bulk}}(1-\phi) + y(2 - P_2 + \eta_0) + \frac{\eta_0}{\varrho} y(C_\Delta - P_2 y).$$

For fixed y we have $y \le \phi$, so $\phi = y$ maximizes F_1. The resulting function decreases in ϱ and as Δ grows. At $(\varrho, \Delta) = (1809/1000, 125)$ its derivative in y is decreasing in y and at $y = 1$ equals $\frac{34097}{1809000} > 0$. Thus the first case is maximized at that corner and $y = \phi = \theta = 1$.

For the second case,

$$\Lambda_2 = 2 - P_3 - \eta_0 + \frac{\eta_0}{\varrho}\left(1 + \frac{2}{\Delta}\right) > 0.$$

First maximize in θ, and then in the affine variable ϕ, to obtain

$$F_2 \le -\varrho + \max\{\Lambda_{\text{bulk}}, \Lambda_2\}.$$

Both $-\varrho + \Lambda_{\text{bulk}}$ and $-\varrho + \Lambda_2$ decrease in ϱ, and the latter also decreases as Δ grows. Hence their maximum occurs at $(\varrho, \Delta) = (1809/1000, 125)$; there $\Lambda_2 - \Lambda_{\text{bulk}} = \frac{61637}{1809000} > 0$, so $\phi = \theta = 1$ is the maximizer at the worst corner. Finally $\Lambda_3 = 226/125 > \Lambda_{\text{bulk}}$, and the same affine maximization gives $F_3 \le -\varrho + \Lambda_3$.

Exact evaluation gives the three gaps

$$\frac{59}{226125}, \qquad \frac{59}{226125}, \qquad \frac{1}{1000},$$

whose minimum is δ_{CV}. The `scalar_closure` certificate in `anc/python/soft_cv_certificate_verifier.py` verifies all three corner values. □

Theorem 5.25 (Geometric soft-CV contraction)**.** *Let q, Δ be positive integers with $\Delta \ge 125$ and $q \ge 1.809\,\Delta$. For every $G \in \mathcal{G}_\Delta$ and pinning τ with $n = |V^\tau| \ge 1$, every $x \in (0,1]$, and every adjacent pair $X, Y \in [q]^{V^\tau}$, there is a coupling (X_1, Y_1) of $K^{G,\tau}_{x,\text{CV}}(X)$ and $K^{G,\tau}_{x,\text{CV}}(Y)$ such that*

$$nq\Big(\mathbb{E}[d_x^{G,\tau}(X_1, Y_1)] - d_x^{G,\tau}(X, Y)\Big) \le -\delta_{\text{CV}}\Delta. \tag{33}$$

Proof. Use the common-coin coupling of Lemma 3.5 and the conditional component coupling of Definition 5.8 and Lemma 5.9. The drift envelope (32), which combines the component charges and geometric-discount estimates, applies to this coupling. Lemma 5.24 bounds its right-hand side by $-\delta_{\text{CV}}\Delta$. At $x = 1$ the metric is Hamming and the active graph is empty. The endpoint $x = 0$ is handled below by continuity of the normalized child laws, not by assigning values to 0^0 in the active representation. □

5.8 From contraction to coupling independence

It remains to compare each child kernel with the common middle kernel and apply stationary comparison in the child metric.

Lemma 5.26 (Soft-CV child–middle row discrepancy)**.** *Let $G \in \mathcal{G}_\Delta$, let τ be a pinning, let $r \in V^\tau$, let $c \in [q]$, and fix $x \in (0,1]$. Define the middle kernel by*

$$K_{\text{mid},x} := K^{G-r,\tau}_{x,\text{CV}}.$$

Its free graph is $G^\tau - r$, with the original boundary counts restricted to $V^\tau \setminus \{r\}$. Thus r and all its incident interactions are deleted without adding the pinned-neighbor constraints created by pinning $r = c$. If the common vertex set $V^\tau \setminus \{r\}$ has $n_0 \ge 1$ vertices, then for every configuration $\sigma \in [q]^{V^\tau \setminus \{r\}}$,

$$W_{1,\text{Ham}}\big(K^{G,\tau^{r=c}}_{x,\text{CV}}(\sigma), K_{\text{mid},x}(\sigma)\big) \le \frac{(1-x)\Delta}{n_0 q} \le \frac{\Delta}{n_0 q}.$$

If $q \ge 1.809\,\Delta$, the same bound holds in the child metric $d_x^{G,\tau^{r=c}} \le \text{Ham}$.

Proof. The child differs from the middle instance by at most Δ pinned-neighbor constraints. Apply the common one-boundary estimate Lemma 3.6 with $\rho_j = P_j$. The CV parameters satisfy $\sup_j jP_j = 1$, so the Hamming bound is $(1-x)\Delta/(n_0 q) \le \Delta/(n_0 q)$. Under the additional parameter assumption, Lemma 5.4 gives $d_x^{G,\tau^{r=c}} \le \mathrm{Ham}$, so the same coupling gives the metric bound. □

Proof of Theorem 5.1. Fix $G \in \mathcal{G}_\Delta$, a pinning τ, a root $r \in V^\tau$, and colors a, b. If $a = b$, the laws coincide. Let $n_0 = |V^\tau \setminus \{r\}|$. If $n_0 = 0$, both children are the empty point mass, so assume $n_0 \ge 1$ and first take $x > 0$.

Let $K_{\mathrm{mid}} := K_{x,\mathrm{CV}}^{G-r,\tau}$, whose stationary law is $\mu_{\mathrm{mid}} := \mu_{G-r,x}^{\tau}$. For $c \in \{a, b\}$, put

$$\mu_c := \mu_{G,x}^{\tau^{r=c}}.$$

Fix such a c, and write $d_c := d_x^{G,\tau^{r=c}}$ for the child metric. Then Theorem 5.25 and Lemma 5.4 gives, for adjacent X, Y,

$$\begin{aligned} W_{1,d_c}\left(K_{x,\mathrm{CV}}^{G,\tau^{r=c}}(X), K_{x,\mathrm{CV}}^{G,\tau^{r=c}}(Y)\right) &\le d_c(X,Y) - \frac{\delta_{\mathrm{CV}}\Delta}{n_0 q} \\ &\le \left(1 - \frac{\delta_{\mathrm{CV}}\Delta}{n_0 q}\right) d_c(X,Y). \end{aligned}$$

By Lemma 5.26, the child–middle row discrepancy is at most $(1-x)\Delta/(n_0 q)$ in this metric. Stationary comparison with contraction gap $\delta_{\mathrm{CV}}\Delta/(n_0 q)$ therefore yields

$$W_{1,d_c}(\mu_c, \mu_{\mathrm{mid}}) \le \frac{1-x}{\delta_{\mathrm{CV}}}.$$

Since $d_c \ge m_0 \,\mathrm{Ham}$,

$$W_{1,\mathrm{Ham}}(\mu_c, \mu_{\mathrm{mid}}) \le \frac{1-x}{m_0 \delta_{\mathrm{CV}}}.$$

Applying this for $c = a, b$ and using the triangle inequality through μ_{mid} gives

$$W_{1,\mathrm{Ham}}(\mu_a, \mu_b) \le \frac{2(1-x)}{m_0 \delta_{\mathrm{CV}}} \le \frac{2}{m_0 \delta_{\mathrm{CV}}}.$$

At $x = 0$, hard feasibility (Lemma 2.2, $q \ge \Delta + 1$) makes both hard child partition sums positive, and Lemma 3.7 passes the uniform bound to their normalized hard limits. This proves (12) also at $x = 0$. □

6 Coupling independence for $q \ge \Delta + 3$ at large girth

Theorem 6.1 (Large-girth coupling independence)**.** *Let q, Δ be integers with $\Delta \ge 3$ and $q \ge \Delta + 3$. There exist $g_* = g_*(q,\Delta) \ge 3$ and $C_* = C_*(q,\Delta) < \infty$ such that, for every $G \in \mathcal{G}_\Delta$ and every pinning τ with $\mathrm{girth}(G^\tau) \ge g_*$, the bound below holds for every $x \in [0,1]$, every $r \in V^\tau$, and all $a, b \in [q]$:*

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}\right) \le C_*.$$

The two laws live on the common free set $V^\tau \setminus \{r\}$. At $x = 0$ they are well defined by Lemma 2.2.

At $x = 0$, the estimate is already contained in [CLMM23, Theorem 5.10 and Lemma 5.13]; it is also recovered below by continuity. To obtain the interval $[0, 1]$, we extend the same tree argument to positive activities. For $0 < x < 1$, the soft recursion is a weighted-color recursion whose scaled cavity messages are subdistributions. Four units of palette slack below the root give a uniform contraction. The contraction implies tree total-influence decay (TID) and relative strong spatial mixing (SSM) after a fixed depth; the root factors are handled by two exact cancellations. The large-girth transfer of [CLMM23] then applies, and finite-state continuity supplies the value at $x = 0$.

6.1 Weighted tree recursion

We first take $0 < x < 1$ and put $s := 1 - x$. Root the free component of a tree of maximum degree at most Δ and condition on a fixed pinning, treating every pinned neighbor as a boundary constraint. At a free vertex v, let d_v be the number of free children and k_v the number of pinned neighbors. As in Section 2, $b_v^\tau(c)$ counts those of color c. Thus

$$d_v + k_v \le \Delta - 1$$

away from the root, while $d_v + k_v \le \Delta$ at the root. Define

$$a_v(c) = x^{b_v^\tau(c)}, \qquad A_v = \sum_{c=1}^{q} a_v(c), \qquad \underline{A}_v = q - sk_v. \tag{34}$$

Bernoulli's inequality gives

$$A_v \ge \underline{A}_v. \tag{35}$$

If p_i is the cavity marginal of free child i, put $m_i = sp_i$. The parent recursion is

$$p_v(c) = \frac{a_v(c) \prod_{i=1}^{d_v} (1 - m_i(c))}{\sum_{b=1}^{q} a_v(b) \prod_{i=1}^{d_v} (1 - m_i(b))}, \qquad m_v = sp_v. \tag{36}$$

Every m_i is a subdistribution, meaning a nonnegative vector of total mass at most one; here its total mass is exactly s. More importantly, for every non-root vertex,

$$\underline{A}_v - d_v = q - sk_v - d_v \ge q - k_v - d_v \ge 4. \tag{37}$$

At the root the same calculation leaves three units of slack. As in the hard proof of [CLMM23], contraction is required only below the root; the first step is retained as a bounded, possibly noncontracting map.

To integrate the Jacobian between two finite messages, we need a convex domain that contains every genuine non-root message and every interpolation point used in potential coordinates. Put

$$\mathcal{D}_s := \left\{ m \in [0, 1/4]^q : \sum_{c \in [q]} m(c) \le s \right\}. \tag{38}$$

This is a convex down-set. The pointwise bound below shows that every genuine non-root scaled message belongs to $\mathcal{D}_s$ and has total mass exactly s; the extra points in (38) occur only along a potential-coordinate interpolation.

We use the elementary product inequality

$$0 < B < 1, \quad 0 \le y_j \le B, \quad \sum_j y_j \le 1 \quad \Longrightarrow \quad \prod_j (1 - y_j) \ge (1 - B)^{1/B}. \tag{39}$$

Indeed, concavity of $\log(1-y)$ on the chord from 0 to B gives $\log(1-y) \geq y \log(1-B)/B$. Since the coefficient is negative,

$$\sum_j \log(1-y_j) \geq \frac{\log(1-B)}{B} \sum_j y_j \geq \frac{\log(1-B)}{B},$$

which proves the claim after exponentiation.

At a non-root vertex, inputs $m_i \in \mathcal{D}_s$ already give the crude invariant bound

$$\begin{aligned} Z_v &:= \sum_b a_v(b) \prod_i (1 - m_i(b)) \\ &\geq A_v - \sum_i \sum_b a_v(b) m_i(b) \geq A_v - d_v \geq \underline{A}_v - d_v \geq 4. \end{aligned} \tag{40}$$

Consequently $p_v(c) \leq 1/4$ and $m_v = sp_v$ belongs to $\mathcal{D}_s$. This proves invariance by induction from the leaves; the sharper estimate below will also provide the coordinate bounds used in the weighted norm.

6.2 Uniform contraction below the root

We choose

$$\phi(t) = 2 \operatorname{arctanh} \sqrt{t}, \qquad \phi'(t) = \frac{1}{\sqrt{t}(1-t)} \tag{41}$$

because its derivative factorizes the transformed Jacobian into one factor depending on the outgoing marginal and one measuring the incoming occupancy of a single color. In the original probability coordinate the same potential is $\psi_s(p) = \phi(sp)$. The map ϕ is increasing and concave on $[0, 1/4]$; write

$$\boldsymbol{\phi}(m) := (\phi(m(c)))_{c \in [q]}.$$

For the local map $(m_1, \ldots, m_{d_v}) \mapsto p_v$, define

$$J^p_{v,i} := D\boldsymbol{\phi}(p_v) \, \frac{\partial p_v}{\partial m_i} \, \big(D\boldsymbol{\phi}(m_i)\big)^{-1}, \qquad J^p_v := (J^p_{v,1} \mid \cdots \mid J^p_{v,d_v}).$$

The superscript p indicates that this auxiliary Jacobian has the unscaled output p_v and the scaled inputs m_i. Let $\widehat{J}_{v,i}$ and $\widehat{J}_v = (\widehat{J}_{v,1} \mid \cdots \mid \widehat{J}_{v,d_v})$ denote the analogous blocks with the scaled output $\boldsymbol{\phi}(m_v)$. At a zero coordinate these matrices mean their continuous extensions from positive coordinates.

For a block Jacobian $J = (J_1, \ldots, J_d)$, a positive parent weight w, and positive child weights w_i, put

$$\|J\|_{\mathbf{w}} = \sup_{(h_i) \neq 0} \frac{w \, \|\sum_i J_i h_i\|_2}{\max_i w_i \, \|h_i\|_2}. \tag{42}$$

For $d = 0$ this empty block norm is zero. Below, $\|J^p_v\|_{2\to 2}$ denotes the spectral norm of the horizontally concatenated q-by-$d_v q$ block Jacobian.

Lemma 6.2 (Exact Jacobian factorization)**.** *Write* $\Sigma_v(c) = \sum_i m_i(c)$ *and*

$$F_{1,v} = \max_{c \in [q]} \big(1 - p_v(c)\big)^{-2}, \qquad F_{2,v} = \max_{c \in [q]} p_v(c) \Sigma_v(c).$$

Then $\|J^p_v\|^2_{2\to 2} \leq F_{1,v} F_{2,v}$.

Proof. Write $p = p_v$ and $\gamma(c) = \sqrt{p(c)}/(1-p(c))$. Differentiating the ϕ-transformed recursion gives

$$(J^p_{v,i})_{c,b} = \gamma(c)\big(p(b) - \mathbf{1}_{[b=c]}\big)\sqrt{m_i(b)}.$$

Hence, for a unit vector h,

$$\|J^p_v\|^2_{2\to 2} = \max_{\|h\|_2=1} \sum_c \Sigma_v(c)\big(\gamma(c)h(c) - p(c)\langle\gamma, h\rangle\big)^2.$$

Set $u(c) = \gamma(c)h(c)$ and $U_1 = \sum_c u(c) = \langle\gamma, h\rangle$. The constraint $\|h\|_2 = 1$ becomes

$$\sum_c \frac{u(c)^2(1-p(c))^2}{p(c)} = 1.$$

Since $p(c)\Sigma_v(c) \le F_{2,v}$, the identity

$$\sum_c \frac{(u(c) - p(c)U_1)^2}{p(c)} = \sum_c \frac{u(c)^2}{p(c)} - U_1^2$$

gives

$$\begin{aligned}\sum_c \Sigma_v(c)(u(c) - p(c)U_1)^2 &\le F_{2,v} \sum_c \frac{(u(c) - p(c)U_1)^2}{p(c)} \\ &\le F_{2,v} \sum_c \frac{u(c)^2}{p(c)} \le F_{1,v}F_{2,v}.\end{aligned}$$

□

We next bound the two factors. The quantity B_v controls each coordinate of p_v, while ζ_v records the loss in the occupancy estimate.

Lemma 6.3 (Pointwise marginal bound)**.** *Consider* (36) *at a non-root vertex v with $m_i \in \mathcal{D}_s$ for every free child. With $d = d_v$ and $\underline{A} = \underline{A}_v$, define*

$$\xi_v = \left(\frac{4}{3}\right)^{4d/(\underline{A}-1)}, \qquad B_v = \frac{\xi_v}{\underline{A} - 1 + \xi_v}. \tag{43}$$

Then

$$\frac{p_v(c)}{1-p_v(c)} \le \frac{\xi_v}{\underline{A}-1}, \qquad p_v(c) \le B_v \le \frac{1}{4}, \qquad m_v(c) \le \frac{1}{4} \tag{44}$$

for every color c. In particular, $m_v \in \mathcal{D}_s$.

Proof. The crude estimate (40) already gives $p_v(c) \le 1/4$. For the sharper recursive estimate, the claim is immediate if $a_v(c) = 0$. Otherwise set $S = A_v - a_v(c) \ge \underline{A} - 1$. Weighted AM–GM and $a_v(b) \le 1$ give

$$\begin{aligned}\sum_{b\neq c} a_v(b) \prod_i (1 - m_i(b)) &\ge S \prod_{b\neq c} \left(\prod_i (1 - m_i(b))\right)^{a_v(b)/S} \\ &\ge S \prod_i \left(\prod_{b\neq c} (1 - m_i(b))\right)^{1/S}.\end{aligned}$$

By (39) with $B = 1/4$, the inner product is at least $(3/4)^4$. The function $S \mapsto S(3/4)^{4d/S}$ is increasing, so

$$\frac{p_v(c)}{1-p_v(c)} \le \frac{1}{S(3/4)^{4d/S}} \le \frac{\xi_v}{\underline{A}-1}.$$

Finally $\underline{A} \ge d+4$ implies $\xi_v/(\underline{A}-1) \le 1/3$. For $d=0$ this is immediate; for $d \ge 1$ it reduces to $(4/3)^{4d/(d+3)} \le (d+3)/3$, whose logarithmic derivative is nonnegative after equality at $d=1$. This proves $B_v \le 1/4$ and hence all of (44). The induction starts at a free leaf, where the same argument has $d=0$. □

For every non-root vertex y, define its local ξ_y, B_y by (43), with v replaced by y, and put

$$\zeta_y = \left(\frac{\underline{A}_y - 1}{\xi_y} + 1\right) \log\left(1 + \frac{\xi_y}{\underline{A}_y - 1}\right) - 1. \tag{45}$$

For a child y_i indexed by i, abbreviate the corresponding quantities by $B_i := B_{y_i}$ and $\zeta_i := \zeta_{y_i}$.

Lemma 6.4 (Occupancy-factor bound)**.** *Under the hypotheses of Lemma 6.3, suppose in addition that each free-child input satisfies $m_i(c) \le B_i$ for all c. Then, for every color c,*

$$p_v(c) \sum_{i=1}^{d_v} m_i(c) \le \frac{1}{e\underline{A}_v} \exp\left\{\frac{1}{\underline{A}_v} \sum_{i=1}^{d_v} (\zeta_i + 1)\right\}. \tag{46}$$

Proof. For fixed c, the (d_v+1)-term AM–GM inequality gives

$$a_v(c) \prod_i (1 - m_i(c)) \sum_i m_i(c) \le e^{-1}.$$

The definition gives $B_i > 0$, while the same slack calculation as in Lemma 6.3 gives $B_i \le 1/4$. Writing $Z_v = \sum_b a_v(b) \prod_i (1-m_i(b))$ for the recursion denominator, weighted AM–GM, $a_v(b) \le 1$, and (39) give

$$\begin{aligned} Z_v &\ge A_v \prod_b \left(\prod_i (1-m_i(b))\right)^{a_v(b)/A_v} \\ &\ge A_v \prod_i \left(\prod_b (1-m_i(b))\right)^{1/A_v} \\ &\ge A_v \prod_i (1-B_i)^{1/(A_v B_i)} \\ &\ge \underline{A}_v \prod_i (1-B_i)^{1/(\underline{A}_v B_i)}. \end{aligned}$$

The last step uses that $t \mapsto t\exp(-H/t)$ is increasing for fixed $H \ge 0$. Since $B_i^{-1}\log(1/(1-B_i)) = \zeta_i + 1$, division proves (46). □

Lemma 6.5 (Weighted Jacobian contraction)**.** *At each non-root vertex set $w_v = (1-\zeta_v)^{-1/2}$, and for a child y_i write $w_i := w_{y_i}$. For every $0 < x < 1$, every fixed pinning, and every collection of free-child inputs $m_i \in \mathcal{D}_s$ satisfying $m_i(c) \le B_i$, the transformed recursion in (36) satisfies*

$$\left\|\widehat{J}_v\right\|_{\mathbf{w}} \le \rho := \exp\left(-\frac{4}{81q}\right) < 1. \tag{47}$$

Moreover

$$1 \le w_v \le e^{1/12}, \tag{48}$$

Proof. First reduce to the unscaled output. Regard p_v in (36) as a recursion with weighted color availability $a_v(c)$ and subdistribution inputs m_i. The calculation below is first made in the relative interior of $\prod_i \mathcal{D}_s$; the displayed formulas extend continuously to the whole closed domain, including when an input coordinate tends to zero. The chain rule with (41) gives

$$\widehat{J}_{v,i} = \operatorname{diag}\left(\frac{\sqrt{s}(1-p_v(c))}{1-sp_v(c)}\right)_{c=1}^{q} J^p_{v,i}. \tag{49}$$

The diagonal factor is at most one because

$$1 - st - \sqrt{s}(1-t) = (1-\sqrt{s})(1+\sqrt{s}\,t) \geq 0 \qquad (0 \leq t \leq 1).$$

Thus it is enough to bound the unscaled Jacobian with Lemma 6.2. That factorization uses only $\sum_c p_v(c) = 1$ and $\Sigma_v(c) \geq 0$, not $\sum_b m_i(b) = 1$; the weighted color availability enters only through Z_v, controlled by Lemmas 6.3 and 6.4. Bounding $F_{1,v} \leq \exp(2\xi_v/(\underline{A}_v - 1))$ by Lemma 6.3 and $F_{2,v}$ by (46), and using $d_v \leq \underline{A}_v - 4$ to produce the term $-4/\underline{A}_v$, yields

$$\|J^p_v\|^2_{2\to 2} \leq \frac{1}{\underline{A}_v} \exp\left\{\frac{2\xi_v}{\underline{A}_v - 1} - \frac{4}{\underline{A}_v} + \frac{1}{\underline{A}_v}\sum_i \zeta_i\right\}. \tag{50}$$

Let $y_v = \xi_v/(\underline{A}_v - 1) \leq 1/3$. The elementary entropy inequality [CLMM23, Fact B.5] used in the weighted coloring recursion gives

$$0 < \zeta_v < 1, \qquad \frac{1}{1-\zeta_v} \leq e^{y_v/2}, \tag{51}$$

and therefore (48). Directly from (42),

$$\|J\|^2_{\mathbf{w}} \leq \|J\|^2_{2\to 2}\, w_v^2 \sum_i w_i^{-2}.$$

Here $w_i^{-2} = 1 - \zeta_i$, so $\sum_i w_i^{-2} = d_v - \sum_i \zeta_i$. Applying this bound to (50) gives

$$\left\|\widehat{J}_v\right\|^2_{\mathbf{w}} \leq \exp\left\{\frac{5\xi_v}{2(\underline{A}_v - 1)} - \frac{8}{\underline{A}_v}\right\}. \tag{52}$$

Indeed, with $\bar{\zeta}_v = \underline{A}_v^{-1}\sum_i \zeta_i$,

$$\left(\frac{d_v}{\underline{A}_v} - \bar{\zeta}_v\right) e^{\bar{\zeta}_v} \leq \frac{d_v}{\underline{A}_v} \leq 1 - \frac{4}{\underline{A}_v} \leq e^{-4/\underline{A}_v}.$$

It remains to retain a strict margin. Since $d_v \leq \underline{A}_v - 4$,

$$\xi_v \leq \frac{256}{81}\exp\left(-\frac{12\log(4/3)}{\underline{A}_v - 1}\right).$$

Writing $u = (\underline{A}_v - 1)^{-1}$ and using $\log(1+u) \leq u \leq 12\log(4/3)u$ shows

$$\xi_v \frac{\underline{A}_v}{\underline{A}_v - 1} \leq \frac{256}{81}.$$

Consequently the exponent in (52) is at most $-8/(81\underline{A}_v) \leq -8/(81q)$. Taking square roots proves (47). If $d_v = 0$, the Jacobian is zero and the conclusion is immediate. □

Lemma 6.6 (From Jacobian contraction to message contraction)**.** *Fix the local color weights at a non-root vertex v and the child structural parameters that determine B_i, w_i. Let $(m_i)_{i=1}^{d_v}$ and $(m_i')_{i=1}^{d_v}$ be two collections in $\mathcal{D}_s$ such that $m_i(c), m_i'(c) \le B_i$ for every i, c, and let m_v, m_v' be their respective outputs under (36). Then*

$$w_v \left\|\boldsymbol{\phi}(m_v) - \boldsymbol{\phi}(m_v')\right\|_2 \le \rho \max_i w_i \left\|\boldsymbol{\phi}(m_i) - \boldsymbol{\phi}(m_i')\right\|_2 . \tag{53}$$

Both sides are interpreted as zero when $d_v = 0$.

Proof. When $d_v = 0$, the two outputs are identical. Otherwise, for $t \in [0,1]$ define the potential-coordinate segment

$$u_i(t) = (1-t)\boldsymbol{\phi}(m_i) + t\boldsymbol{\phi}(m_i'), \qquad m_i(t) = \boldsymbol{\phi}^{-1}(u_i(t)).$$

Write $u(t) = (u_i(t))_{i=1}^{d_v}$ and $m(t) = (m_i(t))_{i=1}^{d_v}$; thus $u'(t) = (\boldsymbol{\phi}(m_i') - \boldsymbol{\phi}(m_i))_{i=1}^{d_v}$. Since ϕ is increasing and concave on $[0, 1/4]$, its inverse is increasing and convex on $[0, \phi(1/4)]$. Coordinatewise,

$$m_i(t,c) \le (1-t)m_i(c) + tm_i'(c) \le B_i, \qquad \sum_c m_i(t,c) \le (1-t)\sum_c m_i(c) + t\sum_c m_i'(c) \le s.$$

Thus the entire segment lies in $\mathcal{D}_s$ and satisfies the sharper coordinate bounds needed in Lemma 6.5. The fundamental theorem of calculus and (47) now give

$$\begin{aligned} w_v \left\|\boldsymbol{\phi}(m_v) - \boldsymbol{\phi}(m_v')\right\|_2 &\le \int_0^1 w_v \left\|\widehat{J}_v(m(t))(u'(t))\right\|_2 dt \\ &\le \rho \max_i w_i \left\|\boldsymbol{\phi}(m_i) - \boldsymbol{\phi}(m_i')\right\|_2 . \end{aligned}$$

If an endpoint has a zero coordinate, approximate it from the relative interior and use continuity. □

6.3 Total-influence decay and spatial mixing

We first fix the orientation and the messages used in the factorization. Fix a tree $\mathcal{T} = (V_{\mathcal{T}}, E_{\mathcal{T}})$ of maximum degree at most Δ, a set $\Lambda \subseteq V_{\mathcal{T}}$, a pinning $\tau\colon \Lambda \to [q]$, and a root $r \in V_{\mathcal{T}} \setminus \Lambda$. Let $\mathcal{T}_\tau$ be the component of r in $\mathcal{T} - \Lambda$, and let $\mathrm{dist}_{\mathcal{T}_\tau}$ denote its graph distance. Root $\mathcal{T}_\tau$ at r. For every $y \in V(\mathcal{T}_\tau) \setminus \{r\}$, let y^- be its parent and let $\mathcal{T}_{y\to y^-}$ be the component containing y after the edge yy^- is removed from $\mathcal{T}_\tau$. Write

$$p_{y\to y^-}$$

for the cavity marginal of y in this component, including all fixed pinnings incident to vertices of the component, and put $m_{y\to y^-} = sp_{y\to y^-}$. At the root, p_r is instead the ordinary marginal in the full rooted component and $m_r = sp_r$. Thus the root object is not being asserted to satisfy the non-root marginal bound. By Lemma 6.3, every cavity message $m_{y\to y^-}$ does satisfy

$$0 \le m_{y\to y^-}(c) \le \frac{1}{4}, \tag{54}$$

because deleting the parent edge leaves at most $\Delta - 1$ child and pinned constraints at y.

For a parent y and its free child w, let $J_{y,w}$ denote the block Jacobian, in the potential $\psi_s(p) = \phi(sp)$, of the appropriate oriented update with respect to the incoming message from w: it outputs the full marginal p_r when $y = r$, and it outputs the cavity marginal $p_{y\to y^-}$ when $y \ne r$. In the scaled coordinate this is one block of $\widehat{J}_y$ when $y \ne r$, and is then controlled by Lemma 6.5;

the root block is handled below. Define, with the same distinction between the root and non-root messages,

$$D_r = \operatorname{diag}(p_r\psi_s'(p_r)) = \operatorname{diag}\frac{\sqrt{m_r}}{1-m_r}, \qquad D_{y\to y^-} = \operatorname{diag}\frac{\sqrt{m_{y\to y^-}}}{1-m_{y\to y^-}}. \tag{55}$$

Products such as $p_r\psi_s'(p_r)$ are coordinatewise. Let I_q denote the q-by-q identity matrix and $\mathbf{1}_q \in \mathbb{R}^q$ the all-ones column vector.

Lemma 6.7 (Bounded root response)**.** *For every free child u of the root,*

$$D_r^{-1}J_{r,u} = -(I_q - \mathbf{1}_q p_r^{\mathsf{T}})\operatorname{diag}(\sqrt{m_{u\to r}}), \qquad \left\|D_r^{-1}J_{r,u}\right\|_{2\to 2} \le K_q := \frac{1+\sqrt{q}}{2}. \tag{56}$$

Thus the first influence factor is bounded without a lower bound on any coordinate of p_r.

Proof. The raw derivative of (36) is

$$\frac{\partial p_r(c)}{\partial p_{u\to r}(b)} = \frac{s p_r(c)[p_r(b) - \mathbf{1}_{[b=c]}]}{1-m_{u\to r}(b)}, \qquad \psi_s'(p_{u\to r}(b))^{-1} = \frac{\sqrt{m_{u\to r}(b)}(1-m_{u\to r}(b))}{s}.$$

The factor $p_r(c)$ cancels against D_r^{-1} before any norm is taken, giving the displayed identity. By (54), $\|\operatorname{diag}\sqrt{m_{u\to r}}\|_{2\to 2} \le 1/2$, while $\left\|I_q - \mathbf{1}_q p_r^{\mathsf{T}}\right\|_{2\to 2} \le 1+\sqrt{q}$. □

For this pinning $\tau\colon \Lambda \to [q]$, write μ_v^τ for the one-site marginal at a free vertex v under the corresponding tree Gibbs law and $\mu_v^{\tau^{r=a}}$ for that marginal after additionally pinning a free vertex r to a. For distinct free vertices r, v, let $\Psi_{r,v}$ be the q-by-q influence block

$$\Psi_{r,v}(a,c) := \mu_v^{\tau^{r=a}}(c) - \mu_v^\tau(c);$$

we suppress its dependence on τ. If deleting the vertices pinned by τ separates r from v, this block is zero. Define the level-ℓ total influence by

$$\mathcal{I}_\ell(r;\tau) := \max_{a,b\in[q]} \sum_{\substack{v\in V(\mathcal{T}_\tau)\\ \operatorname{dist}_{\mathcal{T}_\tau}(r,v)=\ell}} \left\|\mu_v^{\tau^{r=a}} - \mu_v^{\tau^{r=b}}\right\|_{\mathrm{TV}}. \tag{57}$$

If r and v remain connected, let their free path be

$$r = u_0, u_1, \dots, u_\ell = v.$$

The influence/Jacobian identity of [CLMM23, Lemma 8.7], applied in the successive rooted subtrees, gives

$$\Psi_{r,v} = D_r^{-1}J_{u_0,u_1}\Big(\prod_{j=1}^{\ell-1} J_{u_j,u_{j+1}}\Big)D_{v\to u_{\ell-1}}(I_q - \mathbf{1}_q p_{v\to u_{\ell-1}}^{\mathsf{T}}). \tag{58}$$

The product is ordered away from the root, is empty when $\ell = 1$, and the intermediate diagonal matrices telescope. The terminal factors use the *cavity* law in $\mathcal{T}_{v\to u_{\ell-1}}$. The identity is exact, and Lemma 6.7 shows that its first factor extends boundedly to a zero root coordinate.

Lemma 6.8 (Tree total-influence decay)**.** *Uniformly over $0 < x < 1$, every tree $\mathcal{T} = (V_\mathcal{T}, E_\mathcal{T})$ of maximum degree at most Δ, every pinning $\tau\colon \Lambda \to [q]$ with $\Lambda \subseteq V_\mathcal{T}$, every root $r \in V_\mathcal{T} \setminus \Lambda$, and every integer $\ell \ge 1$ satisfy*

$$\mathcal{I}_\ell(r;\tau) \le C_0\rho^{\ell-1} = C_{\mathrm{TID}}\rho^\ell, \tag{59}$$

where one may take

$$C_0 = \frac{\Delta e^{1/12}\sqrt{q}(1+\sqrt{q})^2}{3}, \qquad C_{\mathrm{TID}} = \frac{C_0}{\rho}. \tag{60}$$

The level sum ranges only over free vertices. No positive lower bound on a one-site marginal enters this constant.

Proof. The first factor is bounded by Lemma 6.7; it need not contract. At the terminal end, (54) applies to $m_{v\to u_{\ell-1}}$, and therefore

$$\left\| D_{v\to u_{\ell-1}}(I_q - \mathbf{1}_q p_{v\to u_{\ell-1}}^{\mathsf{T}}) \right\|_{2\to 2} \le \frac{2(1+\sqrt{q})}{3}, \tag{61}$$

because $\|D_{v\to u_{\ell-1}}\|_{2\to 2} \le 2/3$ and $\left\|I_q - \mathbf{1}_q p_{v\to u_{\ell-1}}^{\mathsf{T}}\right\|_{2\to 2} \le 1+\sqrt{q}$.

We now use (58) simultaneously over a level. If the level contains no free vertex, the claimed influence bound is immediate. Otherwise, group the terminals by their first child of r, keep the corresponding first factor (56) intact, and peel the remaining $\ell-1$ levels with the block norm of Lemma 6.5. Every update peeled in this way is a non-root cavity update. Iterating the weighted inequality gives $\rho^{\ell-1}$; the ratio between its terminal and initial weights is at most $e^{1/12}$ by (48). Finally sum the at most Δ first-child blocks by the triangle inequality. Thus, for arbitrary terminal block vectors $h_v \in \mathbb{R}^q$,

$$\frac{\left\|\sum_{\mathrm{dist}_{\mathcal{T}_\tau}(r,v)=\ell} \Psi_{r,v} h_v\right\|_2}{\max_v \|h_v\|_2} \le \Delta e^{1/12} K_q \frac{2(1+\sqrt{q})}{3}\rho^{\ell-1}.$$

Here the product and the contraction exponent are interpreted as empty and 1, respectively, when $\ell = 1$. The standard block L^2-to-L^∞ duality conversion costs at most $\sqrt{q}$. Indeed, if

$$\mathcal{A}(h) := \sum_{\mathrm{dist}_{\mathcal{T}_\tau}(r,v)=\ell} \Psi_{r,v} h_v,$$

the triangle inequality through the unconditioned law makes the two factors $1/2$ in total variation cancel, and hence

$$\mathcal{I}_\ell(r;\tau) \le \max_a \sum_{\mathrm{dist}_{\mathcal{T}_\tau}(r,v)=\ell} \sum_c |\Psi_{r,v}(a,c)| \le \sup_{\max_v \|h_v\|_\infty \le 1} \|\mathcal{A}(h)\|_\infty \le \sqrt{q} \sup_{\max_v \|h_v\|_2 \le 1} \|\mathcal{A}(h)\|_2 .$$

Using the definition of K_q and writing $\rho^{\ell-1} = \rho^{-1}\rho^\ell$ gives exactly (59)–(60). □

Uniform relative SSM cannot start at distance one. Indeed, on a single edge, compare boundary colors a and $b \neq a$. The probability that the free endpoint has color a is respectively

$$\frac{x}{q-1+x} \qquad \text{and} \qquad \frac{1}{q-1+x},$$

so the ratio is x (or $1/x$ in the reverse comparison). The large-girth transfer only needs a relative estimate beyond a fixed depth.

Lemma 6.9 (Relative SSM beyond a fixed depth)**.** *For*

$$C_{\mathrm{rel}} := \frac{5}{8}\Delta q\, e^{1/12} \log 3, \tag{62}$$

the following holds for every $0 < x < 1$ on every tree $\mathcal{T} = (V_{\mathcal{T}}, E_{\mathcal{T}})$ of maximum degree at most Δ. Fix $\Lambda \subseteq V_{\mathcal{T}}$, a pinning $\tau\colon \Lambda \to [q]$, and a root $v \in V_{\mathcal{T}} \setminus \Lambda$. For an integer $K \geq 2$, let

$$S_K(v) := \{u \in V_{\mathcal{T}} : \mathrm{dist}_{\mathcal{T}}(u, v) = K\},$$

and let

$$\omega, \omega'\colon S_K(v) \setminus \Lambda \longrightarrow [q]$$

be two assignments of all unpinned shell vertices. Thus $\tau \cup \omega$ and $\tau \cup \omega'$ are the unions of partial maps with disjoint domains. The two systems have the same fixed pinning τ and differ only on the shell. Write

$$p_v := \mu_v^{\tau \cup \omega}, \qquad p'_v := \mu_v^{\tau \cup \omega'}.$$

Here each right-hand side denotes the one-site marginal at v under the indicated pinning. Then, for every root color c,

$$\left|\log \frac{p_v(c)}{p'_v(c)}\right| \leq C_{\mathrm{rel}} \rho^{K-2}. \tag{63}$$

Define

$$K_1 := \min\{K \in \mathbb{Z}_{\geq 2} : C_{\mathrm{rel}} \rho^{K-2} \leq 1\}.$$

For every integer $K \geq K_1$, the ratio-form relative-SSM error is at most $2C_{\mathrm{rel}} \rho^{K-2}$. The case $x = 0$ will be added after the final coupling bound by finite-state continuity; at $x = 1$ the left side is zero.

Proof. The pinned children adjacent to the root belong to the fixed pinning τ, so their color weights $a_v(c)$ agree. Write

$$P_c = \prod_i (1 - m_i(c)), \qquad p_v(c) = \frac{a_v(c) P_c}{Z}, \qquad Z = \sum_b a_v(b) P_b,$$

and use primes for the second condition. The potentially tiny root color weight cancels exactly:

$$\log \frac{p_v(c)}{p'_v(c)} = \log \frac{P_c}{P'_c} + \log \frac{Z'}{Z}. \tag{64}$$

Because $K \geq 2$, every message entering the root is the output of a non-root recursion and hence is bounded by $1/4$. Therefore

$$\left|\log \frac{P_c}{P'_c}\right| \leq \frac{4}{3} \sum_i |m_i(c) - m'_i(c)|.$$

Moreover telescoping the products gives $|Z - Z'| \leq \sum_i \|m_i - m'_i\|_1$, while the root version of (40) gives $Z, Z' \geq 3$. Consequently

$$\left|\log \frac{p_v(c)}{p'_v(c)}\right| \leq \frac{5}{3} \sum_i \|m_i - m'_i\|_1 . \tag{65}$$

The update adjacent to the shell is not compared by a Jacobian, because its local color weights may change with the shell assignment. Its two output messages nevertheless obey Lemma 6.3. For $1 \leq j \leq K - 1$, let $\mathcal{F}_j$ be the free vertices at distance j from v in the same component after the vertices pinned by τ are deleted. Orient this component toward v, and write y^- for the parent of y. Let

$$E_j := \max_{y \in \mathcal{F}_j} w_y \left\| \boldsymbol{\phi}(m_{y \to y^-}) - \boldsymbol{\phi}(m'_{y \to y^-}) \right\|_2 .$$

We set the maximum over an empty set to zero. The potential diameter and (48) give

$$E_{K-1} \le e^{1/12}\sqrt{q}\,\phi(1/4) = e^{1/12}\sqrt{q}\log 3.$$

The two boundary assignments pin the same set of shell vertices, so the two recursions have the same d_y, k_y, B_y, w_y. Every update strictly inside that shell has the same local color weights on both sides. Thus Lemma 6.6 gives

$$E_j \le \rho E_{j+1} \qquad (1 \le j \le K-2).$$

These are exactly $K-2$ contracting levels. Since $w_i \ge 1$ and $\inf_{[0,1/4]} \phi' = 8/3$, Cauchy–Schwarz gives, for each message entering the root,

$$\|m_i - m_i'\|_1 \le \frac{3}{8}\sqrt{q}\,\|\boldsymbol{\phi}(m_i) - \boldsymbol{\phi}(m_i')\|_2 \le \frac{3}{8} q\, e^{1/12}\log 3\,\rho^{K-2}.$$

Summing over at most Δ root children in (65) proves (63) with (62).

To convert this logarithmic estimate to the relative-error form of [CLMM23, Definition 5.7], put $\varepsilon_K := C_{\mathrm{rel}}\rho^{K-2}$. By the definition of K_1, for $K \ge K_1$,

$$\left|\frac{p_v(c)}{p_v'(c)} - 1\right| \le e^{\varepsilon_K} - 1 \le 2\varepsilon_K.$$

Thus the usual relative-SSM estimate holds uniformly at every $K \ge K_1$; no assertion is made at distance one. □

6.4 Transfer to large-girth graphs

The transfer of [CLMM23] still applies when relative SSM begins at a fixed depth K_0.

Lemma 6.10 (Large-girth transfer from tree decay)**.** *Let $q \ge 1$ and $\Delta \ge 3$ be integers, and let $J \subseteq (0,1]$. Suppose there are $0 < C_{\mathsf{INFL}}, C_{\mathsf{SM}} < \infty$, $\delta \in (0,1)$, and a fixed positive integer K_0 such that, for every $x \in J$ and uniformly over all trees $T \in \mathcal{G}_\Delta$, all pinnings τ of T, and all free roots, the Potts law $\mu^\tau_{T,x}$ has level-ℓ tree total influence at most $C_{\mathsf{INFL}}(1-\delta)^\ell$ for every integer $\ell \ge 1$, and relative SSM in the ratio sense of [CLMM23, Definition 5.7] at distance K at most $C_{\mathsf{SM}}(1-\delta)^K$ for every integer $K \ge K_0$. Then there are constants $g_{\mathsf{tr}} \in \mathbb{Z}_{\ge 3}$ and $C_{\mathsf{tr}} \ge 0$, depending only on $q, \Delta, C_{\mathsf{INFL}}, C_{\mathsf{SM}}, \delta, K_0$, and in particular not on x or J, with the following property. For every $x \in J$, every $G \in \mathcal{G}_\Delta$, every pinning τ with $\mathrm{girth}(G^\tau) \ge g_{\mathsf{tr}}$, every free source vertex $u \in V^\tau$, and all $b, c \in [q]$,*

$$W_{1,\mathrm{Ham}}\left(\mu^{\tau^{u=b}}_{G,x}, \mu^{\tau^{u=c}}_{G,x}\right) \le C_{\mathsf{tr}}. \tag{66}$$

Both laws live on the common free set $V^\tau \setminus \{u\}$.

Proof. Fix $x \in J$, and regard each pinned pair (G,τ) with $G \in \mathcal{G}_\Delta$ as a pairwise spin system on its free graph G^τ, with spin set $[q]$, the Potts edge factors, and external fields

$$\lambda_{v,c} = x^{b_v^\tau(c)}.$$

The family of these systems is closed under further pinnings and deletion of free vertices. Pinning a free vertex r to c removes one free incidence at each neighbor and adds c to its boundary count. Hence, for every remaining vertex v,

$$\deg_{G^{\tau^{r=c}}}(v) + \sum_d b_v^{\tau^{r=c}}(d) = \deg_{G^\tau}(v) + \sum_d b_v^\tau(d) \le \Delta,$$

whereas taking a subgraph only lowers this budget. Since $x > 0$, every configuration has positive weight, so every spin is globally feasible after every pinning and the feasibility requirements of the pairwise transfer hold trivially. Conditioning the source u fixes the scalar $\lambda_{u,c}$, which disappears on normalization, and turns each incident edge into the boundary factor in $\tau^{u=c}$. Thus the conditional exterior laws supplied by the transfer are exactly the two laws in (66).

The pairwise extension follows from [CLMM23, Remark 5.11], and the relative input is the ratio form in [CLMM23, Definition 5.7]. It remains to check that the proof tolerates the restriction $K \geq K_0$. The radius, the cutting depth, and the girth bound chosen below depend only on $q, \Delta, C_{\mathsf{INFL}}, C_{\mathsf{SM}}, \delta, K_0$, so they serve every $x \in J$ at once.

The cutting depth K is chosen after the influence radius R. The error in the proof of [CLMM23, Lemma 5.14] is

$$2C_{\mathsf{SM}}(1-\delta)^K \Delta^R + C_{\mathsf{INFL}}(1-\delta)^R. \tag{67}$$

First choose R, subject to all lower bounds on R in the cited proof, so large that

$$C_{\mathsf{INFL}}(1-\delta)^R \leq \frac{1}{16R\log\Delta}.$$

After this R is fixed, choose K subject to *all* lower bounds on the cutting depth in [CLMM23, Lemmas 5.14 and 5.20], and also

$$K \geq K_0, \qquad K > R, \qquad K \geq \left\lceil \frac{\log C_{\mathsf{SM}}}{\delta} \right\rceil, \qquad 2C_{\mathsf{SM}}(1-\delta)^K \Delta^R \leq \frac{1}{16R\log\Delta}.$$

Such a K exists; with this choice (67) is at most $1/(8R\log\Delta)$, which is the intermediate influence-decay threshold in [CLMM23, Lemma 5.14]. Finally take the girth bound to be the maximum of every girth lower bound in the cited lemmas and $2K+2$. Relative SSM is invoked only at this chosen cutting depth, so the restriction $K \geq K_0$ causes no change later in the argument. Lemmas 5.14 and 5.20 of [CLMM23] therefore supply the intermediate decay condition, and their Lemma 5.13 gives (66) with

$$C_{\mathsf{tr}} = 2\Delta^R.$$

□

Proof of Theorem 6.1. Lemmas 6.8 and 6.9 give, for every $0 < x < 1$, tree TID and relative SSM beyond a fixed depth, with constants and rate uniform in x. Apply Lemma 6.10 with

$$J = (0,1), \qquad C_{\mathsf{INFL}} = C_{\mathrm{TID}}, \qquad C_{\mathsf{SM}} = 2C_{\mathrm{rel}}\rho^{-2}, \qquad \delta = 1-\rho, \qquad K_0 = K_1.$$

It supplies g_{tr} and C_{tr}, depending only on (Δ, q), such that, for every $0 < x < 1$, whenever G^τ has girth at least g_{tr},

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=a}}, \mu_{G,x}^{\tau^{r=b}}\right) \leq C_{\mathsf{tr}}. \tag{68}$$

They live on the common finite state space $[q]^{V^\tau \setminus \{r\}}$. At $x = 0$, the hypothesis $q \geq \Delta + 3$ implies hard feasibility for every normalized child by Lemma 2.2, so both limiting normalizing constants are positive. Their weights are polynomials in x, and Lemma 3.7 passes the same bound to $x = 0$. At $x = 1$ all remaining spins are independent and the distance is zero. Taking $g_* = \max\{3, g_{\mathsf{tr}}\}$ and $C_* = C_{\mathsf{tr}}$ proves the theorem. □

7 Coupling independence at high temperature and large girth

This section turns the tree contraction of Bencs, Berrekkal, and Regts [BBR25] into a coupling bound on sufficiently large-girth graphs. The extra step is tree total-influence decay: strong spatial mixing alone does not provide this input to the large-girth transfer of Chen, Liu, Mani, and Moitra [CLMM23].

Fix integers q, Δ such that

$$q \geq 3, \qquad \frac{\Delta}{q} \geq \frac{e - \frac{1}{2}}{e - 1}. \tag{69}$$

For $(q, \Delta) \neq (3, 4)$, put

$$k := e\frac{\Delta - q/2}{\Delta - q}, \qquad x_0 := 1 - \frac{q}{\Delta}\left(1 - \frac{k}{\Delta}\right)^2 \frac{\Delta - k}{\Delta - k/2}. \tag{70}$$

For $(q, \Delta) = (3, 4)$, put $x_0 := 3/4$. The proof of Theorem 4.6 shows that $0 < x_0 < 1$, and that $0 < k < \Delta$ outside the exceptional case. Throughout this section, $x \in [x_0, 1]$, and

$$\kappa := \frac{\Delta - 1}{\Delta}, \qquad \theta := \sqrt{\kappa} < 1. \tag{71}$$

Write I_q for the q-by-q identity matrix, $\mathbf{1}_q$ for the all-ones vector in $\mathbb{R}^q$, and e_i for the i-th standard basis vector.

7.1 Square-root cavity messages and the BBR certificate

Consider a rooted tree of maximum degree at most Δ with a fixed pinning τ. We work in one component of the free graph and represent each free–pinned edge by an individually labeled pinned leaf. Thus the pinning contributes only unary factors at free vertices. For a free non-root vertex v, let T_v be its descendant cavity tree. The numerator $Z_{T_v, v=i}$ below is the restricted partition-function contribution with $v = i$, retaining the unary factor at v; it is not the normalized child partition function obtained by dropping that scalar. The denominator $Z_{T_v - v}$ deletes v and its incident edges, retaining the unary factors at the remaining vertices. Define

$$R_v(i) := \left(\frac{Z_{T_v, v=i}(x)}{Z_{T_v - v}(x)}\right)^{1/2}, \qquad i \in [q]. \tag{72}$$

We use the same ratio at the global root. All entries are positive because $x \geq x_0 > 0$. For $y \in \mathbb{R}^q_{>0}$, write

$$S(y) := \sum_i y_i^2, \qquad S_i(y) := S(y) - (1 - x)y_i^2, \qquad F_x(y)_i := \sqrt{\frac{S_i(y)}{S(y)}}. \tag{73}$$

If $\mathcal{F}_u$ is the set of free children of u, put $f_u := |\mathcal{F}_u|$. The recursion is

$$R_u = B_u \odot \prod_{v \in \mathcal{F}_u} F_x(R_v), \tag{74}$$

where $B_u(i) = x^{b_u^\tau(i)/2}$ is the product of the square-root unary factors from pinned children. The symbol $\odot$ denotes coordinatewise multiplication, and all products and quotients of vectors in this recursion are coordinatewise.

For two positive messages R, R', let $R_t = tR + (1-t)R'$ and set

$$L_x(R, R') := \max_{\substack{0\le t\le 1\\ i\in[q]}} \frac{\sqrt{S(R_t)}}{S_i(R_t)}. \tag{75}$$

A pair is called *jointly realizable* when its two messages arise at the same cavity vertex of the same rooted tree, with the same pinned domain, after assigning possibly different colors on that domain. This same-tree, same-domain restriction is part of the statement below; no claim is made for arbitrary positive vectors.

Lemma 7.1 (BBR contraction certificate)**.** *Let u be a free non-root cavity vertex with $f \le \Delta - 1$ free child branches. For every $x \in [x_0, 1]$ and every jointly realizable pair R, R' at u at activity x whose two boundary assignments agree on all pinned vertices at distance at most one from u,*

$$f\,\frac{1-x}{e}\,L_x(R, R')^2 \le \kappa. \tag{76}$$

The empty disagreement set is interpreted as having distance $+\infty$. Equivalently, the hypothesis is that the nearest changed pin has distance strictly larger than one from u. This is the distance condition in the BBR lower-bound lemma used below. In particular, at a vertex whose nearest changed pin is at distance at least two, the local pinned factor B_u is the same on the two sides, so the finite-difference recursion may use this certificate there.

Proof. Set $d = \Delta - 1$. If $x = 1$, then the left side of (76) is zero, so the claim is immediate; hence assume $x < 1$ in the remainder. First suppose $(q, \Delta) \ne (3, 4)$. The ratio hypothesis and integrality then imply $\Delta \ge q + 2$, and hence $d \ge q + 1$. Although Proposition 2.6 of Bencs, Berrekkal, and Regts [BBR25] states the stronger hypothesis $d \ge q+2$ for its two conclusions together, the estimate in part (i) only invokes their Lemmas 4.3, 4.4, and 4.6 and Corollary 4.5(i). The hypotheses printed with those results are respectively $x \in (0, 1)$, $q \ge 3$ and $d \ge q + 1$ (for Lemma 4.4), the distance condition in Lemma 4.6, and $q \ge 3$, $d \ge q + 1$ (for Corollary 4.5(i)); part (ii) of the corollary, which is the place where BBR require $d \ge q + 2$, is not used here. Thus the following extraction of the proof of Proposition 2.6(i) is valid for $d \ge q + 1$. In the notation of their (2.13), put

$$\alpha := \frac{(1-x)\Delta}{q}, \qquad \text{so that} \qquad x = 1 - \alpha q/\Delta.$$

Their calculation then gives, for $0 \le f \le d$, the upper bound

$$\frac{\alpha f}{e\Delta}\left(1 - \frac{k}{\Delta}\right)^{-2} k^{(d-f)/d}\left(e\frac{\Delta - k/2}{\Delta - k}\right)^{f/d}.$$

For completeness, this expression is at most its value at $f = d$. Indeed, writing $t = f/d$ and $B = e(\Delta - k/2)/(\Delta - k)$, the ratio to the $f = d$ value is $t(k/B)^{1-t}$. The ratio assumption gives $k \le e^2$ (the function $a \mapsto e(a - 1/2)/(a - 1)$ is decreasing and equals e^2 at $a = (e - 1/2)/(e - 1)$). Therefore

$$\frac{B}{k} = \frac{e}{k}\,\frac{\Delta - k/2}{\Delta - k} \ge \frac{e}{k} \ge e^{-1}.$$

Since $\log t \le t - 1$ for $0 < t \le 1$, we have $t(k/B)^{1-t} \le te^{1-t} \le 1$ (the case $f = 0$ is immediate). Consequently the preceding bound is at most

$$\frac{\alpha d}{\Delta}\left(1 - \frac{k}{\Delta}\right)^{-2}\frac{\Delta - k/2}{\Delta - k}.$$

For $x \geq x_0$, (70) gives

$$\alpha \leq \left(1 - \frac{k}{\Delta}\right)^2 \frac{\Delta - k}{\Delta - k/2}.$$

The preceding bound is therefore at most $d/\Delta = (\Delta - 1)/\Delta = \kappa$.

It remains to treat $(q, \Delta) = (3, 4)$, so $d = 3$ and $\kappa = 3/4$. Every coordinate of a cavity message is a product of at most three factors from $[\sqrt{x}, 1]$. Hence, along the segment between two realizable messages,

$$S(R_t) \geq 3x^3, \qquad S_i(R_t) \geq xS(R_t), \qquad L_x(R, R')^2 \leq \frac{1}{3x^5}.$$

Consequently, for $x \geq x_0 = 3/4$ and $f \leq 3$,

$$f\frac{1-x}{e} L_x(R, R')^2 \leq \frac{1-x}{ex^5} \leq \frac{256}{243e} < \frac{3}{4} = \kappa.$$

The cited BBR bounds, as well as the direct exceptional-case estimate, apply to messages produced by the same cavity tree under two boundary assignments on the same domain, which is exactly the realizability convention above. □

7.2 Differential contraction

Let H_u denote the local map in (74), and let $J_{u\leftarrow v}$ be its Jacobian block with respect to the message of the free child v. Define the pointwise weight

$$\ell_x(R) := L_x(R, R) = \max_i \frac{\sqrt{S(R)}}{S_i(R)}, \qquad a_v := \frac{1-x}{e} \ell_x(R_v)^2. \tag{77}$$

Lemma 7.2 (Differential contraction)**.** *At every collection of child messages arising at a vertex u of a pinned rooted tree at activity x, and for arbitrary vectors $z_v \in \mathbb{R}^q$,*

$$\left\| \sum_{v \in \mathcal{F}_u} J_{u\leftarrow v} z_v \right\|_2^2 \leq \sum_{v \in \mathcal{F}_u} a_v \left\| z_v \right\|_2^2. \tag{78}$$

In particular, if R_v arises as a non-root cavity message and f_v is the number of free children of v, then

$$f_v a_v \leq \kappa. \tag{79}$$

Proof. Put

$$P_v := I_q - \frac{R_v R_v^{\mathsf{T}}}{S(R_v)}.$$

Differentiating (73) gives

$$DF_x(y) = \operatorname{diag}\left(\frac{x-1}{F_x(y)} \odot \frac{y}{S(y)}\right)\left(I_q - \frac{yy^{\mathsf{T}}}{S(y)}\right).$$

Writing $Y = H_u((R_v)_v)$, the product rule therefore gives

$$\left(\sum_v J_{u\leftarrow v} z_v\right)_i = Y_i \sum_v (x-1) \frac{R_v(i)}{S_i(R_v)} (P_v z_v)_i. \tag{80}$$

Set

$$\xi_v = \ell_x(R_v) P_v z_v, \qquad h_v(i) = Y_i(1-x)\frac{R_v(i)}{\ell_x(R_v) S_i(R_v)}.$$

Since $\ell_x(R_v) S_i(R_v) \geq \sqrt{S(R_v)}$ and every coordinate of B_u is at most one,

$$\sum_v h_v(i)^2 \leq \left[\prod_v F_x(R_v)_i^2\right](1-x)^2 \sum_v \frac{R_v(i)^2}{S(R_v)}.$$

With $t_v = (1-x)R_v(i)^2/S(R_v)$, the right side is $(1-x)\prod_v(1-t_v)\sum_v t_v \leq (1-x)/e$. Coordinatewise Cauchy–Schwarz in (80), followed by $\|P_v\|_{2\to 2} \leq 1$, now yields

$$\left\|\sum_v J_{u\leftarrow v} z_v\right\|_2^2 \leq \frac{1-x}{e}\sum_v \|\xi_v\|_2^2 \leq \sum_v a_v \|z_v\|_2^2 .$$

Finally, apply Lemma 7.1 to the identical pair (R_v, R_v) to obtain (79). □

7.3 Cavity derivatives and influence

Fix a tree of maximum degree at most Δ with pinning τ, and let μ_v^τ denote the one-site marginal at a free vertex v. Root the free component containing a free vertex r, and, for every other free vertex v in that component, write

$$\Psi_{r,v}(b,j) := \mu_v^{\tau^{r=b}}(j) - \mu_v^\tau(j)$$

for the q-by-q influence block. In this and the next subsection, distances are measured in the underlying tree. A vertex separated from r by the fixed pinning has zero influence. For every free v in the rooted component, let $D_v := \mathrm{diag}(R_v)$, and let

$$Q_r := D_r^{-1} - \mathbf{1}_q \frac{R_r^{\mathsf{T}}}{S(R_r)}. \tag{81}$$

For a free path from v to r, let $\mathcal{J}_{r\leftarrow v}$ be the ordered product of the Jacobian blocks along that path. For a free vertex v outside the component of r, set $\Psi_{r,v} = 0$.

Lemma 7.3 (Influence–Jacobian identity)**.** *For every free vertex $v \neq r$ in the component of r,*

$$\Psi_{r,v} = Q_r \mathcal{J}_{r\leftarrow v} D_v. \tag{82}$$

Consequently, for every integer $\ell \geq 1$, if vectors $z_v \in \mathbb{R}^q$ are placed on the free vertices in this component at distance ℓ from r, set $y_v = D_v z_v$ there and propagate upward by

$$y_u = \sum_{v\in\mathcal{F}_u} J_{u\leftarrow v} y_v. \tag{83}$$

At each preceding level, take this sum over the branches meeting the sphere; an empty sum is zero. Then

$$\sum_{\substack{\mathrm{dist}(r,v)=\ell \\ v \text{ in the component of } r}} \Psi_{r,v} z_v = Q_r y_r. \tag{84}$$

Proof. Multiply the weight of a configuration by

$$\exp\left(t \sum_{\substack{\operatorname{dist}(r,v)=\ell \\ v \text{ in the component of } r}} z_v(\sigma_v)\right).$$

Write $R_v(t)$ and $p_r(\cdot;t)$ for the cavity messages and root marginal under this reweighted law, and use a dot for differentiation with respect to t at zero. No two vertices on this sphere are ancestors of one another. Hence, at a sphere vertex v,

$$\dot{R}_v(0) = \tfrac{1}{2} D_v z_v.$$

The chain rule through (74) gives $\dot{R}_r(0) = y_r/2$. Since the root marginal is $p_r(i) = R_r(i)^2/S(R_r)$,

$$d \log p_r = 2Q_r \, dR_r.$$

Thus

$$\frac{d}{dt} \log p_r(b;t)\Big|_{t=0} = (Q_r y_r)_b. \tag{85}$$

Differentiating the Gibbs law directly gives the covariance identity

$$\frac{d}{dt} \log p_r(b;t)\Big|_{t=0} = \sum_{\substack{\operatorname{dist}(r,v)=\ell \\ v \text{ in the component of } r}} \sum_j (\mu_v^{\tau^{r=b}}(j) - \mu_v^{\tau}(j)) z_v(j).$$

Comparison with (85), for arbitrary $(z_v)_v$, proves both displayed identities. □

7.4 Tree total-influence decay

Every non-root message contains at most $\Delta - 1$ child factors, each in $[\sqrt{x}, 1]$. Uniformly for $x \in [x_0, 1]$,

$$x_0^{\Delta-1} \le R_v(i)^2 \le 1, \qquad S_i(R_v) \ge x_0 S(R_v), \qquad S(R_v) \ge q x_0^{\Delta-1}. \tag{86}$$

It follows that

$$a_v \le A := \frac{1 - x_0}{e \, q \, x_0^{\Delta+1}}. \tag{87}$$

At the global root there are at most Δ factors, so

$$\|Q_r\|_{2\to 2} \le \left\|D_r^{-1}\right\|_{2\to 2} + \frac{\sqrt{q}}{\sqrt{S(R_r)}} \le 2 x_0^{-\Delta/2}. \tag{88}$$

Proposition 7.4 (Tree total-influence decay)**.** *For every tree of maximum degree at most Δ, every pinning τ, every free root r, all $b, c \in [q]$, every $x \in [x_0, 1]$, and every integer $\ell \ge 1$,*

$$\sum_{\substack{v:\, \operatorname{dist}(r,v)=\ell \\ v \text{ free}}} \left\|\mu_v^{\tau^{r=b}} - \mu_v^{\tau^{r=c}}\right\|_{\mathrm{TV}} \le C_0 \theta^{\ell-1}, \tag{89}$$

where

$$C_0 := \sqrt{\frac{2\Delta(1-x_0)}{e}} \, x_0^{-(\Delta+1/2)}. \tag{90}$$

Thus the tree TID hypothesis of Lemma 6.10 holds with $C_{\mathsf{INFL}} = C_0/\theta$ and decay factor θ.

Proof. If the level contains no free vertex, the claim is immediate. Otherwise, fix vectors z_v on the free part of the level and put $M = \max_v \|z_v\|_2$. Propagate the vectors y by (83). At distance $\ell - 1$ from the root, Lemma 7.2, (87), and $\|D_v\|_{2\to 2} \le 1$ give

$$\frac{\|y_u\|_2^2}{f_u} \le AM^2 \tag{91}$$

whenever the subtree below u meets the perturbed sphere. At each higher non-root vertex, every child v carrying a nonzero perturbation has $f_v \ge 1$, and (78)–(79) give

$$\frac{\|y_u\|_2^2}{f_u} \le \kappa \max_{v:\, y_v \neq 0} \frac{\|y_v\|_2^2}{f_v}.$$

At the root we do not divide by its number of children; there are at most Δ of them. Hence

$$\|y_r\|_2 \le \sqrt{\Delta A}\, \theta^{\ell-1} M. \tag{92}$$

For $\ell = 1$, this is the direct estimate from (78) and (87).

Combining (84), (88), and (92) yields

$$\left\| \sum_{\substack{\operatorname{dist}(r,v)=\ell \\ v \text{ free}}} \Psi_{r,v} z_v \right\|_2 \le 2\sqrt{\frac{\Delta(1-x_0)}{e\, q}}\, x_0^{-(\Delta+1/2)} \theta^{\ell-1} M. \tag{93}$$

For $b = c$ the claim is immediate. Otherwise fix $b \neq c$ and take

$$z_v(j) = \operatorname{sgn}\left(\mu_v^{\tau^{r=b}}(j) - \mu_v^{\tau^{r=c}}(j)\right).$$

Then $\|z_v\|_2 \le \sqrt{q}$, while

$$\sum_{\substack{v:\, \operatorname{dist}(r,v)=\ell \\ v \text{ free}}} \left\| \mu_v^{\tau^{r=b}} - \mu_v^{\tau^{r=c}} \right\|_1 = (e_b - e_c)^{\mathsf{T}} \sum_{\substack{v:\, \operatorname{dist}(r,v)=\ell \\ v \text{ free}}} \Psi_{r,v} z_v.$$

Use $\|e_b - e_c\|_2 = \sqrt{2}$, apply (93), and divide by two to obtain (89). □

7.5 Spatial mixing and transfer to large-girth graphs

The same BBR certificate was designed to control finite differences. We record the precise uniform consequence needed below.

Lemma 7.5 (Uniform relative SSM)**.** *Fix $x \in [x_0, 1]$. Let two boundary assignments on a tree of maximum degree at most Δ have the same pinned domain, and suppose their first possible disagreement from a free root r is at integer distance $K \ge 2$. If p_r, p_r' are the two root marginals and R_r, R_r' their corresponding messages, then*

$$\|R_r - R_r'\|_2^2 \le \Delta q\, \kappa^{K-2}. \tag{94}$$

Consequently, for every $i \in [q]$,

$$\left| \frac{p_r(i)}{p_r'(i)} - 1 \right| \le C_{\mathsf{SM}} \theta^K, \qquad C_{\mathsf{SM}} := 4q\sqrt{\Delta q}\, x_0^{-3\Delta/2} \kappa^{-1}. \tag{95}$$

Proof. The finite-difference inequality of [BBR25, Theorem 2.5] is the analogue of (78) with $\ell_x(R_v)$ replaced by the segment weight $L_x(R_v, R'_v)$. For $K = 2$, the crude bound $\|R_r - R'_r\|_2^2 \le q$ is already at most the right side of (94). Now let $K \ge 3$. At distance $K-2$ from the root, initialize with

$$\frac{\|R_u - R'_u\|_2^2}{f_u} \le q.$$

Indeed, every message that differs there has a free child branch carrying the disagreement, so $f_u \ge 1$. At every vertex contracted above this level, the nearest changed pin is at distance at least two; hence its local pinned factor B_u agrees on the two sides. The branch-normalized finite-difference induction gives a factor κ at each of the $K-3$ non-root levels, and the final root estimate contributes at most $\Delta\kappa$. This proves (94). The same-tree, same-domain quantifiers are exactly those of Lemma 7.1.

For completeness, every coordinate of a root message satisfies $R_r(i)^2 \ge x_0^{\Delta}$, and $S(R_r) \le q$. Thus $p_r(i) \ge x_0^{\Delta}/q$. Along the segment between R_r and R'_r,

$$\left\| \nabla \log \frac{y_i^2}{S(y)} \right\|_2 \le \frac{2}{y_i} + \frac{2}{\sqrt{S(y)}} \le 4x_0^{-\Delta/2}.$$

It follows from (94) that

$$\left| \log \frac{p_r(i)}{p'_r(i)} \right| \le 4\sqrt{\Delta q}\, x_0^{-\Delta/2} \kappa^{-1} \theta^K.$$

If this last bound is at most one, use $e^t - 1 \le 2t$. Otherwise its being larger than one implies $C_{\mathsf{SM}}\theta^K \ge q x_0^{-\Delta}$, which covers the crude bound $|p_r(i)/p'_r(i) - 1| \le q x_0^{-\Delta}$. This proves (95). □

Theorem 7.6 (BBR-certified interval at large girth)**.** *Under* (69)*, there exist* $g_{\mathrm{BBR}} = g_{\mathrm{BBR}}(q, \Delta) \ge 3$ *and* $C_{\mathrm{BBR}} = C_{\mathrm{BBR}}(q, \Delta) < \infty$ *with the following property. For every* $G \in \mathcal{G}_\Delta$*, every pinning* τ *for which* $\mathrm{girth}(G^\tau) \ge g_{\mathrm{BBR}}$*, every free vertex* $r \in V^\tau$*, all* $b, c \in [q]$*, and every* $x \in [x_0, 1]$*,*

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau^{r=b}}, \mu_{G,x}^{\tau^{r=c}}\right) \le C_{\mathrm{BBR}}. \tag{96}$$

The two laws are on the common free set $V^\tau \setminus \{r\}$*.*

Proof. Consider all normalized pinned Potts systems with activity $x \in [x_0, 1]$ and degree budget at most Δ. On the free graph, a boundary multiplicity $b_v^\tau(i)$ contributes the positive unary factor $x^{b_v^\tau(i)}$; equivalently, these are the pinned-leaf factors used in (74). Since $x_0 > 0$, this is the family of pinned Potts laws in Lemma 6.10 with $J = [x_0, 1]$.

Proposition 7.4 and Lemma 7.5 give the two tree inputs to Lemma 6.10, uniformly over the whole interval, with

$$C_{\mathsf{INFL}} = C_0/\theta, \qquad \delta = 1 - \theta, \qquad K_0 = 2,$$

and C_{SM} as in (95). That lemma, which incorporates all cutting-depth requirements from the cited large-girth transfer, supplies constants g_{BBR} and C_{BBR}. Its conditional exterior laws are exactly the two normalized child laws in (96), proving the theorem. □

8 Coupling independence for Edge-Potts at $q \ge 3\Delta$

For a finite simple graph $G = (V, E)$, edge-color assignments are vertex colorings of the line graph $L(G)$. An edge-color pinning $\tau : \Lambda \to [q]$, with $\Lambda \subseteq E$, is likewise a vertex-color pinning of $L(G)$. We write

$$(\mu_{G,x}^{\mathrm{edge}})^\tau := \mu_{L(G),x}^\tau.$$

The Fisher polynomial of this edge-Potts model is simply the vertex-Potts polynomial on $L(G)$; in particular,

$$Z^{\tau}_{L(G)}(z) = z^{m_{L(G)}(\tau)}\widehat{Z}^{\tau}_{L(G)}(z). \tag{97}$$

The analytic argument is the vertex-Potts transfer on line graphs; the additional input is the following coupling bound.

Theorem 8.1 (Edge-Potts coupling independence)**.** *Let q, Δ be positive integers with $q \geq 3\Delta$. For every finite simple graph G of maximum degree at most Δ, every edge-color pinning τ, every $x \in [0,1]$, every free edge $i \in E \setminus \operatorname{dom}\tau$, and all $a, b \in [q]$, put, for $c \in [q]$,*

$$\tau^{i=c} := \tau \cup \{i \mapsto c\}, \qquad (\mu^{\mathrm{edge}}_{G,x})^{\tau,i=c} := (\mu^{\mathrm{edge}}_{G,x})^{\tau^{i=c}}.$$

Then, on the common free edge set $E \setminus (\operatorname{dom}\tau \cup \{i\})$,

$$W_{1,\mathrm{Ham}}\Big((\mu^{\mathrm{edge}}_{G,x})^{\tau,i=a}, (\mu^{\mathrm{edge}}_{G,x})^{\tau,i=b}\Big) \leq \Delta - 1. \tag{98}$$

Corollary 8.2 (Edge-Potts zero-freeness)**.** *Let q, Δ be positive integers with $\Delta \geq 2$ and $q \geq 3\Delta$. There exists $\varepsilon = \varepsilon(q,\Delta) > 0$ such that, for every finite simple graph G of maximum degree at most Δ and every edge-color pinning τ,*

$$\widehat{Z}^{\tau}_{L(G)}(z) \neq 0 \qquad \text{whenever } \operatorname{dist}(z,[0,1]) < \varepsilon.$$

Thus the unnormalized pinned polynomial has no zero in this neighborhood except possibly at $z = 0$; its order of vanishing there is $m_{L(G)}(\tau)$. In particular, $Z_{L(G)}(z) \neq 0$ throughout the same neighborhood.

Proof. The line graph has maximum degree at most $\Delta_L := 2\Delta - 2$. The class of line graphs of simple graphs of maximum degree at most Δ is closed under taking induced subgraphs, since an induced subgraph on an edge set S is the line graph of (V, S). Vertex pinnings in this class are exactly edge-color pinnings, with the normalization in (97). Moreover, $q \geq 3\Delta$ implies $q \geq \Delta_L + 1$, and Theorem 8.1 supplies one coupling-independence constant on $[0,1]$. Apply Theorem 2.4 with degree bound Δ_L. The statement for the unnormalized polynomial follows from (97). □

Proof overview. For $0 < x < 1$, hidden endpoint slots represent the edge-Potts weight by a hard distinct-label constraint. Replacing one occupied endpoint label admits a coupling with expected Hamming distance at most $(\Delta-1)/2$; conditioning i requires one replacement at each endpoint. The resulting coupling is a countable weighted adaptation of the list-edge-coloring coupling in [CWZZ25].

8.1 Exact slot lift

Fix a finite simple graph $G = (V, E)$ of maximum degree at most Δ, and $q \geq 3\Delta$. Fix $0 < x < 1$, a subset $\Lambda \subseteq E$, and an edge-color pinning $\tau\colon \Lambda \to [q]$. Thus $\operatorname{dom}\tau = \Lambda$, and $F_0 = E \setminus \Lambda$ is the initial set of free edges. Fix once and for all an ordering $e = (e^-, e^+)$ of the two endpoints of every $e \in E$. When using slot notation for e, we always write $(u, v) = (e^-, e^+)$, so that $e = uv$ has this fixed ordered meaning. For a vertex v and color c, write

$$b^{\tau}_v(c) := \left|\{e \in \operatorname{dom}\tau : e \ni v,\ \tau(e) = c\}\right|,$$

so here $b^{\tau}_v(c)$ counts pinned edges incident to the base-graph vertex v. For $\phi\colon F_0 \to [q]$, put

$$k_{v,c}(\phi) := \left|\{e \in F_0 : e \ni v,\ \phi(e) = c\}\right|.$$

For an edge coloring $\psi : D \to [q]$, where $D \subseteq E$, put

$$N_{\mathrm{conf}}(\psi) := \left|\{\{e, f\} \subseteq D : e \neq f,\ e \cap f \neq \varnothing,\ \psi(e) = \psi(f)\}\right|.$$

Then $m_{L(G)}(\tau) = N_{\mathrm{conf}}(\tau)$. Because G is simple, two distinct edges share at most one endpoint. Consequently,

$$N_{\mathrm{conf}}(\tau \cup \phi) - m_{L(G)}(\tau) = \sum_{v,c} \left[b_v^\tau(c)\, k_{v,c}(\phi) + \binom{k_{v,c}(\phi)}{2} \right]. \tag{99}$$

The two terms count pinned–free and free–free conflicts, respectively; the pinned–pinned conflicts are precisely the subtracted normalization.

The second term in (99) has the following probabilistic representation.

Lemma 8.3 (Slot representation)**.** *For every $0 < x < 1$, there is a probability distribution $(\kappa_r)_{r \geq 1}$, with every $\kappa_r > 0$, such that for every integer $k \geq 0$,*

$$\sum_{\substack{r_1,\dots,r_k \geq 1 \\ r_1,\dots,r_k \text{ all distinct}}} \prod_{j=1}^{k} \kappa_{r_j} = x^{\binom{k}{2}}. \tag{100}$$

Proof. Define the deformed exponential

$$\mathcal{E}_x(t) := \sum_{k \geq 0} x^{\binom{k}{2}} \frac{t^k}{k!}.$$

By [MFB72, Ise93], all zeros of $\mathcal{E}_x$ are real and negative; see also [WZ18, §1, pp. 312–313]. Write them, with multiplicity, as $-1/\kappa_r$, where $\kappa_r > 0$. If a_k is the coefficient of t^k, then

$$\log \frac{1}{|a_k|} = |\log x| \binom{k}{2} + \log k!,$$

so $\mathcal{E}_x$ is entire of order zero. Its genus-zero Hadamard factorization is

$$\mathcal{E}_x(t) = \prod_{r \geq 1} (1 + \kappa_r t).$$

There is no exponential factor, and the constant is fixed by $\mathcal{E}_x(0) = 1$. In particular, $\sum_r \kappa_r < \infty$, so the product and its coefficient expansions converge absolutely. Comparison of the linear coefficients gives $\sum_r \kappa_r = 1$, and comparison at degree k gives

$$\sum_{1 \leq r_1 < \cdots < r_k} \kappa_{r_1} \cdots \kappa_{r_k} = \frac{x^{\binom{k}{2}}}{k!}.$$

Multiplication by $k!$ gives (100). □

Thus k independent slots with law (κ_r) are all distinct with probability $x^{\binom{k}{2}}$. Apply this identity separately at every vertex–color pair.

For a free edge $e = uv$, use the lifted state

$$\xi = (c, r, s) \in [q] \times \mathbb{N}_{\geq 1}^2$$

with activity

$$w_e(c, r, s) := x^{b_u^\tau(c) + b_v^\tau(c)} \kappa_r \kappa_s. \tag{101}$$

These activities always use the original color pinning τ and remain fixed during later lifted-state pinnings; the latter are recorded only in the occupied-label sets defined below. At u this state exposes the endpoint label $\ell_{e,u}(\xi) = (c, r)$, and at v it exposes $\ell_{e,v}(\xi) = (c, s)$. A lifted assignment is admissible when all labels exposed at any one vertex are distinct.

After some lifted edge states have been pinned, let $F \subseteq F_0$ denote the current free-edge set and let $B_v \subseteq [q] \times \mathbb{N}_{\geq 1}$ be a finite *set* of occupied labels at v, write $B = (B_v)_{v \in V}$, and put

$$\deg_F(v) := |\{e \in F : v \in e\}|.$$

An assignment

$$\omega = (\omega_e)_{e \in F} \in \left([q] \times \mathbb{N}_{\geq 1}^2\right)^F$$

on the current free-edge set is admissible when the labels in B_v, together with those exposed by the incident free edges, are pairwise distinct at every v. Its weight is

$$W_{F,B}(\omega) := \prod_{e \in F} w_e(\omega_e),$$

and $\nu_{F,B}$ denotes the corresponding normalized law on admissible assignments whenever its partition sum is positive.

For a current free edge $e = uv$, define

$$d_F(e) := |\{f \in F \setminus \{e\} : f \cap e \neq \varnothing\}|$$

and let

$$M_e(B) := \sum_{\substack{\xi = (c,r,s) \\ \ell_{e,u}(\xi) \notin B_u,\, \ell_{e,v}(\xi) \notin B_v}} w_e(\xi)$$

be the activity compatible with the occupied boundary labels. The required bounds, for every $v \in V$ and $e \in F$ as applicable, are

$$\deg_F(v) + |B_v| \leq \Delta, \tag{102}$$

$$\sum_{\xi:\, \ell_{e,y}(\xi) = \zeta} w_e(\xi) \leq 1 \quad (y \in e,\ \zeta \in [q] \times \mathbb{N}_{\geq 1}), \tag{103}$$

$$M_e(B) - d_F(e) \geq \Delta + 2. \tag{104}$$

Pinning an admissible positive-activity state ξ on $e = uv$ produces $F' = F \setminus \{e\}$ and the occupied-label family B' given by

$$B'_u = B_u \cup \{\ell_{e,u}(\xi)\}, \qquad B'_v = B_v \cup \{\ell_{e,v}(\xi)\}, \qquad B'_y = B_y \quad (y \notin \{u, v\}).$$

Lemma 8.4 (Exact lift and inherited bounds)**.** *Assume $q \geq 3\Delta$. For the initial data $(F_0, \varnothing)$, where $\varnothing$ denotes the family of empty occupied-label sets, color projection $(c, r, s) \mapsto c$ sends the lifted law exactly to the normalized pinned edge-Potts law $(\mu^{\mathrm{edge}}_{G,x})^\tau$.*

The initial slot instance and every boundary obtained by successively pinning admissible positive-activity states satisfy (102)–(104). More generally, any instance (F, B) with finite occupied-label sets that uses the endpoint-label maps and activities in (101) and satisfies these three bounds has a finite, positive partition sum.

Proof. Fix a color assignment $\phi\colon F_0 \to [q]$ and sum over its slot fiber. The sum factorizes over the pairs (v,c). If $k = k_{v,c}(\phi)$, the activities contribute $x^{b_v^\tau(c)k}$, while admissibility requires the k endpoint slots to be distinct. By (100), their total slot weight is $x^{\binom{k}{2}}$. Hence the fiber weight is

$$\prod_{v,c} x^{b_v^\tau(c)k_{v,c}(\phi)+\binom{k_{v,c}(\phi)}{2}} = x^{N_{\mathrm{conf}}(\tau\cup\phi)-m_{L(G)}(\tau)}$$

by (99). This proves exact projection.

Initially, (102) follows from $\deg_{F_0}(v) \le \Delta$. For a fixed endpoint label, say (c,r) at u,

$$\sum_s w_e(c,r,s) = x^{b_u^\tau(c)+b_v^\tau(c)}\kappa_r \sum_s \kappa_s \le 1,$$

and the other endpoint is symmetric. This proves (103).

For the slack, let

$$b_e := \sum_c (b_u^\tau(c) + b_v^\tau(c)),$$

the number of pinned edges adjacent to $e = uv$. Since $\sum_r \kappa_r = 1$,

$$M_e(\varnothing) = \sum_c x^{b_u^\tau(c)+b_v^\tau(c)} \ge q - b_e.$$

Here we used $x^m \ge 1 - m$ for nonnegative integers m and $0 < x < 1$. Simplicity gives

$$d_{F_0}(e) + b_e = |\{f \in E \setminus \{e\} : f \cap e \ne \varnothing\}| \le 2\Delta - 2,$$

and therefore

$$M_e(\varnothing) - d_{F_0}(e) \ge q - d_{F_0}(e) - b_e \ge 3\Delta - (2\Delta - 2) = \Delta + 2.$$

It remains to check one pinning step for current data (F,B). Suppose e is pinned and a remaining edge f meets e. Simplicity makes the common endpoint unique. The new occupied label removes from M_f at most one endpoint fiber, whose activity is at most one by (103); deleting e decreases $d_F(f)$ by exactly one. Thus

$$M_f(B') - d_{F\setminus\{e\}}(f) \ge (M_f(B) - 1) - (d_F(f) - 1) \ge \Delta + 2.$$

Nonadjacent edges are unchanged. At each endpoint of e, the free degree drops by one and one genuinely new occupied label is added, so (102) is preserved. The activities, and hence (103), do not change.

Finally, the three bounds imply feasibility by greedy exposure. At every step an unassigned edge has at least $M_e(B) - d_F(e) \ge \Delta + 2 > 0$ compatible activity, and the preceding pinning argument preserves the same conclusion. The partition sum is finite because there are finitely many edges and $\sum_\xi w_e(\xi) \le q$ for each edge. □

8.2 Coupling under a boundary-label replacement

Consider two occupied-label systems with the same free graph, endpoint-label maps, and edge activities. Fix a vertex v, a finite $C_v \subseteq [q] \times \mathbb{N}_{\ge 1}$ and distinct $\alpha, \beta \in ([q] \times \mathbb{N}_{\ge 1}) \setminus C_v$. Their occupied-label families satisfy $B_y^\alpha = B_y^\beta$ for $y \ne v$, while at v,

$$B_v^\alpha = C_v \cup \{\alpha\}, \qquad B_v^\beta = C_v \cup \{\beta\}.$$

Both systems satisfy (102)–(104). These three bounds, rather than derivation from the initial lift, are the hypotheses because the two-step comparison uses mixed endpoint boundaries.

For two lifted assignments on F, let

$$\mathrm{Ham}_F(\omega,\omega') := \bigl|\{e \in F : \omega_e \neq \omega'_e\}\bigr|,$$

and write $W_{1,F}$ for the infimum of the expected Ham_F-cost over couplings of two laws on the resulting countable state space.

For each integer $n \geq 0$, let T_n be the supremum of $W_{1,F}(\nu_{F,B^\alpha}, \nu_{F,B^\beta})$ over all data described in the preceding paragraph that satisfy the three displayed bounds and have at most n free edges. The laws exist by Lemma 8.4, $0 \leq \mathsf{T}_n \leq n$, and $\mathsf{T}_0 = 0$.

Lemma 8.5 (One-label coupling bound)**.** *One has* $\mathsf{T}_0 = 0$*, and for every integer* $n \geq 1$*,*

$$\mathsf{T}_n \leq \frac{\Delta-1}{2}\left[1 - \left(\frac{\Delta-1}{\Delta}\right)^n\right] \leq \frac{\Delta-1}{2}. \tag{105}$$

Proof. Only $n \geq 1$ requires proof. Fix an admissible comparison with $1 \leq |F| \leq n$; the case $F = \varnothing$ has zero cost. Write $\mu_\alpha := \nu_{F,B^\alpha}$, $\mu_\beta := \nu_{F,B^\beta}$, and let $N = \{j \in F : j \ni v\}$. Under μ_α, define

$$E_j^\beta := \{\omega : \ell_{j,v}(\omega_j) = \beta\}, \qquad E_0^\beta := \{\omega : \text{no edge of } N \text{ exposes } \beta \text{ at } v\}.$$

Define E_j^α, E_0^α symmetrically under μ_β. Distinctness of endpoint labels makes the label-exposure events $(E_j^\beta)_{j\in N}$ disjoint, and together with E_0^β they partition the α-side state space; the same holds on the other side.

The avoidance events have positive probability. Fix either current boundary $B \in \{B^\alpha, B^\beta\}$. For that side, greedily expose the free edges while additionally forbidding the missing label at v. For an incident edge this removes at most one further endpoint fiber, so the available activity is at least

$$M_e(B) - d_F(e) - 1 \geq \Delta + 1 > 0.$$

At each exposure, deleting the chosen edge lowers the remaining free edge-degrees by the same amount as the new occupied endpoint labels can lower their allowed activities, so this positive margin persists at every later step. Set

$$\rho_j := \frac{\mu_\alpha(E_j^\beta)}{\mu_\alpha(E_0^\beta)}, \qquad \sigma_j := \frac{\mu_\beta(E_j^\alpha)}{\mu_\beta(E_0^\alpha)},$$

with a zero numerator interpreted as zero.

To bound these label-exposure odds, start from a configuration in E_j^β, keep every state off j and replace the state of j by a compatible state that does not expose β at v. The other adjacent free edges exclude at most $d_F(j)$ endpoint fibers, and avoiding β excludes one more. Thus the replacement activity is at least $\Delta + 1$. Route the source weight among the replacements proportionally to their activities. Conversely, for a fixed target, summing all source states whose endpoint color is β gives total activity $x^{b_u^\tau(\beta)+b_v^\tau(\beta)} \leq 1$, which is also the bound supplied by (103). Summing this flow gives

$$\sum_{\omega\in E_j^\beta} W_{F,B^\alpha}(\omega) \leq \frac{1}{\Delta+1} \sum_{\omega\in E_0^\beta} W_{F,B^\alpha}(\omega),$$

and hence

$$\rho_j \leq \frac{1}{\Delta+1}.$$

The same argument gives $\sigma_j \le 1/(\Delta+1)$. Since one boundary label is already occupied at v, (102) implies $|N| \le \Delta - 1$. Hence

$$R := \sum_{j \in N} \max\{\rho_j, \sigma_j\} \le \frac{\Delta - 1}{\Delta + 1}. \tag{106}$$

Write

$$\nu_0^\alpha := \mu_\alpha(\,\cdot \mid E_0^\beta), \qquad \nu_j^\alpha := \mu_\alpha(\,\cdot \mid E_j^\beta),$$

and define ν_0^β, ν_j^β symmetrically. A conditional law with zero coefficient below is omitted. We next record the three conditional comparisons:

$$W_{1,F}(\nu_0^\alpha, \nu_0^\beta) = 0, \tag{107}$$

$$W_{1,F}(\nu_j^\alpha, \nu_j^\beta) \le 1 + \mathsf{T}_{n-1}, \tag{108}$$

$$\max\Big\{W_{1,F}(\nu_j^\alpha, \mu_\beta), W_{1,F}(\mu_\alpha, \nu_j^\beta)\Big\} \le 1 + 2\mathsf{T}_{n-1}. \tag{109}$$

For (107), the two avoidance events produce the same boundary $C_v \cup \{\alpha, \beta\}$, so the conditional laws are identical.

For the other two bounds, condition further on the complete state of $j = vw$, remove j, and put its endpoint labels into the boundary. For each pair of conditioned states, the remaining-edge laws are compared through the corresponding residual boundary; mixing these conditional couplings over the state of j preserves the stated uniform cost. For (108), the boundaries at v then agree, and the two systems differ by at most one label replacement at w. The residual cost is at most T_{n-1}, and restoring j costs at most one.

For (109), compare the two residual laws through the boundary that uses the first conditioned system's labels at v and the second conditioned system's labels at w. This requires at most one label replacement at each endpoint. After j is removed, simplicity ensures that no remaining edge meets both v and w. Every remaining edge therefore sees exactly the endpoint restrictions of one of the two conditioned systems, so the intermediate boundary satisfies (102)–(104) and is feasible. Its residual cost is at most $2\mathsf{T}_{n-1}$, and restoring j again costs at most one. The conditioning mixtures are countable, but all these costs are uniform. For each $\eta > 0$, choose every conditional coupling with expected cost within η of its Wasserstein infimum. Tonelli's theorem, followed by $\eta \downarrow 0$, gives (108)–(109).

It remains to couple the exposure blocks. The following table specifies a nonnegative mixture of couplings. Every displayed mass is divided by $1 + R$; the first row is used once and the other rows once for each $j \in N$. Zero-mass rows are omitted, and $(t)_+ := \max\{t, 0\}$:

numerator	α-side	β-side
1	ν_0^α	ν_0^β
$\min\{\rho_j, \sigma_j\}$	ν_j^α	ν_j^β
$(\rho_j - \sigma_j)_+$	ν_j^α	μ_β
$(\sigma_j - \rho_j)_+$	μ_α	ν_j^β

$(j \in N)$.

To check the first marginal, note that

$$\nu_0^\alpha + \sum_j \rho_j \nu_j^\alpha = \left(1 + \sum_j \rho_j\right) \mu_\alpha$$

and

$$R = \sum_j \rho_j + \sum_j (\sigma_j - \rho_j)_+.$$

Thus the rows on the α-side sum to $(1+R)\mu_\alpha$; the other marginal is symmetric. Combining the table with (107)–(109) gives

$$W_{1,F}(\mu_\alpha,\mu_\beta) \le \frac{\sum_j \min\{\rho_j,\sigma_j\}(1+\mathsf{T}_{n-1}) + \sum_j |\rho_j-\sigma_j|\,(1+2\mathsf{T}_{n-1})}{1+R}$$
$$\le \frac{R}{1+R}(1+2\mathsf{T}_{n-1}) \le \frac{\Delta-1}{2\Delta}(1+2\mathsf{T}_{n-1}),$$

where the last step uses (106). Taking the supremum over all such comparisons gives

$$\mathsf{T}_n \le \frac{\Delta-1}{2\Delta}(1+2\mathsf{T}_{n-1}).$$

Solving this affine recursion from $\mathsf{T}_0 = 0$ yields (105). □

8.3 From slot couplings to coupling independence

Proof of Theorem 8.1. If $a = b$, the two normalized children are identical. Assume $a \neq b$, and write the root edge as $i = uv$.

Case $x = 1$. Under every pinning, the remaining edge colors are independent and uniform, so the root-excluded distance is zero.

Case $0 < x < 1$. Construct the single slot lift associated with the original pinning τ. By Lemma 8.4, its color projection is exactly $(\mu^{\mathrm{edge}}_{G,x})^\tau$. Conditioning the root edge i to color a disintegrates the lifted child over the positive-mass root states (a,r,s); the b-child is the analogous mixture over (b,r',s'). Both are conditionals of this same lift.

For $c \in \{a,b\}$, let Π_c denote the law of ω_i under $\nu_{F_0,\varnothing}$, conditional on the color coordinate of ω_i being c. Thus every $\xi \in \operatorname{supp}\Pi_c$ has the form (c,r,s). For such a state ξ, delete i and define the occupied-label family on the remaining system by

$$B^\xi_y := \begin{cases} \{\ell_{i,y}(\xi)\}, & y \in i, \\ \varnothing, & y \notin i. \end{cases}$$

Set

$$\nu^\xi := \nu_{F_0\setminus\{i\},B^\xi}.$$

Thus the root-excluded lifted c-child is the Π_c-mixture of the laws ν^ξ.

Fix $\xi \in \operatorname{supp}\Pi_a$ and $\xi' \in \operatorname{supp}\Pi_b$, and delete i. Each conditioned law satisfies (102)–(104). Compare the two laws through the boundary that takes the first root state's label at u and the second root state's label at v. Since G is simple, no remaining edge meets both u and v; hence this intermediate boundary inherits the three bounds from the two conditioned systems. The two comparisons change at most one occupied label each. Therefore

$$W_{1,F_0\setminus\{i\}}(\nu^\xi,\nu^{\xi'}) \le 2\mathsf{T}_{|F_0|-1} \le \Delta-1$$

by Lemma 8.5.

Couple the two countable root-state laws by the product law $\Pi_a \otimes \Pi_b$. For each $\eta > 0$ and every positive-mass pair (ξ,ξ'), choose a coupling of ν^ξ and $\nu^{\xi'}$ whose expected cost is within η of the corresponding Wasserstein infimum, and then let $\eta \downarrow 0$. Tonelli's theorem preserves the same uniform bound. Color projection cannot increase Hamming distance, so the two edge-Potts children have root-excluded distance at most $\Delta-1$.

Case $x = 0$. Work in the finite color space. Every normalized hard child exists: after fixing i, greedily color the remaining vertices of the line graph, equivalently the remaining free edges of G. Each sees at most $2\Delta - 2$ already colored neighbors, whereas $q \geq 3\Delta$.

For $c \in \{a, b\}$, put

$$t_c := b_u^\tau(c) + b_v^\tau(c).$$

Then

$$m_{L(G)}(\tau^{i=c}) = m_{L(G)}(\tau) + t_c,$$

and for every coloring ϕ of the remaining edges and $x > 0$,

$$x^{N_{\mathrm{conf}}(\tau^{i=c} \cup \phi) - m_{L(G)}(\tau)} = x^{t_c}\, x^{N_{\mathrm{conf}}(\tau^{i=c} \cup \phi) - m_{L(G)}(\tau^{i=c})}.$$

The first factor is independent of ϕ, so ordinary conditioning for $x > 0$ is exactly the normalized child. After this scalar is removed, the two families of weights are finite polynomials in x, and their partition sums are positive at $x = 0$ by the preceding greedy argument. Lemma 3.7 therefore passes the bound $\Delta - 1$ to the normalized hard children. □

9 Coupling independence at girth five for $q \geq (1 + \delta)\Delta$

9.1 Proof of coupling independence

The proof combines a two-layer disintegration, a covariance bound for the insertion residual, and a weighted response recursion. The spectral-gap input is proved in Section 9.2.

Throughout this section, q, Δ are positive integers, $0 < \delta \leq 1$, $x \in [0, 1]$, and $s := 1 - x$. In this subsection, assume $q \geq (1 + \delta)\Delta$. A residual instance $I = (H, \mathbf{b})$ consists of a finite simple free graph H and nonnegative integer boundary counts $b_v(c)$ satisfying

$$\deg_H(v) + k_H(v) \leq \Delta, \qquad k_H(v) := \sum_{c \in [q]} b_v(c). \tag{110}$$

These counts agree with $b_v^\tau(c)$ from Section 2. Write μ_I for the normalized Gibbs law at the fixed activity x, $\mathbb{E}_I$ for its expectation, and $p_{I,v}$ for its marginal at v. The instance $I - v$ deletes v and its incident edges but retains the boundary counts at the remaining vertices; pinning a vertex instead adds the corresponding boundary counts at its free neighbors. For $x > 0$, $\mu_{I,w}^{v=c}$ denotes the marginal at w after conditioning v to color c. At $x = 0$ it denotes the w-marginal of the normalized child law from Section 2, even when c has zero probability in the parent law. All distances and spheres refer to the stated free graph; in particular, $S_2^H(v) := \{w : \mathrm{dist}_H(v, w) = 2\}$. In color space, I_q is the identity matrix and $\mathbf{1} \in \mathbb{R}^q$ is the all-ones vector.

Set

$$m := q - \Delta, \qquad B := m^{-1}, \qquad L_\Delta := \frac{1 + \sqrt{q/(m+1)}}{m}. \tag{111}$$

The color hypothesis gives $m \geq \delta\Delta$; the estimates below include equality throughout. We use the threshold and gap constant

$$\Delta_0(\delta) := \left\lceil \frac{4096(1+\delta)e^{2/\delta}}{\delta^4} \right\rceil, \qquad \gamma_\delta := \frac{\delta}{4(2+\delta)}. \tag{112}$$

We assume $\Delta \geq \Delta_0(\delta)$, so in particular $B < 1$. For a free site w, write $\mathrm{Var}_w f := \mathrm{Var}(f \mid \sigma_{V(H)\setminus\{w\}})$ under the current Gibbs law. The spectral-gap input, proved in Theorem 9.7, is

$$\mathrm{Var}_{\mu_I}(f) \leq \frac{1}{\gamma_\delta} \sum_{w \in V(H)} \mathbb{E}_{\mu_I}[\mathrm{Var}_w f]. \tag{113}$$

It applies after every further pinning and deletion that preserves (110) and girth at least five.

Local geometry.

Lemma 9.1 (Second-layer disintegration)**.** *Let $I = (H, \mathbf{b})$ be a residual instance, let v be free, and put $U = N_H(v)$, $S = S_2^H(v)$, and $\nu = \mu_{I-v}$. If $\mathrm{girth}(H) \geq 5$, then U is independent, each $w \in S$ has a unique neighbor in U, and, with $O = V(H) \setminus (\{v\} \cup U \cup S)$, the following holds for every ξ with $\nu(\sigma_S = \xi) > 0$:*

$$\nu(d\sigma_U, d\sigma_O \mid \sigma_S = \xi) = \left(\bigotimes_{u \in U} \pi_u^\xi(d\sigma_u) \right) \otimes \nu_O^\xi(d\sigma_O). \tag{114}$$

Here π_u^ξ and ν_O^ξ are the conditional marginals of σ_u and σ_O, respectively, under ν given $\sigma_S = \xi$. In particular, all edges inside S and all edges inside O are retained; the marginal law on S is not assumed to be a product law.

Proof. There are no edges inside U by the absence of triangles. Every neighbor of $u \in U$ other than v lies in S, and uniqueness follows because two neighbors in U would give a four-cycle through v. After S is fixed, all factors incident with S become unary factors and the remaining Gibbs weight separates into the isolated vertices U and the induced exterior graph O. Normalization gives (114). □

The disintegration isolates the neighbors of the root once the second layer is revealed. We next control how changing one second-layer coordinate changes that root-neighbor contribution in the color direction.

Lemma 9.2 (One-edge color operator)**.** *Assume $m \geq \delta\Delta$ and let r be a cavity color law obtained after deleting one incident constraint. For $h \in \mathbb{R}^q$, define*

$$(T_r h)(t) = \frac{\langle r, h \rangle - sr(t)h(t)}{1 - sr(t)}, \qquad t \in [q].$$

Then $r(c) \leq (m+1)^{-1}$ and, for the Euclidean projection $\Pi := I_q - q^{-1}\mathbf{1}\mathbf{1}^\mathsf{T}$ onto the subspace orthogonal to constants,

$$\|\Pi T_r \Pi\|_{2\to 2} \leq L_\Delta.$$

If the updated one-site law ϱ has atoms at most B, then

$$\mathrm{Var}_{t\sim\varrho}((T_r h)(t)) \leq B L_\Delta^2 \|h\|_2^2. \tag{115}$$

The estimate remains valid when h depends on all other coordinates but not on the color currently updated.

Proof. Writing $\vartheta_c := sr(c)/(1 - sr(c))$ and $\vartheta = (\vartheta_c)_{c\in[q]}$ gives

$$T_r = \mathbf{1}r^\mathsf{T} - \mathrm{diag}(\vartheta)(I_q - \mathbf{1}r^\mathsf{T}).$$

The first term vanishes after projection, while $\max_c \vartheta_c \leq m^{-1}$ and $\|I_q - \mathbf{1}r^\mathsf{T}\|_{2\to 2} \leq 1 + \sqrt{q/(m+1)}$. This proves the quotient bound. Finally $\mathrm{Var}_\varrho(g) = \min_{a\in\mathbb{R}} \sum_j \varrho_j (g_j - a)^2 \leq B\|\Pi g\|_2^2$, which gives (115). □

Color covariance defect. For the quantitative bounds below set

$$a_B = (1-B)^{-1}, \quad V_\pi = \frac{\Delta B L_\Delta^2}{\gamma_\delta}, \quad V_M = \Delta V_\pi,$$

$$D_\Delta = \frac{\Delta B^2 a_B}{2}, \quad E_\Delta = D_\Delta + \tfrac{1}{2} e^{\Delta B} V_M, \quad b_\Delta = a_B^2 \sqrt{V_\pi} + a_B e^{\Delta B + 2D_\Delta}(\sqrt{V_M} + D_\Delta),$$

$$g_\Delta = \sqrt{V_M}\, b_\Delta, \qquad \ell_\Delta = 2\Delta B b_\Delta, \qquad K_\Delta = \Delta g_\Delta, \qquad \widetilde{\epsilon}_\Delta = \sqrt{B}\, K_\Delta. \tag{116}$$

For fixed δ, these satisfy $b_\Delta = O_\delta(\Delta^{-1/2})$, $g_\Delta = O_\delta(\Delta^{-1})$, $K_\Delta = O_\delta(1)$, and $\widetilde{\epsilon}_\Delta = O_\delta(\Delta^{-1/2})$.

The next lemma is the key girth-five estimate: the insertion residual has only a small color-direction covariance, even though the second layer itself need not be a product measure.

Lemma 9.3 (Color covariance of the insertion residual)**.** *Assume $0 < \delta \le 1$, $\Delta \ge \Delta_0(\delta)$, $q \ge (1+\delta)\Delta$, and $0 < x < 1$. Suppose that every residual instance used below satisfies the degree budget, so that (113) applies. Under the hypotheses of Lemma 9.1, sums and products over neighbors range over U unless stated otherwise. Fix a color c and put $X_{u,c} = \mathbf{1}_{\{\sigma_u = c\}}$, $p_{u,c} = \mathbb{E}_\nu X_{u,c}$,*

$$F_c = \prod_{u \in U}(1 - sX_{u,c}), \quad T_c = \mathbb{E}_\nu F_c, \quad Q_c = \prod_{u \in U}(1 - sp_{u,c}), \quad \alpha_{u,c} = \frac{s}{1 - sp_{u,c}}.$$

Let

$$R_c = F_c/T_c - 1 + \sum_u \alpha_{u,c}(X_{u,c} - p_{u,c}), \qquad G_c = \mathbb{E}_\nu[R_c \mid \sigma_S].$$

For $\beta_{u,c}$ defined by

$$\beta_{u,c} = \alpha_{u,c} - \frac{s}{T_c} \prod_{j \ne u}(1 - s\pi_{j,c}), \qquad \pi_{u,c} = \mathbb{E}_\nu[X_{u,c} \mid \sigma_S],$$

the coefficient $\beta_{u,c}$ does not depend on the configuration σ_{W_u}, where $W_u = N_H(u) \setminus \{v\}$. Moreover, for every deterministic $z \in \mathbb{R}^q$,

$$\mathrm{Var}_w\left(\sum_c z_c G_c\right) \le B L_\Delta^2 \sum_c z_c^2 \beta_{u,c}^2 \qquad (w \in W_u). \tag{117}$$

The insertion-quotient expansion proved below gives $\|\beta_{u,c}\|_{L^2(\nu)} \le b_\Delta$ uniformly in u, c, where $b_\Delta = O_\delta(\Delta^{-1/2})$. Consequently,

$$\|\operatorname{Cov}_\nu(G_1, \dots, G_q)\|_{2\to 2} \le V_M b_\Delta^2 =: g_\Delta^2, \qquad g_\Delta = O_\delta(\Delta^{-1}). \tag{118}$$

Write $G = (G_c)_{c \in [q]}$ as a column vector; its covariance matrix has entries $\operatorname{Cov}_\nu(G_c, G_d)$. For every probability vector r with $r_c \le B$, it follows that

$$\operatorname{Cov}_\nu(\operatorname{diag}(\sqrt{r})G) = \operatorname{diag}(\sqrt{r}) \operatorname{Cov}_\nu(G) \operatorname{diag}(\sqrt{r}), \qquad \|\operatorname{Cov}_\nu(\operatorname{diag}(\sqrt{r})G)\|_{2\to 2} \le B g_\Delta^2. \tag{119}$$

Proof. The affine formula for G_c gives the stated expression for $\beta_{u,c}$. Changing σ_w changes only π_u, so (115), applied to the vector $(z_c \beta_{u,c})_c$, gives (117).

We now record the insertion estimates used below. For a fixed color c, set

$$M_c = \sum_{u \in U}(\pi_{u,c} - p_{u,c}), \qquad Y_c(\xi) = \prod_{u \in U}(1 - s\pi_{u,c}(\xi)). \tag{120}$$

Conditional independence gives $T_c = \mathbb{E}_\nu Y_c$. Applying (115) in each disjoint set W_u and then (113) gives

$$\mathrm{Var}_\nu(\pi_{u,c}) \le V_\pi, \qquad \mathrm{Var}_\nu(M_c) \le V_M. \tag{121}$$

For $0 \le y \le B$, set $\mathfrak{r}(y) := -\log(1-y) - y$. The bound $0 \le \mathfrak{r}(y) \le y^2/[2(1-B)]$ gives

$$\zeta_c := \sum_{u \in U} \big(\mathfrak{r}(sp_{u,c}) - \mathfrak{r}(s\pi_{u,c})\big), \qquad \log \frac{Y_c}{Q_c} = -sM_c + \zeta_c, \qquad |\zeta_c| \le D_\Delta. \tag{122}$$

Indeed, each of the two remainder sums lies in $[0, D_\Delta]$. Since $\mathbb{E}_\nu M_c = 0$ and $|M_c| \le \Delta B$, the exponential Taylor bound gives $1 \le \mathbb{E}_\nu e^{-sM_c} \le 1 + \frac{1}{2} e^{\Delta B} V_M$. It follows that

$$\left|\log \frac{T_c}{Q_c}\right| \le E_\Delta, \qquad T_c \ge e^{-\Delta B - D_\Delta}. \tag{123}$$

Finally,

$$\beta_{u,c} = s\left(\frac{1}{1-sp_{u,c}} - \frac{1}{1-s\pi_{u,c}}\right) + \frac{s}{1-s\pi_{u,c}}\left(1 - \frac{Y_c}{T_c}\right). \tag{124}$$

To bound the variance, put

$$\Phi_c := \exp\left(-s \sum_{u \in U} \pi_{u,c}\right) = \exp\left(-s \sum_{u \in U} p_{u,c}\right) e^{-sM_c}.$$

Then $Y_c = \Phi_c \exp(-\sum_u \mathfrak{r}(s\pi_{u,c}))$, so $0 \le Y_c \le \Phi_c \le 1$ and $|Y_c - \Phi_c| \le D_\Delta$. As a function of M_c on its range, Φ_c is s-Lipschitz. Thus (121) gives $\mathrm{Var}_\nu(\Phi_c) \le s^2 \mathrm{Var}_\nu(M_c) \le V_M$. Minkowski therefore gives $\sqrt{\mathrm{Var}_\nu(Y_c)} \le \sqrt{V_M} + D_\Delta$. Together with $1/(1-s\pi_{u,c}) \le a_B$, we obtain

$$\|\beta_{u,c}\|_{L^2(\nu)} \le a_B^2 \sqrt{V_\pi} + a_B e^{\Delta B + D_\Delta}(\sqrt{V_M} + D_\Delta) \le b_\Delta. \tag{125}$$

Since the coordinate blocks W_u are disjoint (we do not assume that the corresponding spins are mutually independent) and $\sum_u |W_u| \le \Delta^2$, (113) now gives

$$\mathrm{Var}_\nu\left(\sum_c z_c G_c\right) \le V_M b_\Delta^2 \|z\|_2^2.$$

This is (118), and the weighted covariance estimate (119) follows by conjugation. □

We now use these estimates to bound the response to a root-color change uniformly over residual instances.

Theorem 9.4 (Unrestricted girth-five coupling independence)**.** *For every fixed $0 < \delta \le 1$ there is a threshold $\Delta_5(\delta) \ge \Delta_0(\delta)$ such that, whenever $\Delta \ge \Delta_5(\delta)$ and $q \ge (1+\delta)\Delta$, every graph $G \in \mathcal{G}_\Delta$ and pinning τ with $\mathrm{girth}(G^\tau) \ge 5$ satisfies*

$$W_{1,\mathrm{Ham}}\left(\mu_{G,x}^{\tau, r=c}, \mu_{G,x}^{\tau, r=d}\right) \le C_5(q, \Delta, \delta) < \infty$$

for every $r \in V^\tau$, all $c, d \in [q]$, and every $x \in [0,1]$. The same constant applies to all graphs and pinnings in this class, and both laws are on $V^\tau \setminus \{r\}$.

Proof. For the response estimates, let $I = (H, \mathbf{b})$ be a residual instance of girth at least five satisfying the degree budget, and fix $\chi \ge 1$. Choose a positive weight vector $\mathbf{a} = (a_w)_{w \in V(H)}$ with $\chi^{-1} \le a_u/a_w \le \chi$ on every free edge. Let $f_I = \sum_w f_w(\sigma_w)$, where $f_w : [q] \to \mathbb{R}$ and $|f_w(c)| \le a_w$ for every color c. In a smaller residual instance J, f_J denotes the sum of the remaining source terms. All bounds below are uniform over further pinnings, deletions, and $x \in (0,1)$. We first derive the exact response identity, then prove uniform weighted 2- and ∞-norm bounds by induction, and finally convert those bounds into a root-color Hamming coupling.

Response recursion. For a free vertex v, put $U = N_H(v)$, $S = S_2^H(v)$, $O = V(H) \setminus (\{v\} \cup U \cup S)$, and $\nu = \mu_{I-v}$. Use F_c, R_c, G_c from Lemma 9.3 for this instance. Put $d = |U|$ and enumerate U as $u_1, \dots, u_d$. Write

$$h_{I,v}(c) = \frac{\sqrt{p_{I,v}(c)}}{1 - s p_{I,v}(c)} \left(\mathbb{E}_I[f_I \mid \sigma_v = c] - \mathbb{E}_I f_I\right). \tag{126}$$

For $n \geq 1$, let U_n and V_n be the suprema of $\|h_{I,v}\|_2/a_v$ and $\|h_{I,v}\|_\infty/(\sqrt{B}a_v)$ over all such instances with at most n free vertices, all admissible weights and sources, all free roots, and all $x \in (0,1)$. Set $U_0 = V_0 = 0$. These suprema are finite at each fixed n: components not containing v cancel from the centered response, and weights in the root component differ from a_v by at most χ^{n-1}. When analyzing a particular I, write $n = |V(H)|$.

For a real parameter t, define the tilted law and its marginals by

$$d\mu_{J,t} := \frac{e^{tf_J}}{\mathbb{E}_J[e^{tf_J}]}\, d\mu_J, \qquad p_{J,w}(c;t) := \mu_{J,t}(\sigma_w = c).$$

Thus $p_{J,w}(c;0) = p_{J,w}(c)$. Set $\nu_t := \mu_{I-v,t}$. Under this perturbation, the exact marginal identity is

$$p_{I,v}(c;t) \propto x^{b_v(c)} e^{t f_v(c)} \prod_{u \sim v} (1 - s p_{I-v,u}(c;t)) e^{e_{v,c}(t)},$$

where

$$e_{v,c}(t) = \log \mathbb{E}_{\nu_t} F_c - \sum_{u \sim v} \log(1 - s\mathbb{E}_{\nu_t} X_{u,c}), \qquad e'_{v,c}(0) = \mathrm{Cov}_\nu(f_{I-v}, R_c).$$

Set $D_v = \mathrm{diag}(\sqrt{p_{I,v}}/(1 - s p_{I,v}))$ and $e'_v = (e'_{v,c}(0))_{c \in [q]}$, with coordinatewise operations inside the diagonal. Write $f := f_{I-v}$ and $f_A := \sum_{w \in A} f_w(\sigma_w)$ for $A \in \{U, S, O\}$. Then $f = f_U + f_S + f_O$. Setting $F_S = \mathbb{E}_\nu[f \mid \sigma_S]$, the disintegration gives the exact source decomposition

$$e'_{v,c}(0) = \underbrace{\mathbb{E}_\nu[\mathrm{Cov}_\nu(f_U, R_c \mid \sigma_S)]}_{A_c} + \underbrace{\mathrm{Cov}_\nu(F_S, G_c)}_{C_c}. \tag{127}$$

The local term satisfies

$$|A_c| \leq 2\Delta B b_\Delta \chi a_v = \ell_\Delta \chi a_v, \tag{128}$$

using $|f_u| \leq a_u \leq \chi a_v$. To control the second term, reveal S one spin at a time. At one step, in the resulting smaller conditional instance J, let

$$g(c) = \mathbb{E}_J[f_J \mid \sigma_w = c] - \mathbb{E}_J f_J.$$

The conditional variance of this Doob increment is

$$\sum_c p_{J,w}(c) g(c)^2 = \sum_c (1 - s p_{J,w}(c))^2 h_{J,w}(c)^2 \leq \|h_{J,w}\|_2^2 \leq a_w^2 U_{n-1}^2. \tag{129}$$

The Doob increments are orthogonal, so summing over $w \in S$ gives

$$\mathrm{Var}_\nu(F_S) \leq \sum_{w \in S} a_w^2 U_{n-1}^2 \leq \Delta^2 \chi^4 a_v^2 U_{n-1}^2. \tag{130}$$

Differentiating the marginal identity at $t = 0$ gives

$$h_{I,v} = \sum_{u \sim v} J_{v,u} h_{I-v,u} + D_v(I_q - \mathbf{1}p_{I,v}^\mathsf{T})(f_v + e'_v), \quad J_{v,u} = -sD_v(I_q - \mathbf{1}p_{I,v}^\mathsf{T})\,\mathrm{diag}\,\sqrt{p_{I-v,u}}. \tag{131}$$

Set $t_\delta = (1+\delta)^{-1}$ and $\ell_B = -\log(1-B)/B$. Here is the weighted AM–GM calculation used below. Put $k_v = \sum_c b_v(c)$ and let $C_0 = \{c : b_v(c) = 0\}$, so that $Q := |C_0| \ge q - k_v \ge m + d$. Write $y_i(c) = sp_{I-v,u_i}(c) \in [0, B]$ and $A_c = \prod_i (1 - y_i(c))$. Define the reference root marginal

$$\bar{p}(c) := \frac{x^{b_v(c)} A_c}{\sum_d x^{b_v(d)} A_d}.$$

The insertion quotient bound above gives

$$p_{I,v}(c) = \frac{\bar{p}(c) e^{e_{v,c}(0)}}{\sum_d \bar{p}(d) e^{e_{v,d}(0)}}.$$

Thus $e^{-2E_\Delta} \le p_{I,v}(c)/\bar{p}(c) \le e^{2E_\Delta}$. Since the factors $x^{b_v(c)}$ are at most one, the normalizing denominator of $\bar{p}$ is at least $\sum_{c \in C_0} A_c$. For each i, $-\log(1 - y_i(c)) \le \ell_B y_i(c)$; hence AM–GM over C_0 gives

$$\sum_{c \in C_0} A_c \ge Q \left(\prod_{c \in C_0} A_c \right)^{1/Q} \ge Q \exp\!\left(-\frac{d \ell_B}{Q} \right).$$

For a fixed color c and $S_c = \sum_i y_i(c)$, ordinary AM–GM gives $A_c \le (1 - S_c/d)^d$ (the case $d = 0$ is void), while $S(1 - S/d)^d \le (d/(d+1))^{d+1} = d^{d+1}/(d+1)^{d+1} \le e^{-1}$ for $0 \le S \le d$, with equality at $S = d/(d+1)$. Therefore, with $\theta = d/Q$,

$$\bar{p}(c) S_c \le Q^{-1} e^{\theta \ell_B - 1}, \qquad d \max_c \bar{p}(c) S_c \le \theta e^{\theta \ell_B - 1} \le t_\delta e^{t_\delta \ell_B - 1},$$

because $Q \ge m + d$, $m \ge \delta d$, and $\theta \le (1+\delta)^{-1} = t_\delta$. Together with the insertion defect bound this yields

$$r_\Delta = e^{E_\Delta} \frac{\sqrt{t_\delta \exp(t_\delta \ell_B - 1)}}{1 - B},$$

which tends to $e^{-\lambda_\delta} < 1$, where $\lambda_\delta = \frac{1}{2}(\log(1+\delta) + \delta/(1+\delta))$. For completeness, write $p = p_{I,v}$, $r_i = p_{I-v,u_i}$, $\mathbf{z} = \sqrt{p}$, and $\Lambda = \max_c p(c) \sum_i r_i(c)$. Then

$$J_i = -s \operatorname{diag}\!\left(\frac{1}{1 - sp} \right) (I_q - \mathbf{z}\mathbf{z}^{\mathsf{T}}) \operatorname{diag}(\sqrt{p\, r_i}),$$

so the projection identity and Cauchy–Schwarz give

$$\left\| \sum_i J_i h_i \right\|_2 \le \frac{s\sqrt{d\Lambda}}{1 - B} \max_i \|h_i\|_2.$$

The weighted AM–GM bound above and $e^{-2E_\Delta} \le p(c)/\bar{p}(c) \le e^{2E_\Delta}$ turn this coefficient into r_Δ: indeed, if $\Lambda = \max_c p(c) \sum_i r_i(c)$, then

$$s^2 d\Lambda \le e^{2E_\Delta} \big(d \max_c \bar{p}(c)\, s \sum_i r_i(c) \big) \le e^{2E_\Delta} t_\delta e^{t_\delta \ell_B - 1},$$

using $s \le 1$. Taking the square root is why the insertion loss is e^{E_Δ} rather than e^{2E_Δ}. For the infinity norm, write coordinatewise

$$\left(\sum_i J_i h_i \right)_c = -\frac{s\sqrt{p(c)}}{1 - sp(c)} \left[\sum_i \sqrt{r_i(c)} h_i(c) - \sum_d p(d) \sum_i \sqrt{r_i(d)} h_i(d) \right].$$

The first (diagonal) term is bounded by $s\sqrt{d\Lambda}(1-B)^{-1}\max_i \|h_i\|_\infty$. For the second term, Cauchy–Schwarz in the joint indices (i,d) gives

$$\sum_{i,d} p(d)\sqrt{r_i(d)}|h_i(d)| \le \sqrt{d\Lambda}\max_i \|h_i\|_2, \qquad \sum_{i,d} p(d)^2 r_i(d) \le \Lambda,$$

and multiplying by $\sqrt{p(c)} \le \sqrt{B}$ yields

$$\sqrt{p(c)}\sum_d p(d)\sum_i \sqrt{r_i(d)}|h_i(d)| \le \sqrt{B}\,\sqrt{d\Lambda}\max_i \|h_i\|_2.$$

Multiplying by $s/(1-B)$ and applying the preceding coefficient bound gives the analogous infinity-norm estimate. These Jacobian blocks consequently satisfy

$$\left\|\sum_i J_{v,u_i}h_i\right\|_2 \le r_\Delta \max_i \|h_i\|_2, \qquad \left\|\sum_i J_{v,u_i}h_i\right\|_\infty \le r_\Delta \max_i \|h_i\|_\infty + r_\Delta\sqrt{B}\max_i \|h_i\|_2. \tag{132}$$

When $d=0$, the empty block sums and their norms are zero; maxima of nonnegative quantities over an empty neighbor set are also taken to be zero. The source term is controlled by the elementary bounds

$$\|D_v(I_q - \mathbf{1}p_{I,v}^{\mathsf{T}})g\|_2 \le a_B\sqrt{\mathrm{Var}_{p_{I,v}}(g)}, \qquad \|D_v(I_q - \mathbf{1}p_{I,v}^{\mathsf{T}})g\|_\infty \le 2a_B\sqrt{B}\|g\|_\infty. \tag{133}$$

Contractive induction. We now bound the previously defined U_n, V_n by induction on n. The full source $f_U + f_S + f_O$ is retained, including the terms outside the second layer. Revealing the spins in S one at a time gives Doob increments which are exactly scores in successively pinned instances satisfying the degree budget; hence the induction only uses smaller instances. Lemma 9.3, the two source bounds, and (132) give

$$U_n \le (r_\Delta\chi + a_B\widetilde{\epsilon}_\Delta\chi^2)U_{n-1} + a_B(1+\ell_\Delta\chi), \tag{134}$$

$$V_n \le r_\Delta\chi V_{n-1} + (r_\Delta\chi + 2a_BK_\Delta\chi^2)U_{n-1} + 2a_B(1+\ell_\Delta\chi). \tag{135}$$

Choose $\chi = e^{\lambda_\delta/4}$. For sufficiently large Δ, $A_V = r_\Delta\chi < 1$ and $A_U = r_\Delta\chi + a_B\widetilde{\epsilon}_\Delta\chi^2 < 1$. Thus

$$U^* = \frac{a_B(1+\ell_\Delta\chi)}{1-A_U}, \qquad V^* = \frac{(A_V + 2a_BK_\Delta\chi^2)U^* + 2a_B(1+\ell_\Delta\chi)}{1-A_V} \tag{136}$$

are finite supersolutions, and $U_n \le U^*$, $V_n \le V^*$ for all n.

From response bounds to coupling independence. Define M_n as the supremum, over the same instances, pinnings, weights, sources, free roots, and n-vertex size range used for U_n, V_n, of

$$\frac{|\mathbb{E}_I[f_I \mid \sigma_v = c] - \mathbb{E}_I[f_I \mid \sigma_v = d]|}{a_v}, \qquad c,d \in [q]. \tag{137}$$

To recover this unweighted root-color response, take differences of two logarithmic coordinates in (131) before dividing by a root marginal. This gives

$$M_n \le 2(1 + \ell_\Delta\chi + K_\Delta\chi^2 U_{n-1}) + 2\Delta B\chi V_{n-1},$$

so

$$M^* = 2(1 + \ell_\Delta\chi + K_\Delta\chi^2 U^*) + 2\Delta B\chi V^* < \infty. \tag{138}$$

Choosing source signs at the fixed real base point and applying (138) yields, for every admissible $\mathbf{a}$ and every pair of root colors,

$$\sum_{w\neq v} a_w \,\|\mu_{I,w}^{v=c} - \mu_{I,w}^{v=d}\|_{\mathrm{TV}} \le \frac{M^*}{2} a_v. \tag{139}$$

It remains only to perform the standard coupling conversion. Work in the connected component of the root, since other components have identical laws under both root pinnings. For a current residual graph H, first take $a_w = \chi^{\mathrm{dist}_H(v,w)}$. For the fixed-ambient condition in [CLMM23, Lemma 5.13], apply (139) after every further pinning, using the restriction of $a_w = \chi^{\mathrm{dist}_{H_0}(v,w)}$ from the original free graph H_0. These are separate applications; no comparison of the two distance functions is used. Choose R so that $M^*\chi^{-R}/2 \le (8R\log\Delta)^{-1}$. Restricting (139) to the fixed-ambient sphere and applying [CLMM23, Lemma 5.13] gives a graph-size-independent Hamming coupling. The endpoint $x = 0$ follows by finite-state continuity for the normalized child laws (proper list colorings exist because $q \ge \Delta + 1$); at $x = 1$ the root-excluded laws coincide directly. □

9.2 Spectral gap used in the CI proof

This subsection proves (113) on every residual instance satisfying (110) and having girth at least five. We use $\Delta_0(\delta)$ and γ_δ from (112), under the same hypothesis $q \ge (1+\delta)\Delta$. Throughout, P_v is conditional expectation given all spins except σ_v, and $Q_v := I - P_v$. Heat-bath updates occur at rate one per free vertex. All operator inequalities are on the supported L^2 space of the current Gibbs law: $A \succeq B$ means that $A - B$ is positive semidefinite. We write $\|f\|_\mu^2 := \mathbb{E}_\mu|f|^2$ and similarly for ρ. Unsubscripted function norms are in $L^2(\mu)$, and unsubscripted operator norms are the induced Hilbert-space norms. Finite color matrices instead carry the subscript $2\to 2$. The identity I acts on the space in question, and $\{A, B\} := AB + BA$ denotes the anticommutator.

Lemma 9.5 (Star bounds and singleton complement)**.** *Fix $0 < \delta \le 1$, $\Delta \ge \Delta_0(\delta)$, $q \ge (1+\delta)\Delta$, and $x \in [0,1]$, and put $s = 1 - x$. Consider a supported conditional star from a residual instance satisfying (110) at every free vertex. Its center has d free leaves and k fixed neighbors, with $d + k \le \Delta$. Write*

$$\mu(c,\sigma) = \frac{a_c}{Z}\prod_{i=1}^{d}\left[p_i(\sigma_i)(1 - s\mathbf{1}_{\{\sigma_i = c\}})\right], \qquad a_c = x^{b(c)}, \quad \sum_c b(c) = k,$$

where $b(c)$ counts fixed neighbors of color c and each p_i is the cavity law obtained after deleting the center edge. Here $c \in [q]$ is the center color and $\sigma = (\sigma_1, \dots, \sigma_d)$ is the leaf configuration. The normalizing constant is $Z = \sum_c a_c \prod_i (1 - sp_i(c))$. Let ν be the center marginal and $\rho = \bigotimes_i p_i$ the product cavity law on the leaves; μ_{leaves} denotes the actual leaf marginal. Set

$$B_{\mathrm{SG}} = \frac{1}{\delta\Delta}, \qquad \kappa = \frac{B_{\mathrm{SG}}^2}{(1 - B_{\mathrm{SG}})^2}, \qquad \omega_{\mathrm{SG}} = \frac{sB_{\mathrm{SG}}}{(1 - B_{\mathrm{SG}})^2}, \qquad c_\delta = \frac{\delta}{1+\delta}.$$

Then, uniformly over all supported stars (including $x = 0$),

$$sp_i(c) \le B_{\mathrm{SG}}, \quad s\nu(c) \le B_{\mathrm{SG}}, \quad \mathcal{R} := \frac{d\mu_{\mathrm{leaves}}}{d\rho} \ge c_\delta. \tag{140}$$

Let P be the center heat-bath projection and $Q = I - P$, and let P_i and $Q_i = I - P_i$ denote the heat-bath projection and its complement at leaf i. Decompose the mean-zero space into the conditional leaf Hoeffding sectors $K_0 \oplus K_1 \oplus K_{\ge 2}$: K_0 consists of mean-zero functions of the center, and K_1 is the orthogonal sum of the spaces of functions $g_i(c, \sigma_i)$ with $\mathbb{E}[g_i \mid c] = 0$, and $K_{\ge 2}$ is the sum of

higher conditional Hoeffding sectors, involving at least two distinct conditionally centered leaf factors and allowing coefficients that depend on c. Write $K_+ := K_1 \oplus K_{\geq 2}$. For a closed subspace $\mathcal{S}$, $\Pi_{\mathcal{S}}$ denotes orthogonal projection in $L^2(\mu)$. Block subscripts indicate the target and source sectors; for example, $P_{ab} := \Pi_{K_a} P|_{K_b}$ for $a, b \in \{0, +\}$. Put

$$C = P_{+0}(I - P_{00})^{-1} P_{0+}, \qquad H = I - P_{++} - C,$$

whenever $r := \|P_{00}\| < 1$, and let H_{11} be the compression of H to K_1. If $\mathcal{A}$ is the subspace generated by leaf-additive residuals $h_i(\sigma_i) - \mathbb{E}[h_i(\sigma_i) \mid c]$, and $\mathcal{B} = K_1 \ominus \mathcal{A}$, then

$$r \leq r_* := \frac{d\kappa(1 + \omega_{\mathrm{SG}})^{d-1}}{c_\delta}, \qquad H_{11}|_{\mathcal{B}} \succeq (1 - \eta) I, \quad \eta := r_* + \varepsilon_1,$$

where

$$\varepsilon_1 = \begin{cases} \dfrac{2(d-1)\kappa(1 + \omega_{\mathrm{SG}})^{d-2}}{c_\delta(1 - B_{\mathrm{SG}})}, & d \geq 2, \\ 0, & d \leq 1. \end{cases} \tag{141}$$

Moreover, for $d \geq 1$, put $\theta_0 = (1 + \delta/2)^{-1}$. Then

$$\Pi_{\mathcal{A}} C \Pi_{\mathcal{A}} \preceq \frac{\theta_0}{d} \Pi_{\mathcal{A}}. \tag{142}$$

Proof. If $d = 0$, then $\mathcal{A} = \{0\}$ and there is no additive sector to control. All estimates below containing $1/d$, including (142), are used only when $d \geq 1$. At a free vertex with k fixed and d free neighbors, let Z_{loc} be the sum of its q conditional color weights after the other spins are fixed. The degree budget gives $Z_{\mathrm{loc}} \geq q - s(k + d) \geq \delta\Delta$. Hence every one-site marginal satisfies $sp_i(c), s\nu(c) \leq B_{\mathrm{SG}}$. If $A_0 = \sum_c a_c$, then $A_0 \geq q - sk \geq s(d + \delta\Delta)$ and

$$\mathcal{R}(\sigma) = \frac{\sum_c a_c \prod_i (1 - s\mathbf{1}_{\{\sigma_i = c\}})}{\sum_c a_c \prod_i (1 - sp_i(c))} \geq \frac{A_0 - sd}{A_0} \geq \frac{\delta\Delta}{d + \delta\Delta} \geq c_\delta.$$

At $s = 0$ this is interpreted by continuity. At $x = 0$, center-color coordinates and sums weighted by ν may be restricted to $\mathcal{C}_* = \{c : \nu(c) > 0\}$; zero-$\nu$ terms vanish. Each leaf expectation and sum retains the full support of its own cavity law p_i.

For the complement estimate define

$$l_i^c = \frac{1 - s\mathbf{1}_{\{\sigma_i = c\}}}{1 - sp_i(c)}, \qquad \xi_i^c = l_i^c - 1, \qquad \mathcal{R} = \sum_c \nu(c) \prod_i l_i^c.$$

The one-leaf Gram matrix is

$$\Psi_i(c, c') = \mathbb{E}_{p_i}[\xi_i^c \xi_i^{c'}] = \frac{s^2(\mathbf{1}_{c=c'} p_i(c) - p_i(c)p_i(c'))}{(1 - sp_i(c))(1 - sp_i(c'))}.$$

It is positive semidefinite and

$$\|\operatorname{diag}(\sqrt{\nu})\Psi_i \operatorname{diag}(\sqrt{\nu})\|_{2\to 2} \leq \kappa, \qquad \max_c \Psi_i(c, c) \leq \omega_{\mathrm{SG}}. \tag{143}$$

The Schur product norm bound for positive semidefinite matrices therefore implies, for every nonempty $J \subseteq [d]$, the following bound, where $\circ$ denotes entrywise (Schur) multiplication:

$$\Big\| \operatorname{diag}(\sqrt{\nu}) \Big(\mathop{\circ}_{j \in J} \Psi_j \Big) \operatorname{diag}(\sqrt{\nu}) \Big\|_{2\to 2} \leq \kappa \omega_{\mathrm{SG}}^{|J|-1}. \tag{144}$$

If $g = \sum_i g_i \in K_1 \ominus \mathcal{A}$, then orthogonality to every $h_i(\sigma_i) - \mathbb{E}[h_i \mid c]$, together with $\mathbb{E}[g_i \mid c] = 0$, gives $\mathbb{E}_\mu[g_i \mid \sigma_i] = 0$. Bayes' formula therefore gives $\sum_c \nu(c) g_i(c, \sigma_i) l_i^c = 0$; this is the vanishing constant term in the following $L^2(\rho)$-orthogonal expansion (and $\mathbb{E}_{p_i}[g_i(c, \cdot) l_i^c] = 0$ ensures that the displayed terms indexed by distinct I are in distinct product Hoeffding sectors):

$$(Pg)\mathcal{R} = \sum_{\substack{I \subseteq [d] \\ |I| \geq 2}} \sum_{i \in I} \sum_c \nu(c) g_i(c, \sigma_i) l_i^c \prod_{j \in I \setminus \{i\}} \xi_j^c. \tag{145}$$

For fixed I, i, (144), followed by $(l_i^c)^2 \leq l_i^c/(1 - B_{\mathrm{SG}})$, yields

$$\left\| \sum_c \nu(c) g_i(c, \sigma_i) l_i^c \prod_{j \in I \setminus \{i\}} \xi_j^c \right\|_\rho^2 \leq \frac{\kappa \omega_{\mathrm{SG}}^{|I|-2}}{1 - B_{\mathrm{SG}}} \|g_i\|_\mu^2. \tag{146}$$

Summing the orthogonal Hoeffding components and using $\| \sum_{i \in I} u_i \|^2 \leq |I| \sum_i \|u_i\|^2$ gives

$$\|(Pg)\mathcal{R}\|_\rho^2 \leq \frac{2(d-1)\kappa(1 + \omega_{\mathrm{SG}})^{d-2}}{1 - B_{\mathrm{SG}}} \|g\|_\mu^2.$$

Since $\|Pg\|_\mu^2 = \mathbb{E}_\rho[((Pg)\mathcal{R})^2/\mathcal{R}]$, (140) gives $\|Pg\|_\mu^2 \leq \varepsilon_1 \|g\|_\mu^2$ with ε_1 as in (141). Together with $C \preceq rI$ below, this yields $H_{11}|_{\mathcal{B}} \succeq (1 - r_* - \varepsilon_1) I$.

For a mean-zero function h of the center alone, the expansion of $(Ph)\mathcal{R}$ starts in degree one; (144) and the same calculation give $\|Ph\|^2 \leq r_* \|h\|^2$, which is the bound on r. The projection identities $P_{0+}P_{+0} = P_{00} - P_{00}^2$ and $P_{++}P_{+0} = P_{+0}(I - P_{00})$ imply $H^2 = H$ and $0 \preceq C \preceq rI$.

It remains to prove (142). For a leaf-additive function $h = \sum_i h_i$, write $\bar{h} = \mathbb{E}[h \mid c]$ and $a = h - \bar{h}$. Conditional independence gives $a \in \mathcal{A}$ and $\|a\|^2 = \sum_i \mathbb{E} \operatorname{Var}(h_i \mid c)$. The exact block relations for $Ph = h$ imply

$$(I - P_{00})\bar{h} = P_{0+}a, \qquad \langle a, Ca \rangle = \|Q\bar{h}\|^2. \tag{147}$$

After subtracting constants from each h_i, normalize $\mathbb{E}_{p_i} h_i = 0$. Then $\bar{h}_i(c) := \mathbb{E}[h_i(\sigma_i) \mid c] = -s p_i(c) h_i(c)/(1 - s p_i(c))$ and a direct channel computation gives

$$\mathcal{D}_i := \mathbb{E} \operatorname{Var}(h_i \mid c) = \sum_c p_i(c) h_i(c)^2 \left[A_i - \frac{s\nu(c)}{(1 - s p_i(c))^2} \right], \quad A_i = \sum_c \frac{\nu(c)}{1 - s p_i(c)}.$$

Thus $\mathcal{D}_i \geq \kappa_{B_{\mathrm{SG}}} \sum_c p_i(c) h_i(c)^2$, where $\kappa_{B_{\mathrm{SG}}} = 1 - B_{\mathrm{SG}}/(1 - B_{\mathrm{SG}})^2 \geq 1 - 2B_{\mathrm{SG}}$. Cauchy–Schwarz in the color coordinate gives

$$\|Q\bar{h}\|^2 \leq \frac{s^2}{(1 - B_{\mathrm{SG}})^2} \max_c \left[\nu(c) \sum_i p_i(c) \right] \sum_{i,c} p_i(c) h_i(c)^2. \tag{148}$$

Let $u = sd/A_0$. Since $A_0 \geq s(d + \delta\Delta)$, $u \leq (1 + \delta)^{-1}$. Weighted AM–GM gives

$$Z \geq A_0 \exp[-u/(1 - B_{\mathrm{SG}})], \qquad a_c \sum_i s p_i(c) \prod_i (1 - s p_i(c)) \leq e^{-1}.$$

Consequently

$$\frac{ds^2}{(1 - B_{\mathrm{SG}})^2} \max_c \left[\nu(c) \sum_i p_i(c) \right] \leq \frac{u}{(1 - B_{\mathrm{SG}})^2} e^{u/(1 - B_{\mathrm{SG}}) - 1} \leq \frac{1}{1 + 3\delta/4}. \tag{149}$$

(The last elementary inequality follows from $B_{\mathrm{SG}} \le \delta/[8(1+\delta)]$.) Combining (148)–(149) yields

$$d\|Q\bar{h}\|^2 \le \frac{1}{(1+3\delta/4)\kappa_{B_{\mathrm{SG}}}}\|a\|^2 \le \frac{1}{1+\delta/2}\|a\|^2,$$

which, by (147), is exactly (142). All statements use only supported coordinates, so the endpoint $x = 0$ is included. □

Lemma 9.6 (Unequal-incidence star inequality)**.** *Under the hypotheses of Lemma 9.5, put*

$$\theta = \frac{1+\delta/4}{1+\delta/2} < 1.$$

If $\Delta \ge \Delta_0(\delta)$, then for every directed incidence choice $0 < \beta_i < 2$ and every supported conditional star,

$$T_\beta := Q + \tfrac{1}{2}\{Q, W\} + D \succeq \frac{1}{2}Q - \frac{\theta}{2d}\sum_i \beta_i^2 Q_i, \quad W = \sum_i \beta_i Q_i, \quad D = N^2 - N,\ N = \sum_i Q_i, \tag{150}$$

where θ is as above. When $d = 0$, the sum on the right is empty and the assertion means simply $T_\beta = Q \succeq \frac{1}{2}Q$; hence the displayed inequality is understood for $d \ge 1$.

Proof. We give the Schur-complement argument, including the constants. On K_+, put $K = \frac{1}{2}H + \frac{1}{2}(HW + WH) + D$. Let $\mathcal{A}$ and $\mathcal{B}$ be as above. The assertion $H\mathcal{A} = 0$ uses the special form of $\mathcal{A}$, not merely the fact that H is a projection. If $a = h - \mathbb{E}[h \mid c] \in \mathcal{A}$ with $h = \sum_i h_i(\sigma_i)$, then $Ph = h$ because h is independent of the center spin. Remove the global constant from h (which does not change a), and write $h = h_0 + a$ with $h_0 = \mathbb{E}[h \mid c] \in K_0$, the K_0 and K_+ blocks of $Qh = 0$ are

$$(I - P_{00})h_0 = P_{0+}a, \qquad (I - P_{++})a = P_{+0}h_0.$$

Consequently $Ca = P_{+0}(I - P_{00})^{-1}P_{0+}a = P_{+0}h_0$, and hence $Ha = (I - P_{++} - C)a = 0$. Moreover, $\mathcal{A} = \bigoplus_i \mathcal{A}_i$ with $\mathcal{A}_i = \{h_i(\sigma_i) - \mathbb{E}[h_i \mid c]\}$, so Q_i is the identity on $\mathcal{A}_i$ and vanishes on $\mathcal{A}_j$ for $j \ne i$. Thus W is diagonal on $\mathcal{A}$; since it is self-adjoint, it also preserves $\mathcal{A}^\perp = \mathcal{B} \oplus K_{\ge 2}$. On $\mathcal{B}$, writing $H_{11} = I - E$, $0 \preceq E \preceq \eta I$, one has

$$K_{11} \succeq (1/2 - 5\eta/2)I. \tag{151}$$

For $f \in K_{\ge 2}$, completing the square gives

$$\frac{1}{2}\langle f, Hf\rangle + \Re\langle Hf, Wf\rangle \ge -\frac{1}{6}\|(W - I)f\|^2.$$

Indeed, with $u = Hf$ and $b = (W - I)f$, the projection property $H^* = H^2 = H$ gives

$$\frac{1}{2}\langle f, Hf\rangle + \Re\langle Hf, Wf\rangle = \frac{3}{2}\|u\|^2 + \Re\langle u, b\rangle = \frac{3}{2}\left\|u + \frac{1}{3}b\right\|^2 - \frac{1}{6}\|b\|^2,$$

which proves the displayed bound. On a conditional Hoeffding sector of degree $n \ge 2$, each Q_i is either zero or the identity, so $0 \le W = \sum_i \beta_i Q_i \le 2nI$. Therefore $(W-I)^2 \preceq (2n-1)^2 I \preceq \frac{9}{2}n(n-1)I$, since the scalar difference is $\frac{9}{2}n(n-1) - (2n-1)^2 = \frac{1}{2}(n-2)(n+1) \ge 0$ for $n \ge 2$. Since W and D preserve these orthogonal sectors, this proves $(W - I)^2 \preceq \frac{9}{2}D$ on all of $K_{\ge 2}$. Applying the completed-square estimate to an arbitrary $f \in K_{\ge 2}$ therefore gives

$$\langle f, Kf\rangle \ge \langle f, Df\rangle - \frac{1}{6}\|(W - I)f\|^2 \ge \frac{1}{4}\langle f, Df\rangle.$$

Thus $K_{\ge 2,\ge 2} \succeq D/4$, including all cross-sector terms of H. The projection identity $H_{1,\ge 2}H_{\ge 2,1} = H_{11} - H_{11}^2$ gives $\|H_{1,\ge 2}\| \le \sqrt{\eta}$. Since $D \ge 2I$ on the higher sectors,

$$\|K_{1,\ge 2}D^{-1/2}\| \le \frac{1}{2}(3/\sqrt{2} + 2\sqrt{2})\sqrt{\eta} = \frac{7}{2\sqrt{2}}\sqrt{\eta}.$$

To record the numerical margin, set $X = K_{1,\ge 2}D^{-1/2}$. Since $\|X\|^2 \le (49/8)\eta$ and the higher block is at least $D/4$, retaining $D/8$ leaves a Schur-complement loss at most $8\|X\|^2 \le 49\eta$. The singleton block left after this elimination is therefore bounded below by

$$\left(\tfrac{1}{2} - \tfrac{5}{2}\eta\right) - \tfrac{1}{4} - 49\eta = \tfrac{1}{4} - \tfrac{103}{2}\eta \ge 0 \qquad (\eta \le 5/2048).$$

Hence

$$K \succeq \frac{1}{4}\Pi_{\mathcal{B}} + \frac{1}{8}D. \tag{152}$$

On the other hand, for $f \in K_+$ write $f = a + y$, where $a \in \mathcal{A}$ and $y \in K_+ \ominus \mathcal{A}$. Positivity of C and Young's inequality imply

$$\langle Wf, CWf\rangle \le (1+\epsilon)\langle Wa, CWa\rangle + (1+\epsilon^{-1})r\|Wy\|^2. \tag{153}$$

Because $W^2 = \sum_i \beta_i^2 Q_i$ on $\mathcal{A}$, (142) bounds $\langle Wa, CWa\rangle$ by $\theta_0\langle a, W^2 a\rangle/d$. Furthermore $\|Wy\|^2 \le 4\|y_1\|^2 + 8\langle y, Dy\rangle$, where $y_1 := \Pi_{\mathcal{B}}y$ is the singleton component of y. Take $\epsilon = \delta/4$. The threshold gives

$$r \le R_\delta := \frac{2(1+\delta)e^{2/\delta}}{\delta^3\Delta} \le \frac{\delta}{2048}, \qquad \eta \le 5/2048,$$

so $(1+\epsilon^{-1})r \le 1/32$. We now verify the claimed combination of (152) and (153). To make the Schur cancellation explicit, set $E_\beta = (\theta/d)\sum_i \beta_i^2 Q_i$ and consider $A := T_\beta - Q/2 + E_\beta/2$ on $K_0 \oplus K_+$. Its blocks are

$$\begin{aligned} A_{00} &= \tfrac{1}{2}(I - P_{00}), & A_{0+} &= -\tfrac{1}{2}P_{0+}(I+W), \\ A_{++} &= \tfrac{1}{2}(I - P_{++}) + W - \tfrac{1}{2}\{P_{++}, W\} + D + \tfrac{1}{2}E_\beta. \end{aligned}$$

Since $A_{00}^{-1} = 2(I - P_{00})^{-1}$, eliminating K_0 subtracts $\frac{1}{2}(I+W)C(I+W)$. Using $H = I - P_{++} - C$ and expanding the anticommutator, the resulting block is

$$\tfrac{1}{2}H + \tfrac{1}{2}\{H, W\} + D - \tfrac{1}{2}WCW + \tfrac{1}{2}E_\beta = K - \tfrac{1}{2}WCW + \tfrac{1}{2}E_\beta.$$

$$K - \tfrac{1}{2}WCW + \frac{\theta}{2d}\sum_i \beta_i^2 Q_i \succeq 0. \tag{154}$$

Eliminating the degree-zero block (whose Schur complement is $-\frac{1}{2}WCW$) proves (150). This step is valid because the correction vanishes on K_0. □

Theorem 9.7 (Uniform spectral gap under the degree budget)**.** *Fix $0 < \delta \le 1$, $\Delta \ge \Delta_0(\delta)$ and $q \ge (1+\delta)\Delta$. Every finite Potts instance on a simple graph with $\deg_H(v) + k_H(v) \le \Delta$ at each free vertex, after any normalized pinning whose free residual graph has girth at least five, and for every $x \in [0,1]$, has the nonnegative rate-one Glauber Laplacian $\mathcal{L} := \sum_v (I - P_v)$ satisfying*

$$\mathcal{L}^2 \succeq \gamma_\delta \mathcal{L}, \qquad \gamma_\delta = \frac{\delta}{4(2+\delta)}. \tag{155}$$

Proof. For each free vertex v let d_v be its current free degree and choose, on an edge vu,

$$\beta_{vu} = \frac{2d_v}{d_v + d_u}, \qquad W_v := \sum_{u\sim v} \beta_{vu} Q_u, \qquad N_v := \sum_{u\sim v} Q_u, \qquad T_v := Q_v + \tfrac{1}{2}\{Q_v, W_v\} + N_v^2 - N_v.$$

Girth at least five implies that every distance-two pair has a unique common neighbor and that adjacent vertices have no common neighbor. Therefore

$$\mathcal{L}^2 = \sum_v T_v + 2 \sum_{\{u,w\}:\, \mathrm{dist}_H(u,w)\geq 3} Q_u Q_w. \tag{156}$$

The last sum is positive semidefinite because nonadjacent heat-bath projections commute. For an edge vu, both endpoint degrees are positive, and

$$\frac{\beta_{vu}^2}{d_v} = \frac{1}{d_u}(1 - (\beta_{vu} - 1)^2), \qquad \sum_{v\sim u} \frac{\beta_{vu}^2}{d_v} \leq 1. \tag{157}$$

For an isolated vertex u, the neighbor sum is empty. Applying (150) in every conditional-star fiber and summing with (157) shows

$$\mathcal{L}^2 \succeq \frac{1-\theta}{2}\mathcal{L} = \frac{\delta}{4(2+\delta)}\mathcal{L}.$$

At $x > 0$ the supported chain is irreducible. At $x = 0$, remove from the color list of a free vertex v the colors used by its k_v fixed neighbors, and denote the resulting list by L_v and its size by ℓ_v. Since at most k_v colors are removed,

$$\ell_v - d_v \;\geq\; q - k_v - d_v \;\geq\; q - \Delta \;\geq\; \delta\Delta \;\geq\; 2,$$

where the last inequality is part of the choice of $\Delta_0(\delta)$. Thus $\ell_v \geq d_v + 2$ for every free vertex. For completeness, connectivity follows by induction on the number of free vertices. Given two proper list-colorings α, β, choose a vertex v and put $c = \beta(v)$. For every neighbor u of v with current color $\alpha(u) = c$, recolor u (one at a time) to a color in L_u different from c and from all colors currently used by the neighbors of u; at most $d_u + 1$ colors are forbidden and $\ell_u \geq d_u + 2$. Now recolor v to c. Keep v fixed at c and apply the induction hypothesis to $H - v$ with the reduced lists $L'_u = L_u \setminus \{c\}$ for neighbors of v and $L'_u = L_u$ otherwise. These lists satisfy $|L'_u| \geq \deg_{H-v}(u) + 2$, and the current and target restrictions are proper list-colorings. The base case has one vertex. Thus the proper-list-coloring support is connected, and (155) is the ordinary spectral-gap bound at the endpoint as well. □

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